



(12) **United States Patent**
Babaei

(10) **Patent No.:** **US 12,402,126 B2**
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(54) **PUCCH CELL SWITCHING AND CSI REPORTING**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 3 days.
This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/382,691**

(22) Filed: **Oct. 23, 2023**

(65) **Prior Publication Data**
US 2024/0057077 A1 Feb. 15, 2024

Related U.S. Application Data
(63) Continuation of application No. 18/075,056, filed on Dec. 5, 2022, now Pat. No. 11,832,246.
(60) Provisional application No. 63/287,324, filed on Dec. 8, 2021.
(51) **Int. Cl.**
H04W 72/1268 (2023.01)
H04L 27/26 (2006.01)
H04W 72/54 (2023.01)
(52) **U.S. Cl.**
CPC ... **H04W 72/1268** (2013.01); **H04L 27/26025** (2021.01); **H04W 72/54** (2023.01)
(58) **Field of Classification Search**
CPC H04W 72/1268; H04W 27/26025; H04W 72/54
See application file for complete search history.

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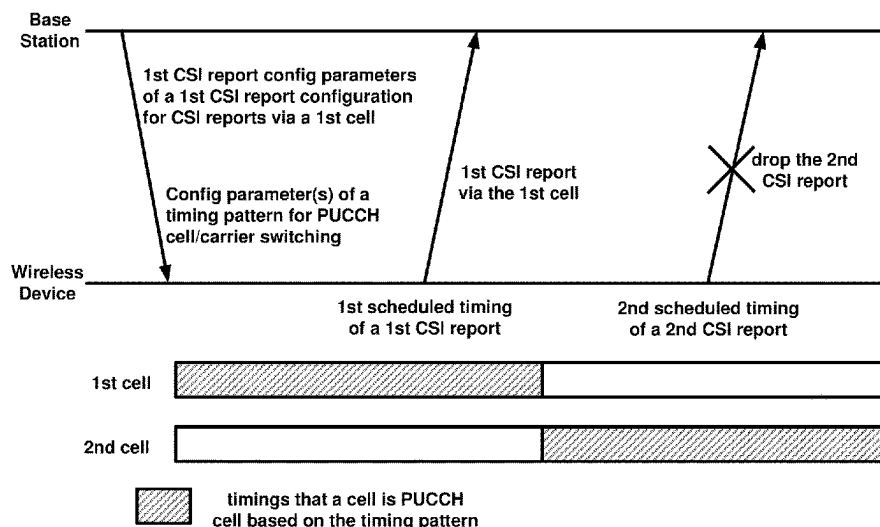
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Primary Examiner — Parth Patel
(74) *Attorney, Agent, or Firm* — Alireza Babaei

(57) **ABSTRACT**

A base station may transmit CSI report configuration parameters for CSI reports via a first cell. The base station may further transmit a time pattern configuration parameter indicating first timings that the first cell is a PUCCH cell and second timings that a second cell is the PUCCH cell. The base station may receive a CSI report, scheduled for transmission in a report timing via the first cell, based on the report timing being within the first timings and based on the report timing not being within the second timings.

20 Claims, 38 Drawing Sheets



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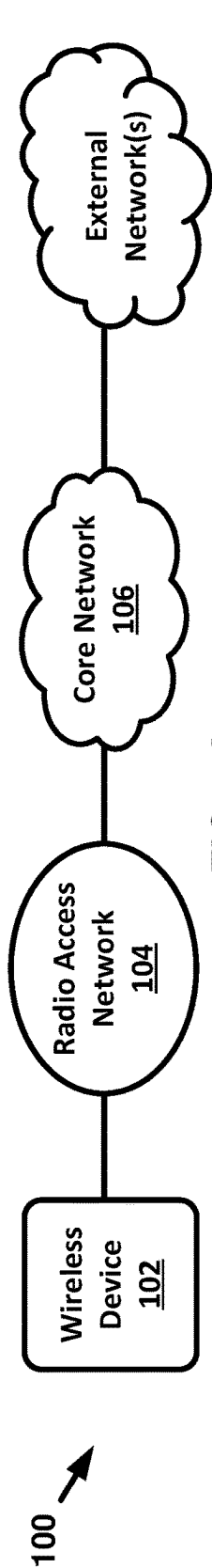


FIG. 1A

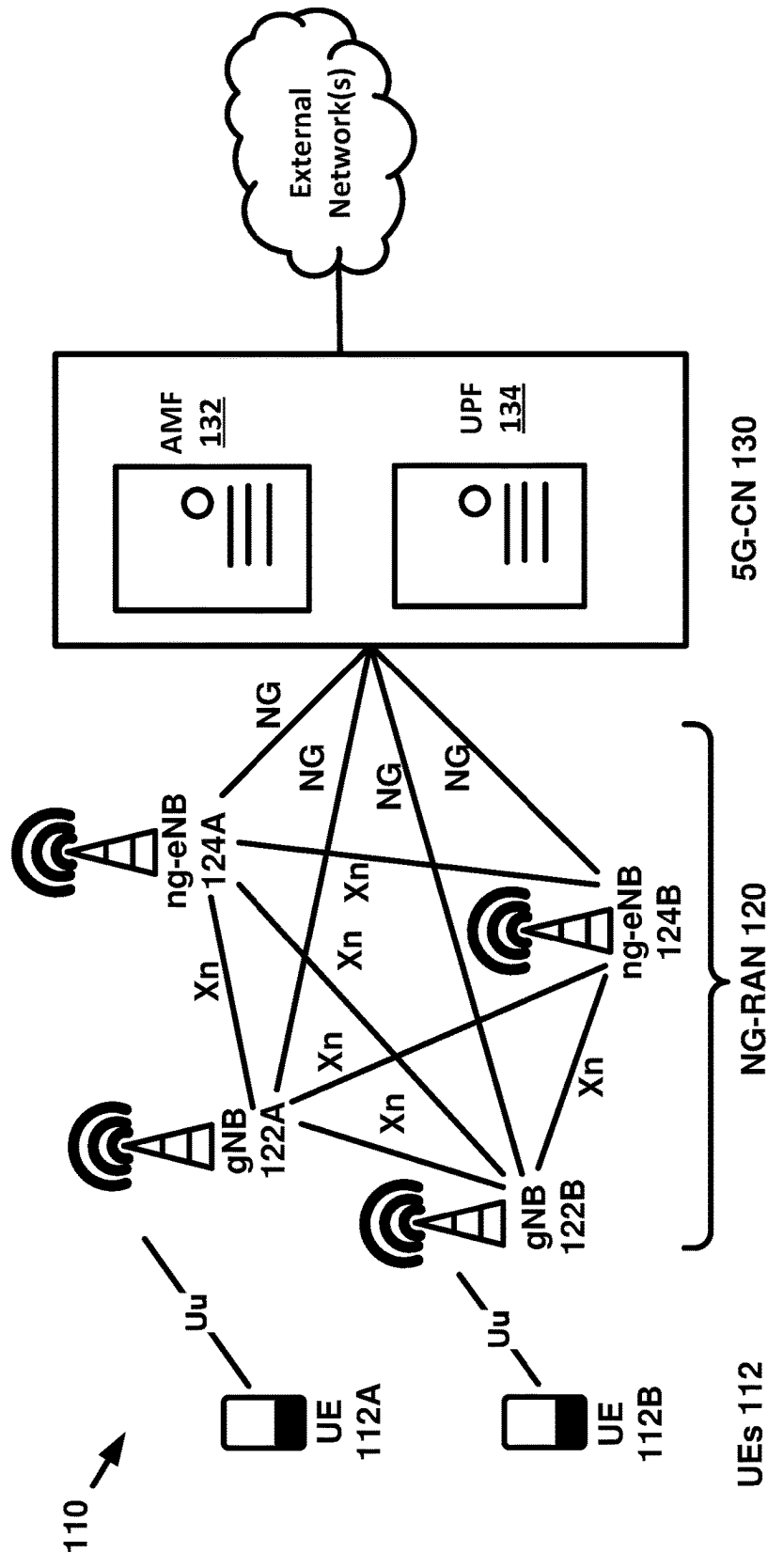


FIG. 1B

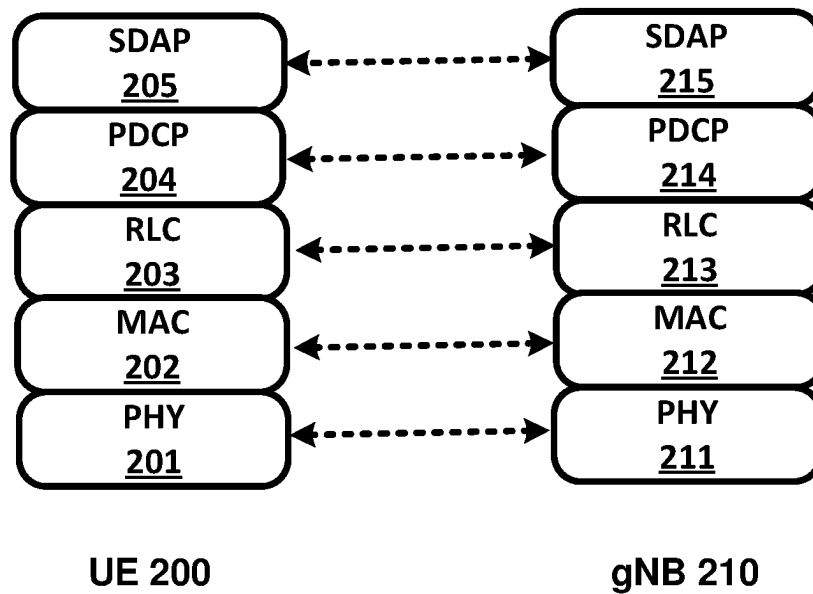


FIG. 2A

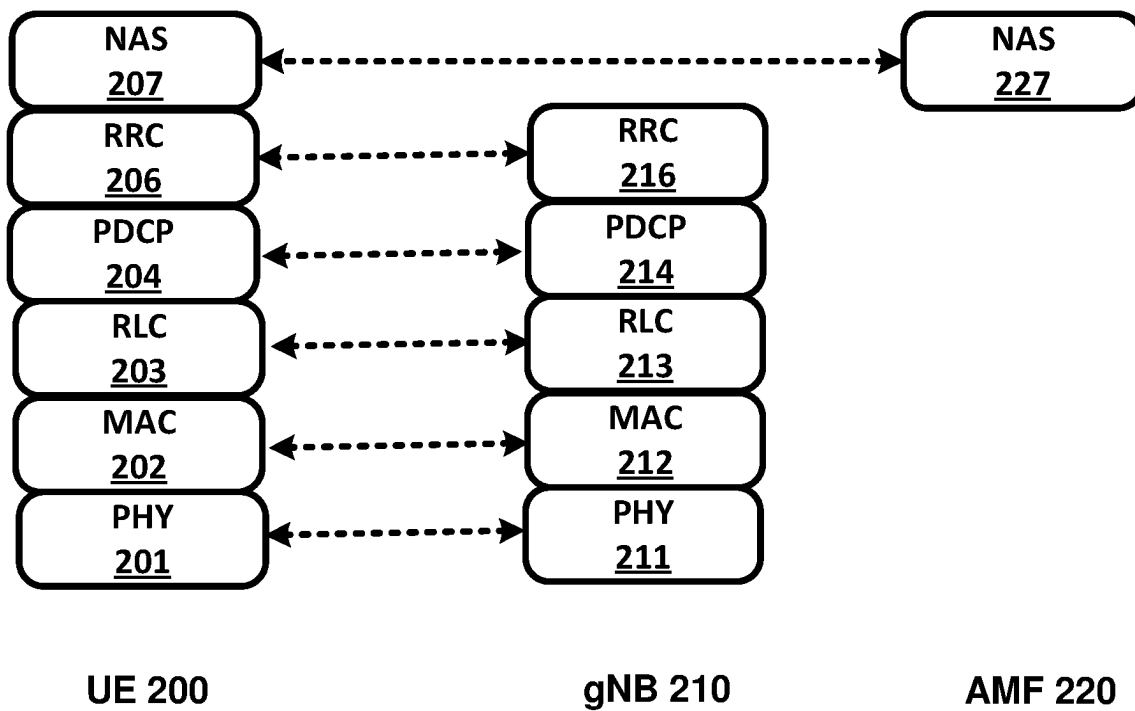


FIG. 2B

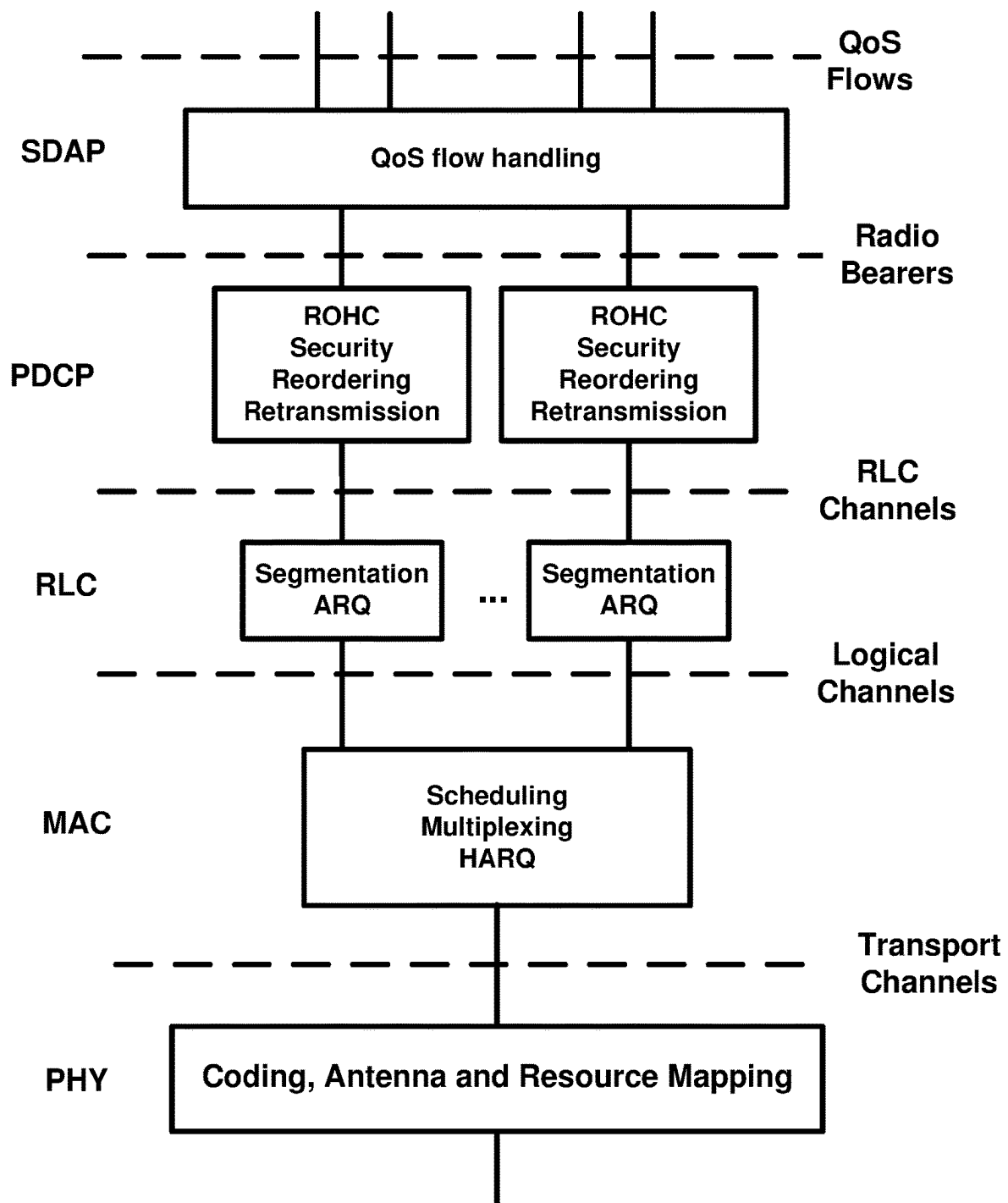


FIG. 3

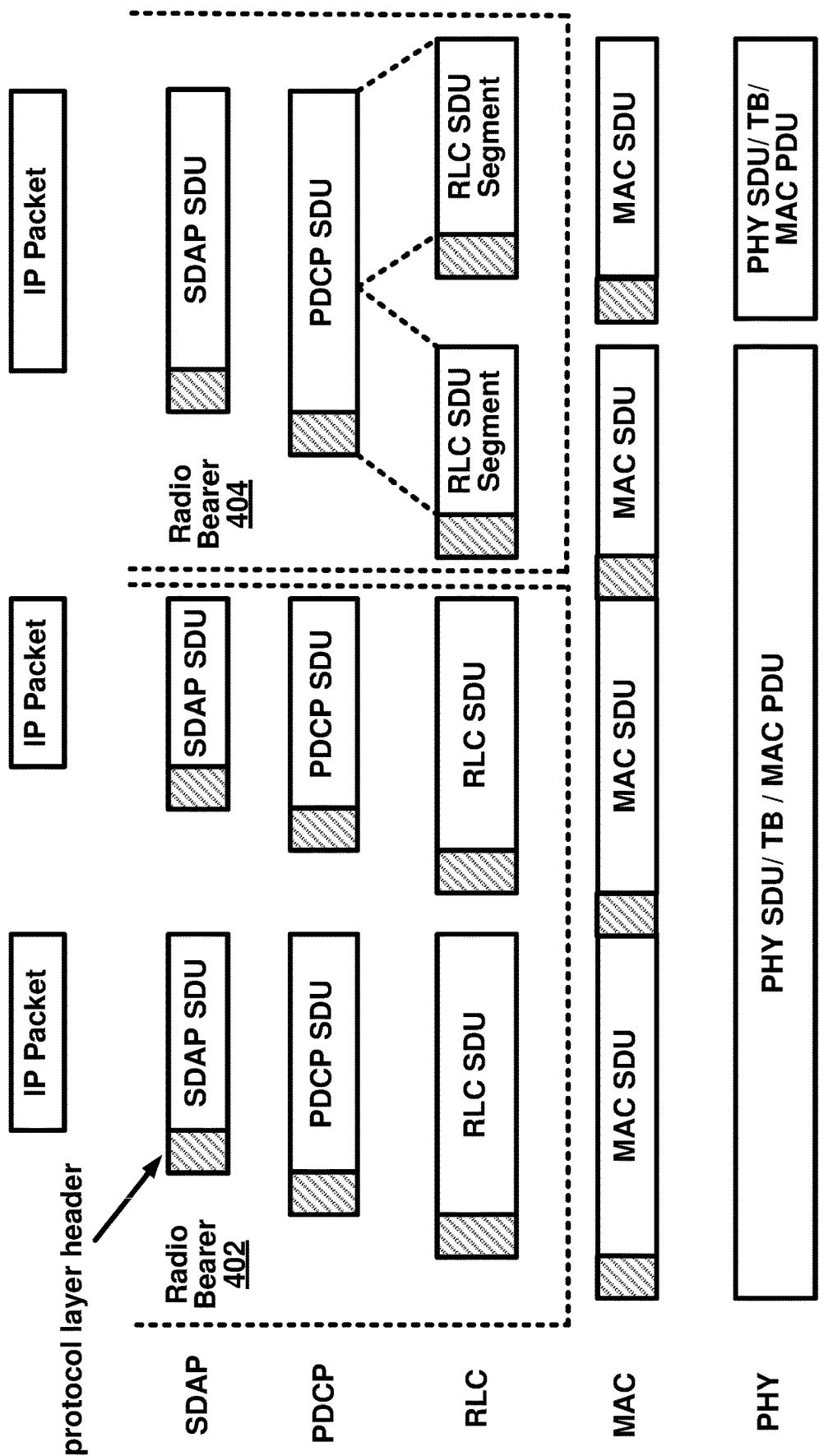


FIG. 4

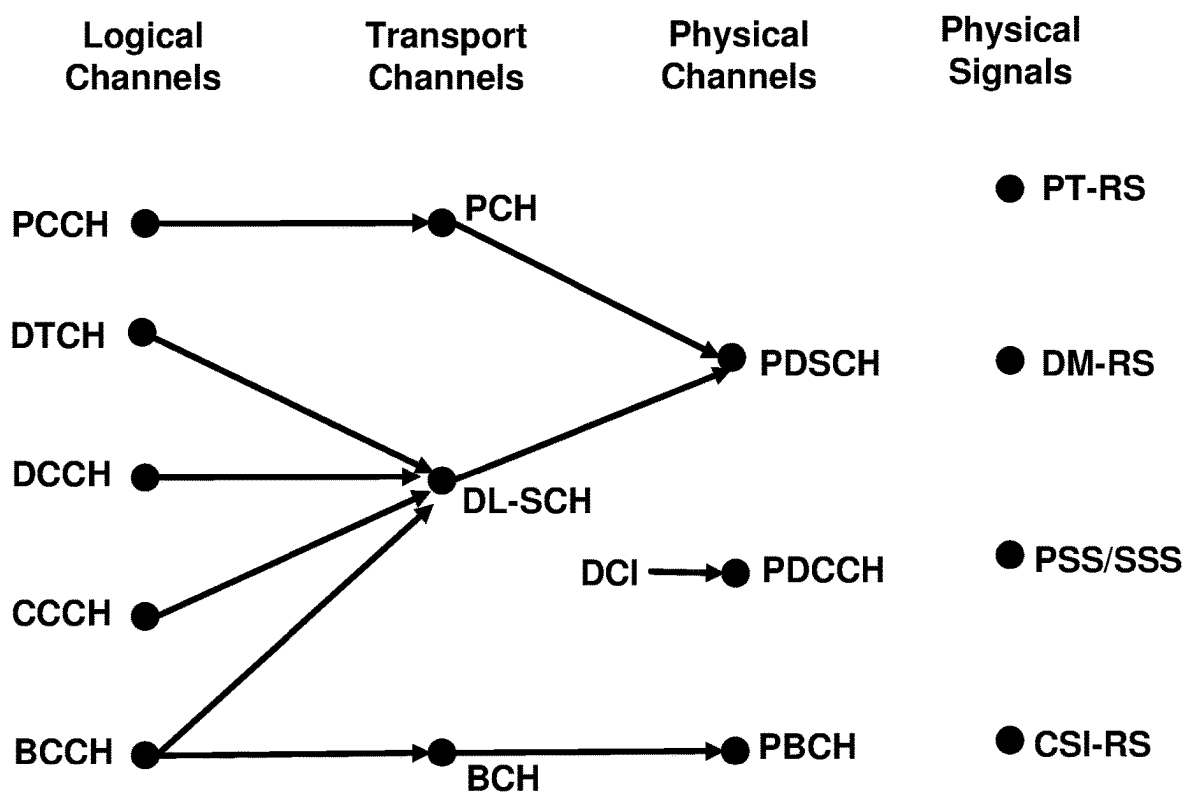


FIG. 5A

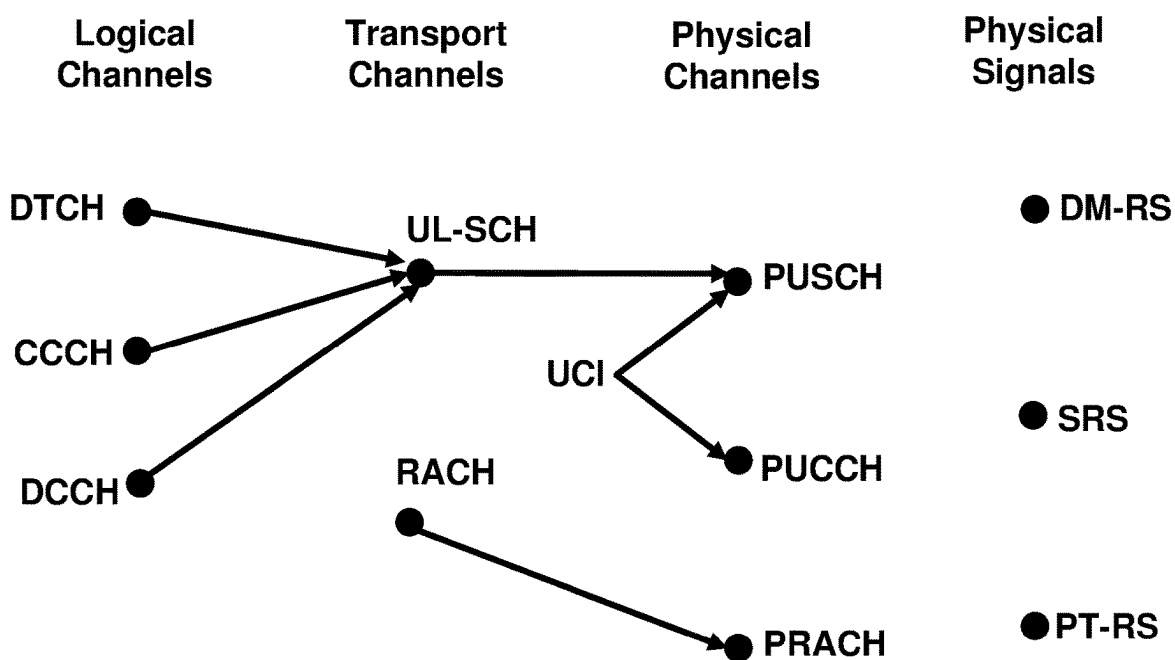


FIG. 5B

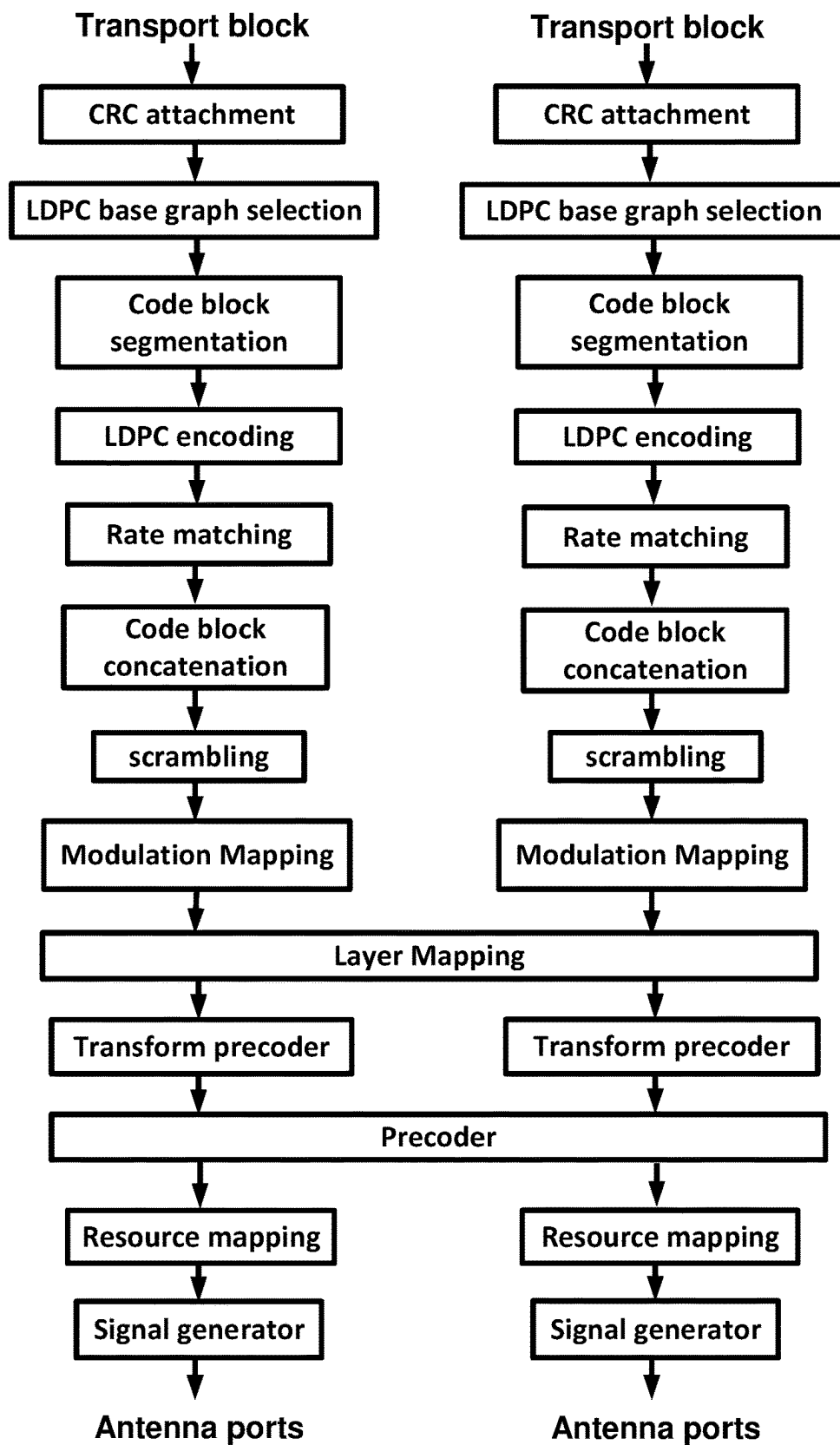


FIG. 6

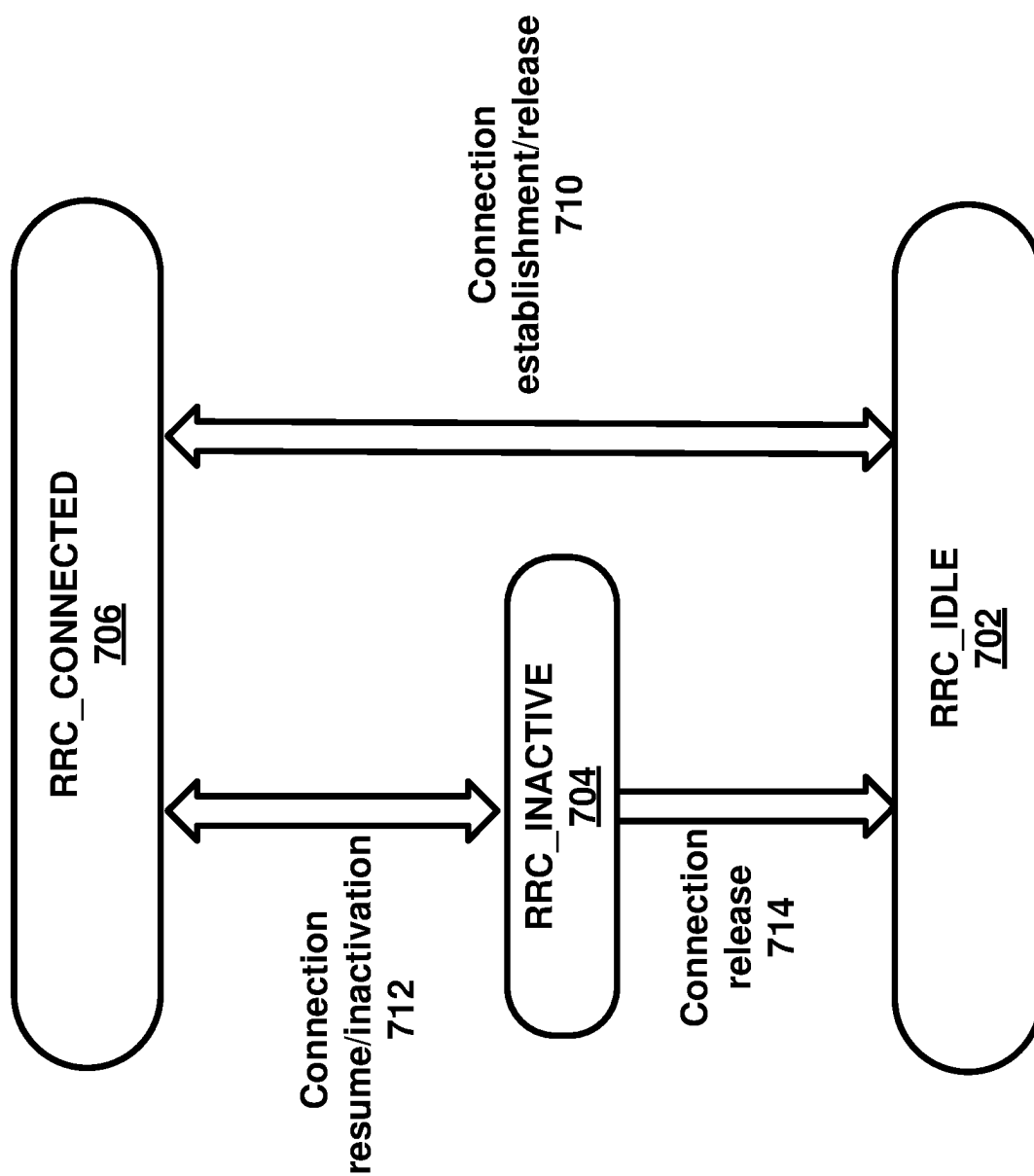


FIG. 7

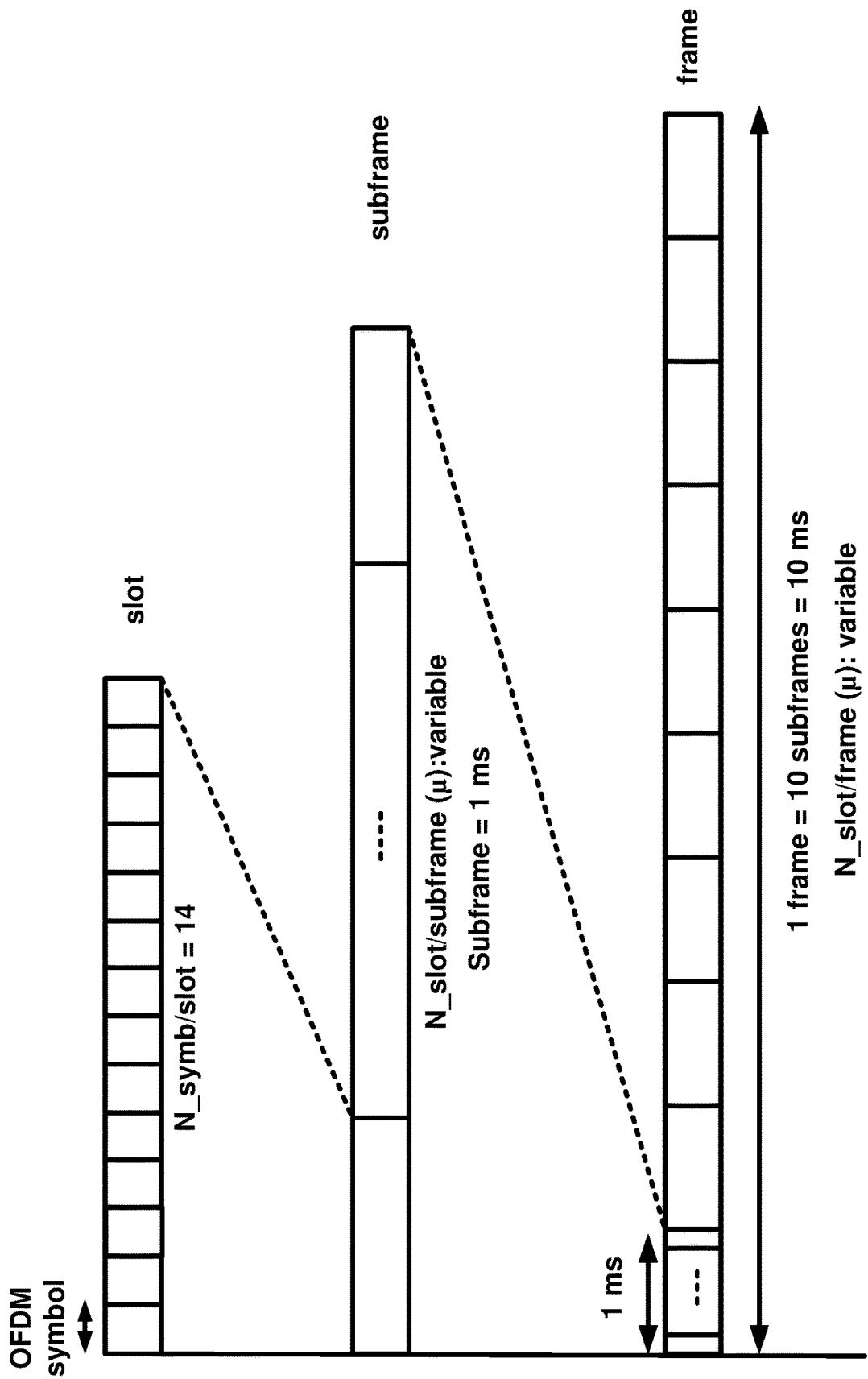


FIG. 8

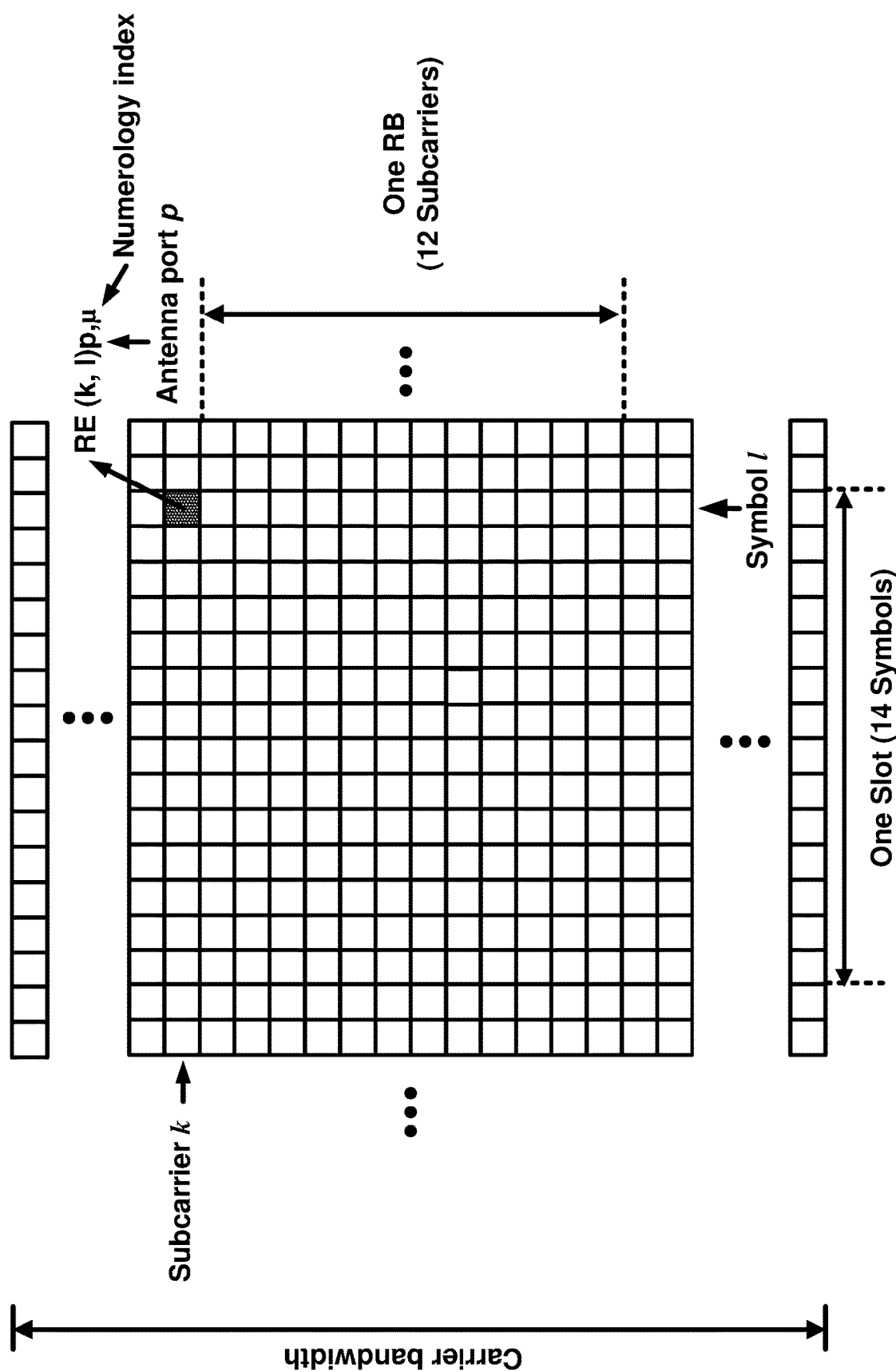


FIG. 9

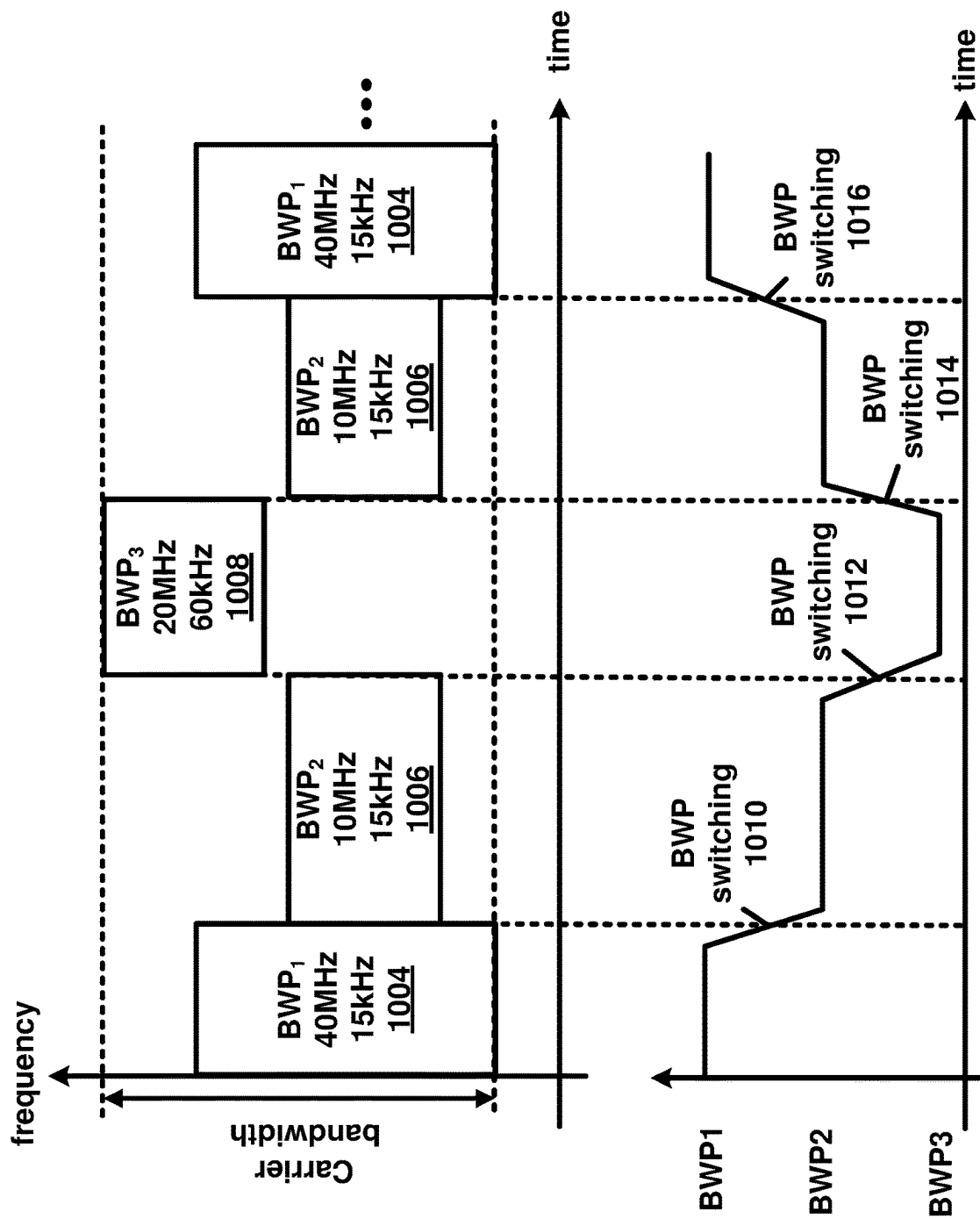


FIG. 10

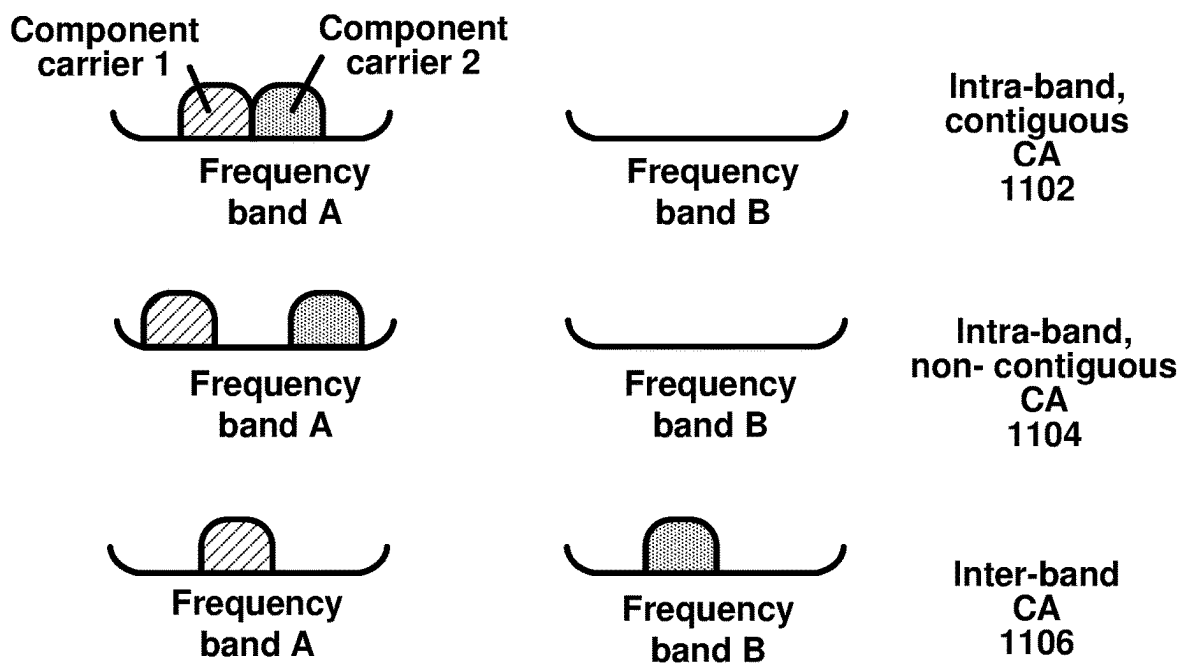


FIG. 11A

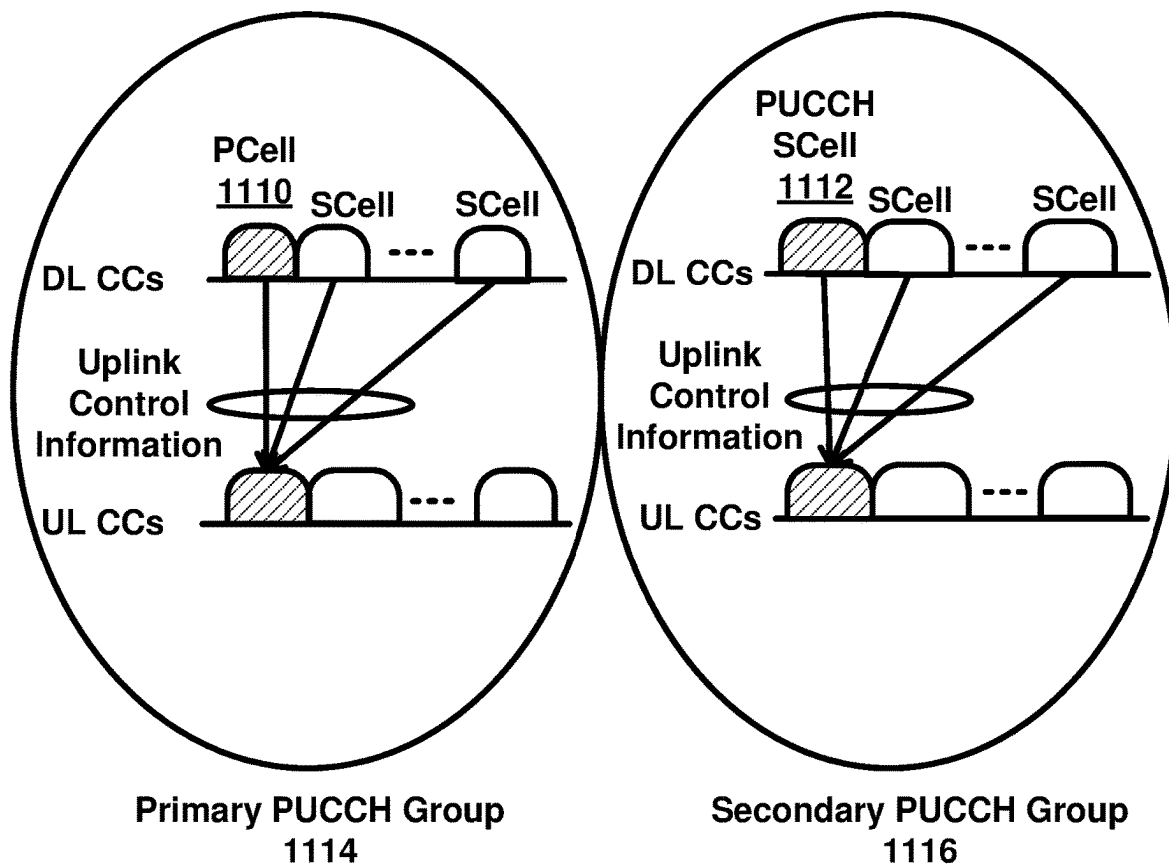


FIG. 11B

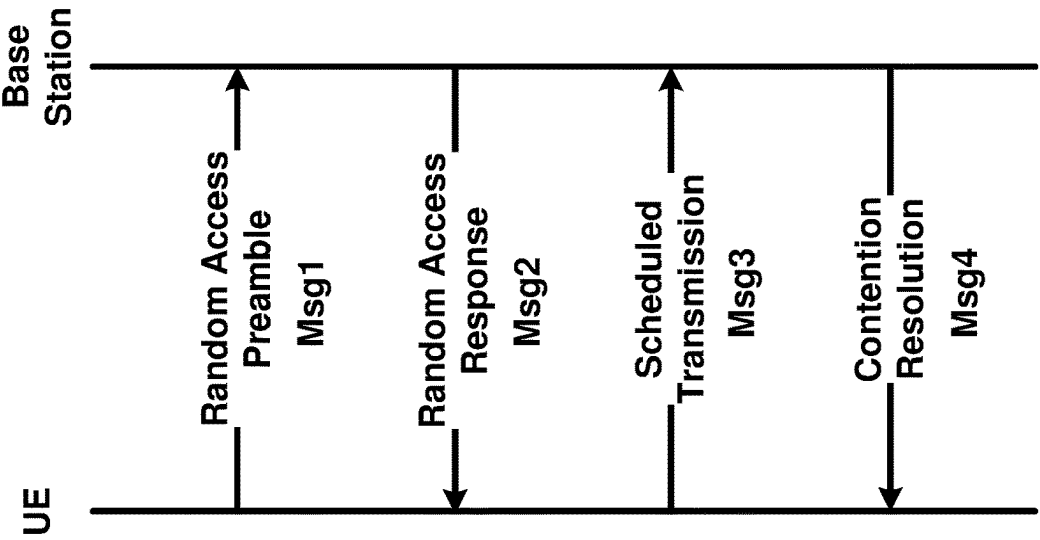


FIG. 12A

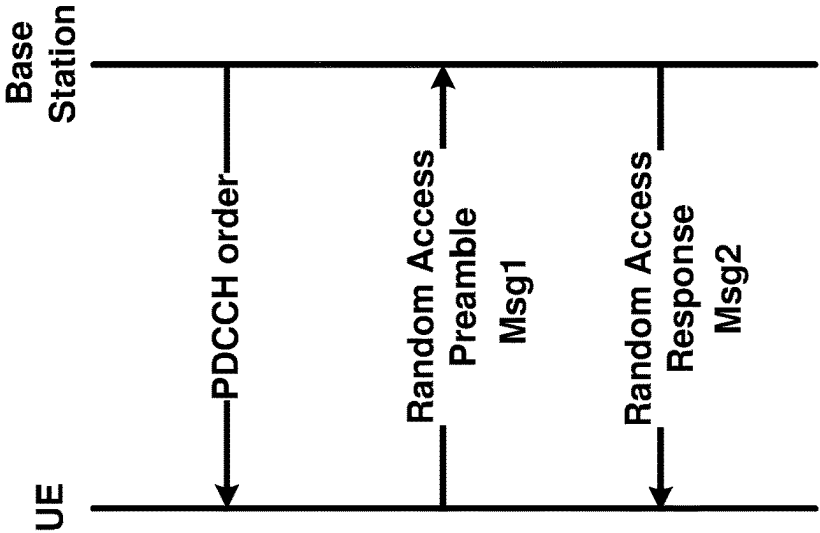


FIG. 12B

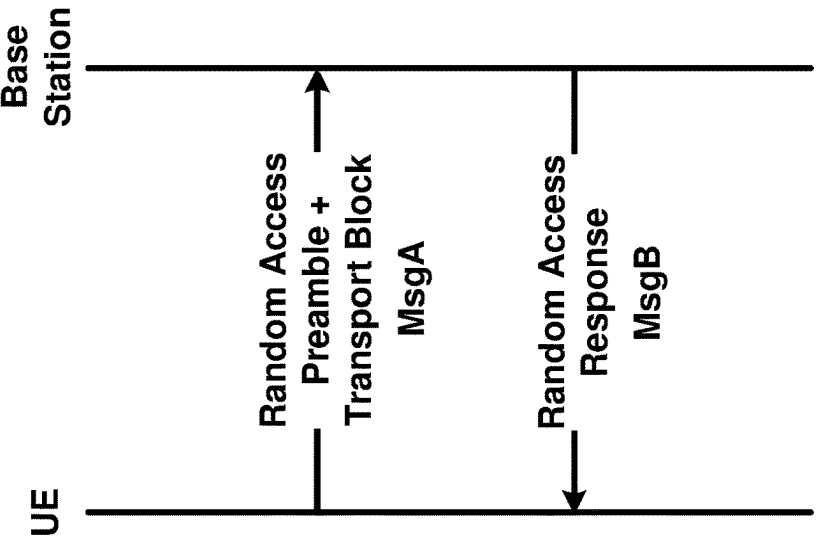


FIG. 12C

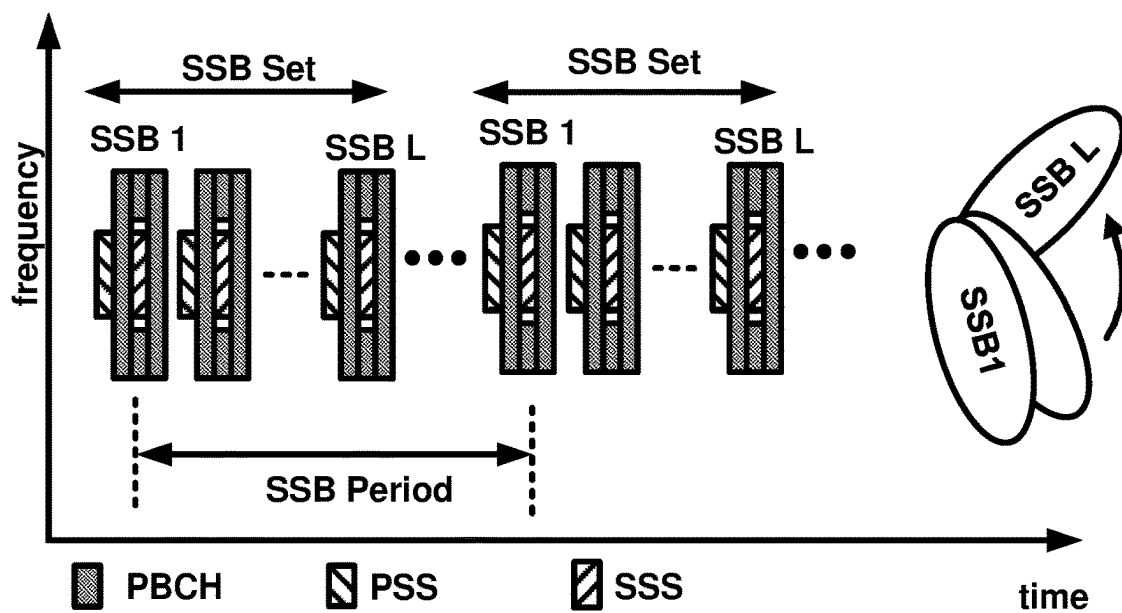


FIG. 13A

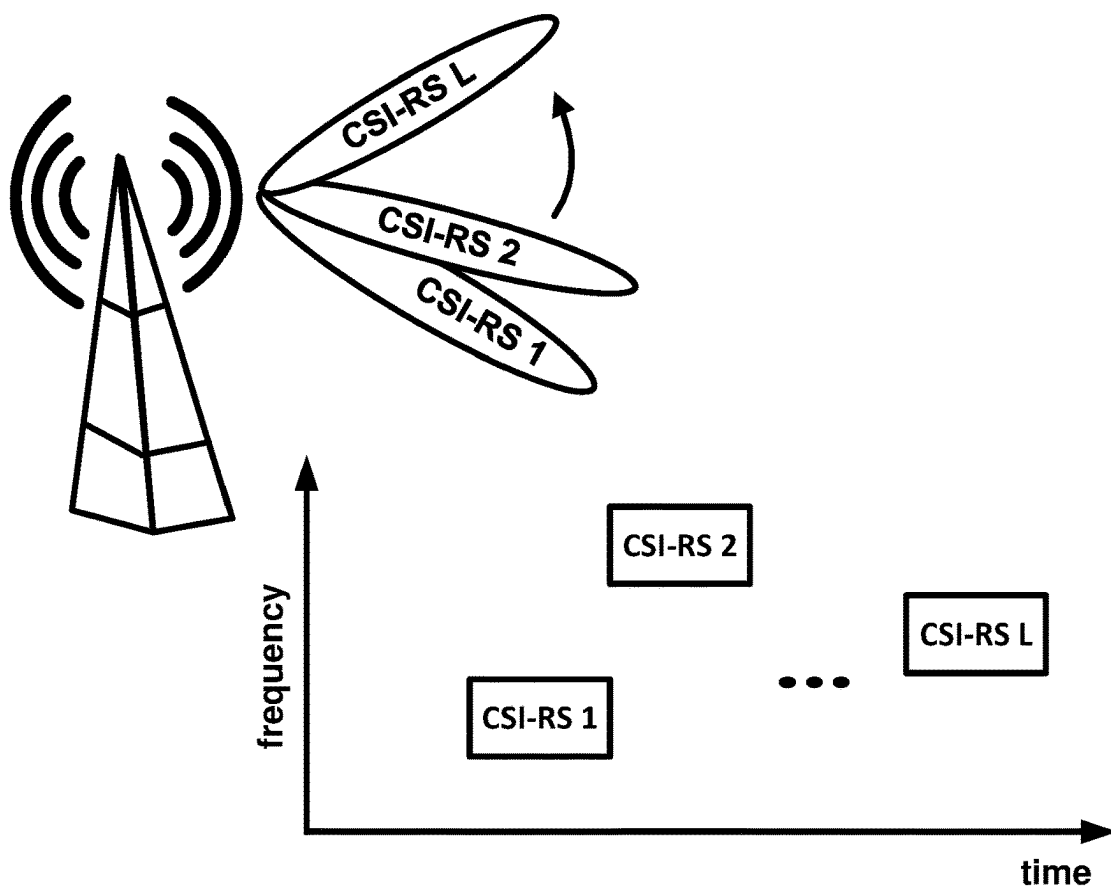


FIG. 13B

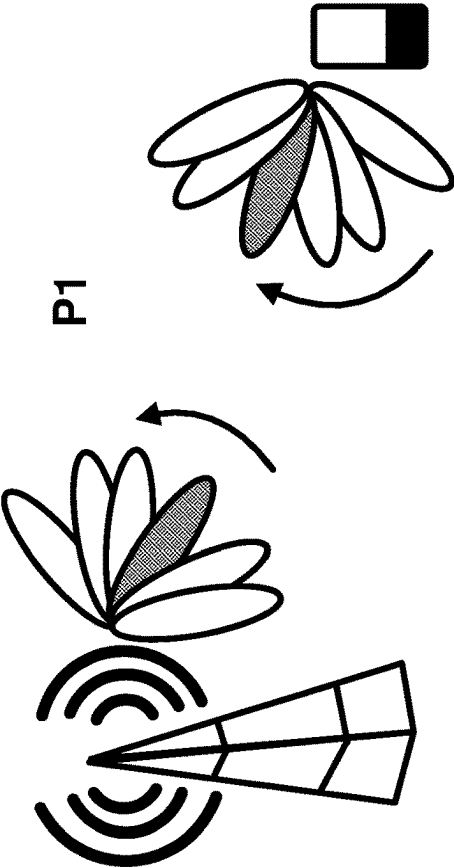


FIG. 14A

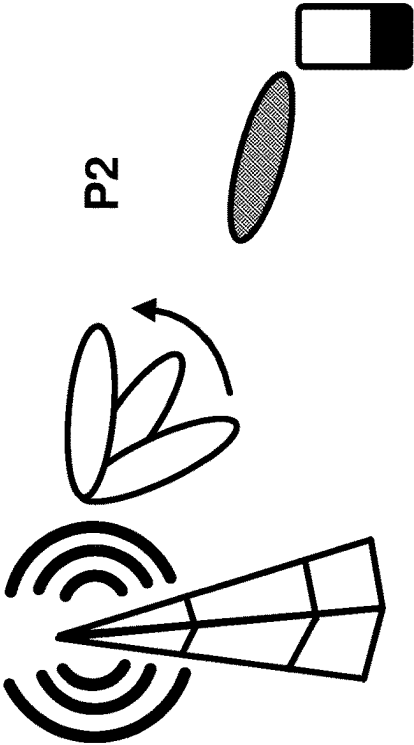


FIG. 14B

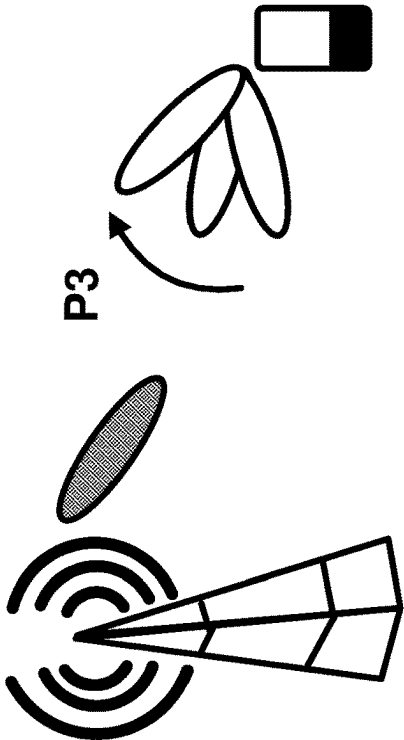


FIG. 14C

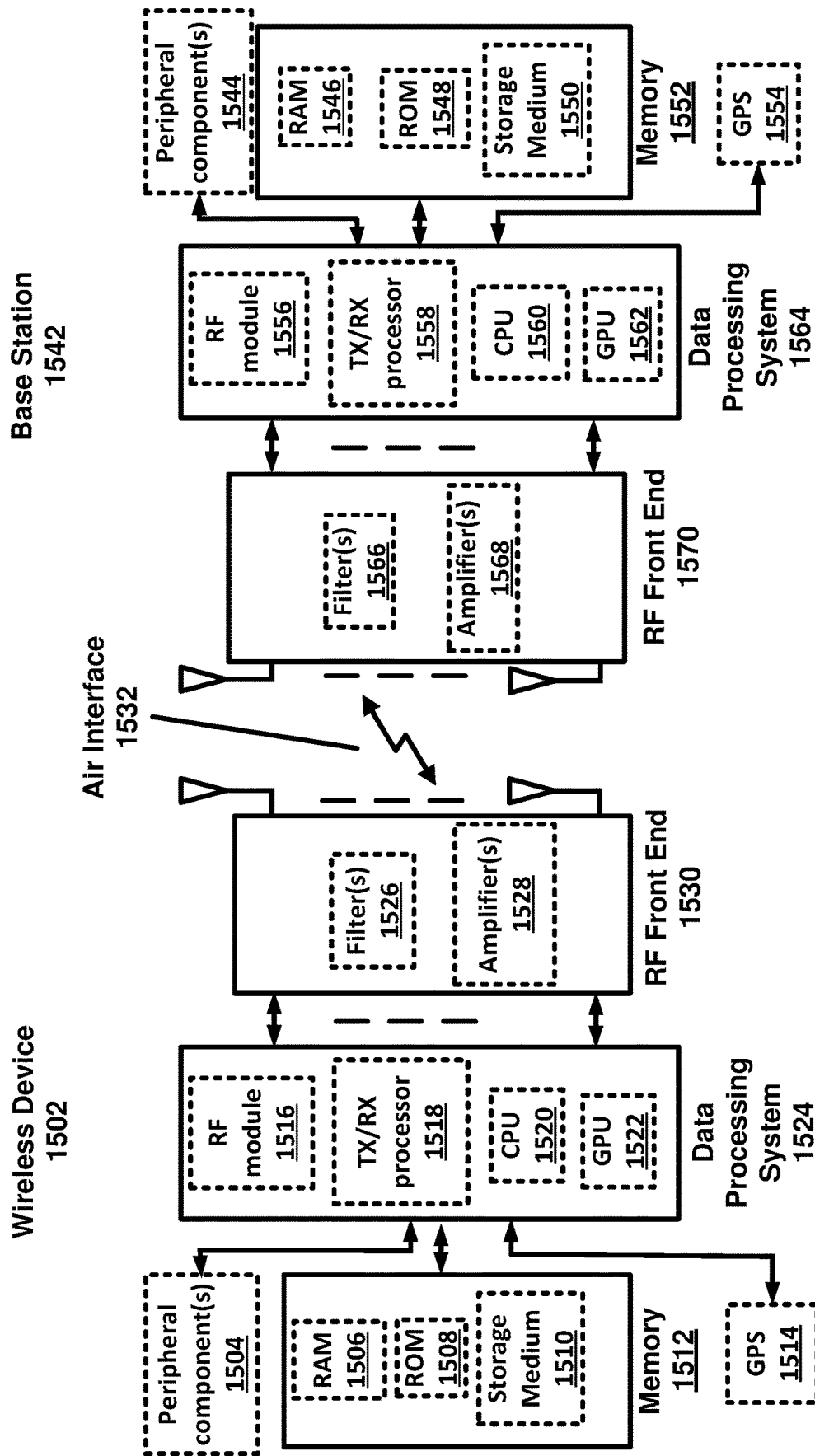


FIG. 15

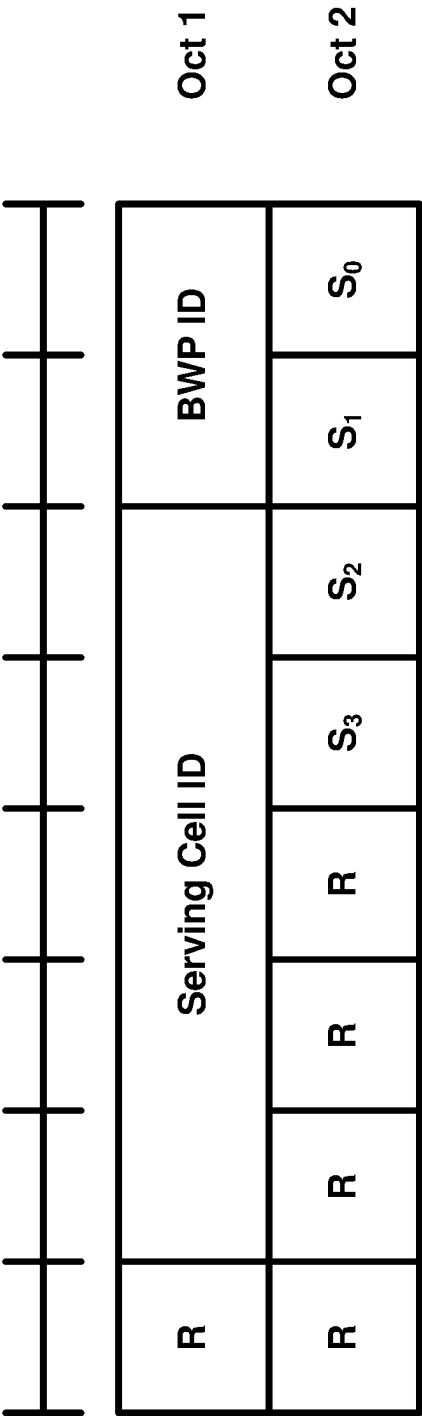


FIG. 16

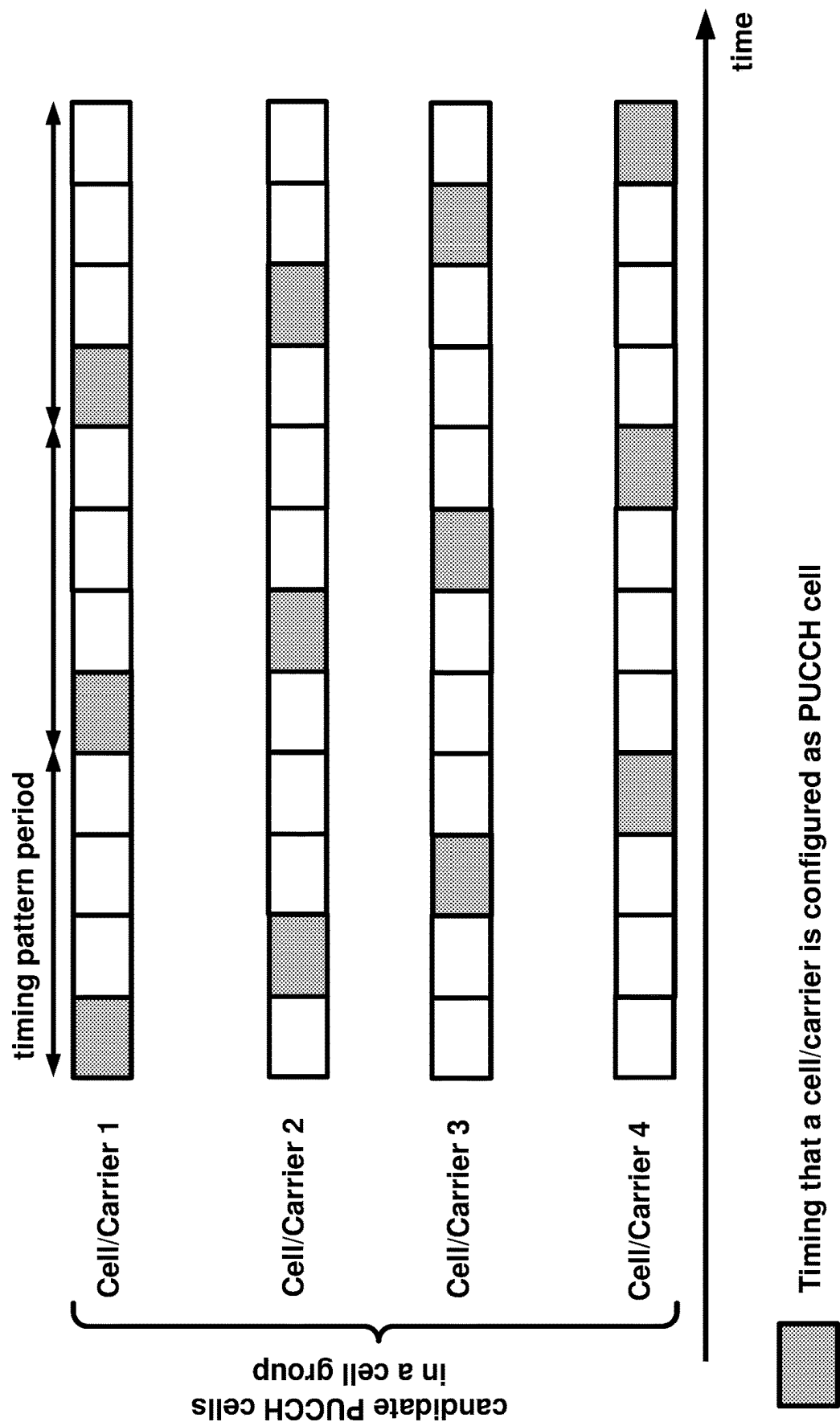


FIG. 17

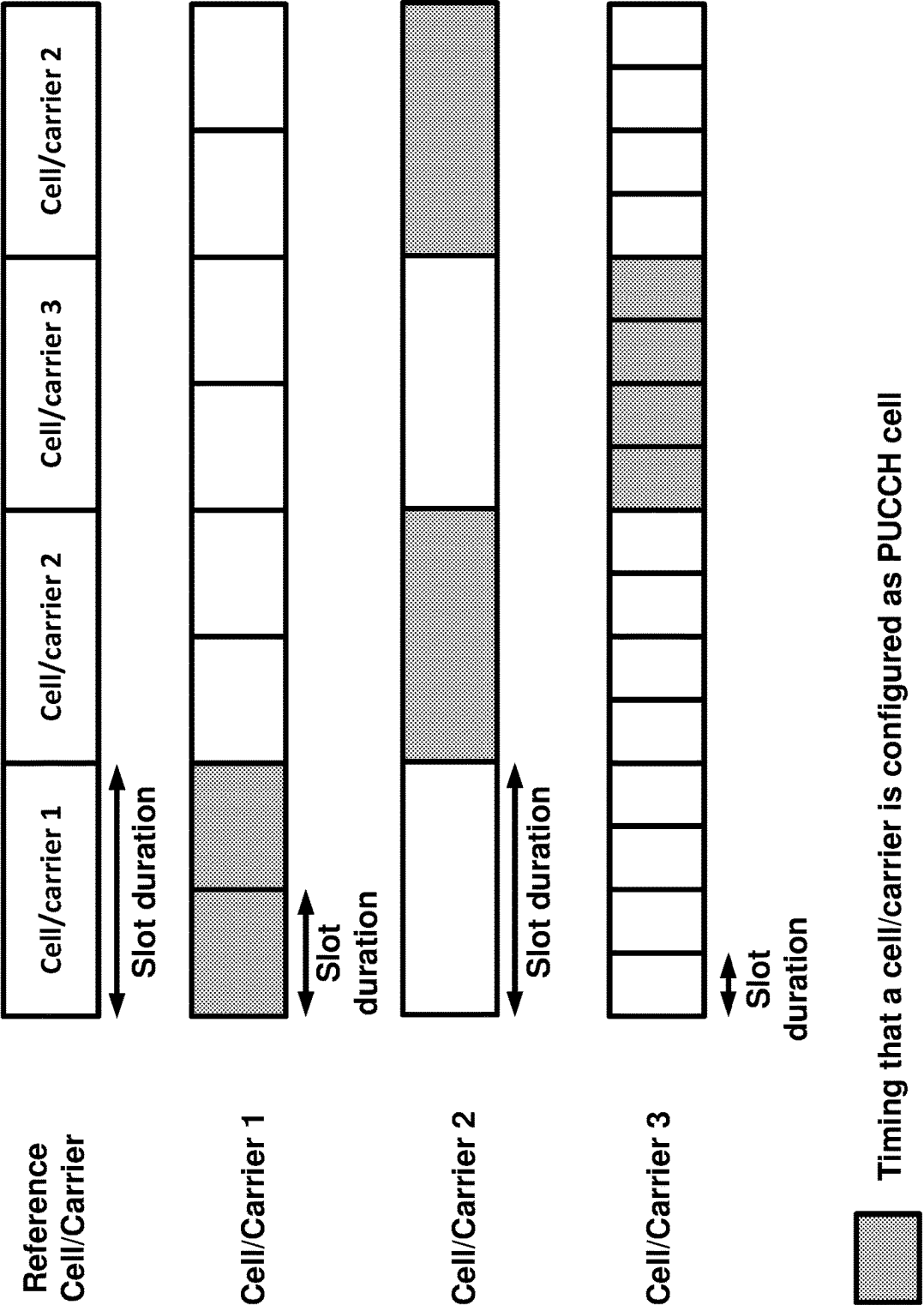


FIG. 18

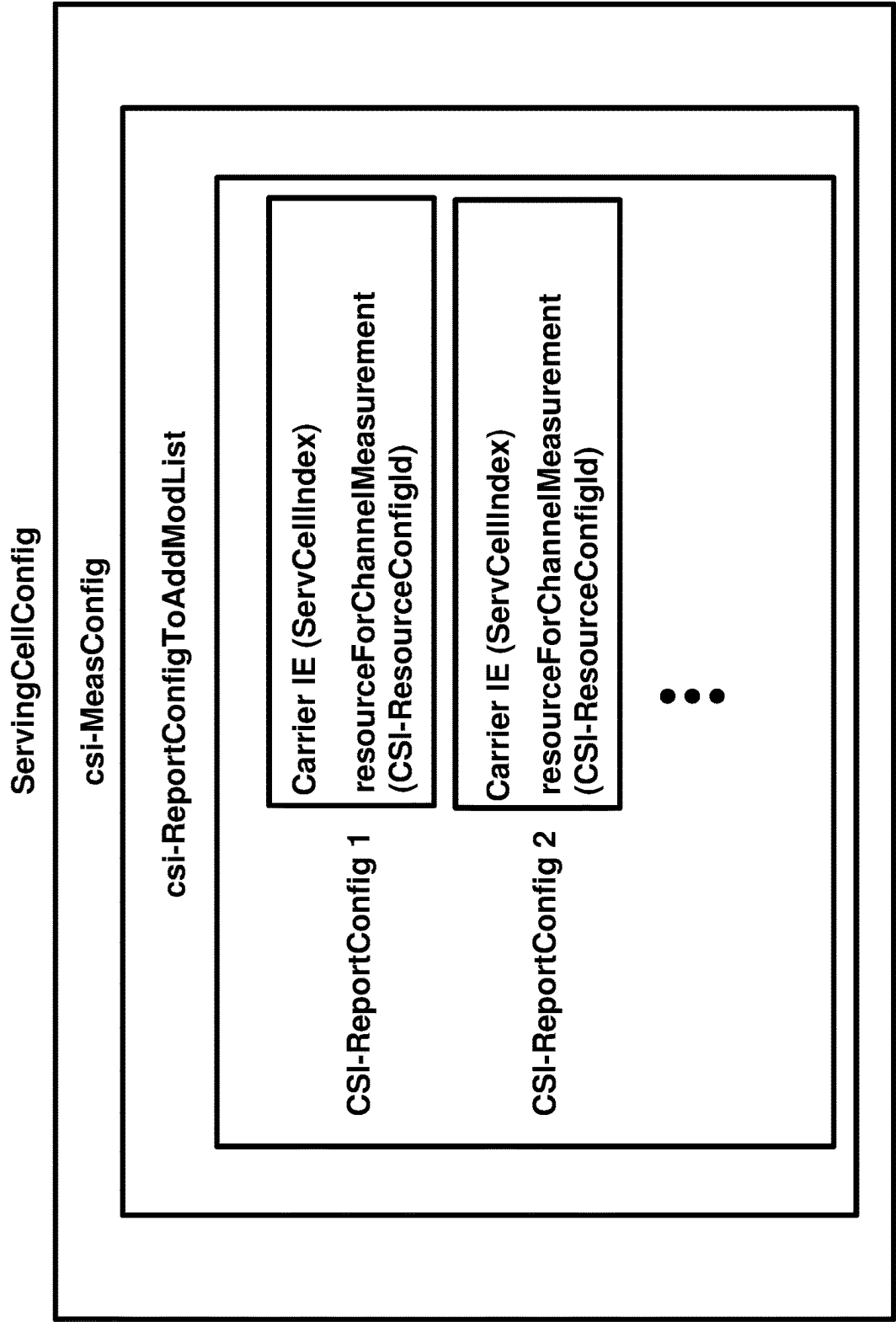


FIG. 19

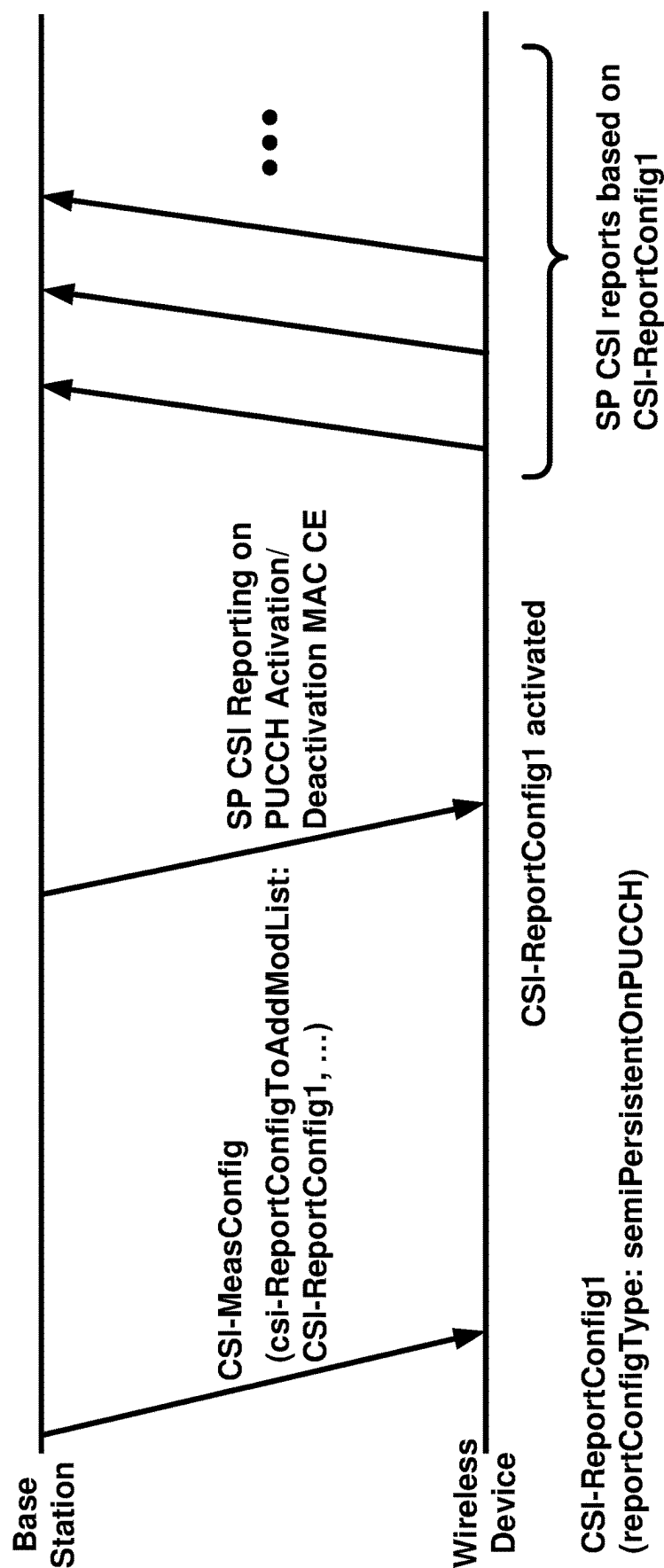


FIG. 20

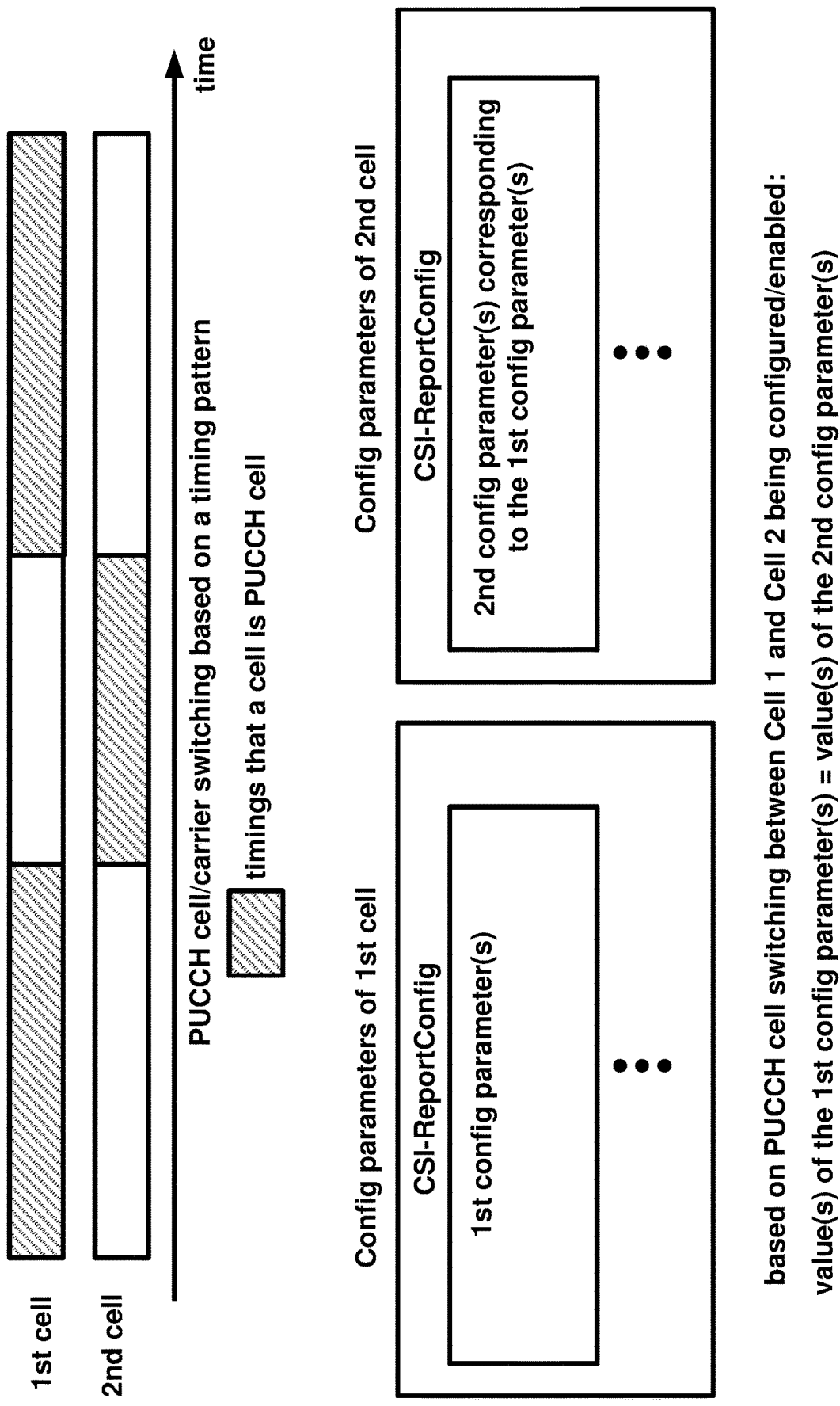


FIG. 21

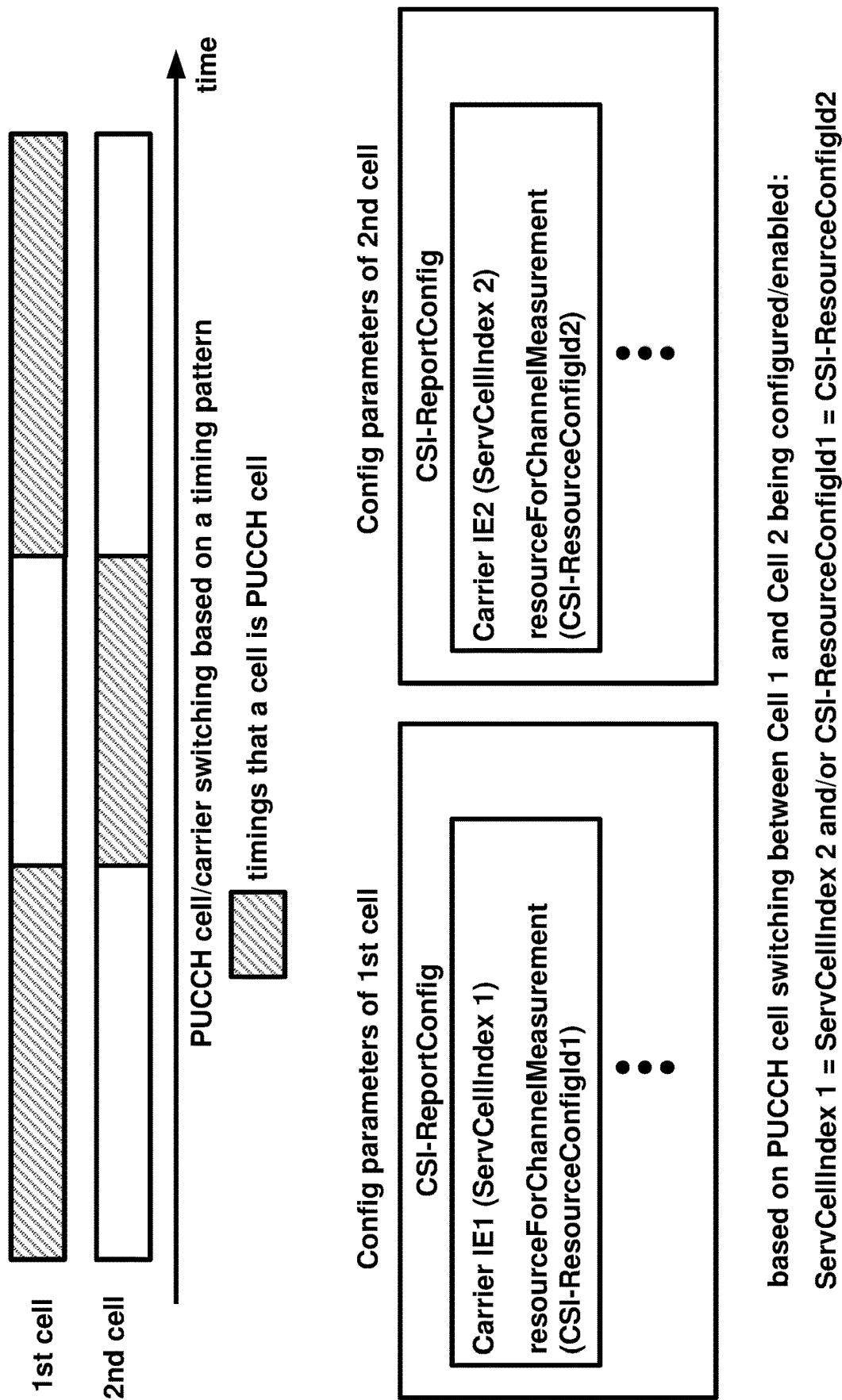


FIG. 22

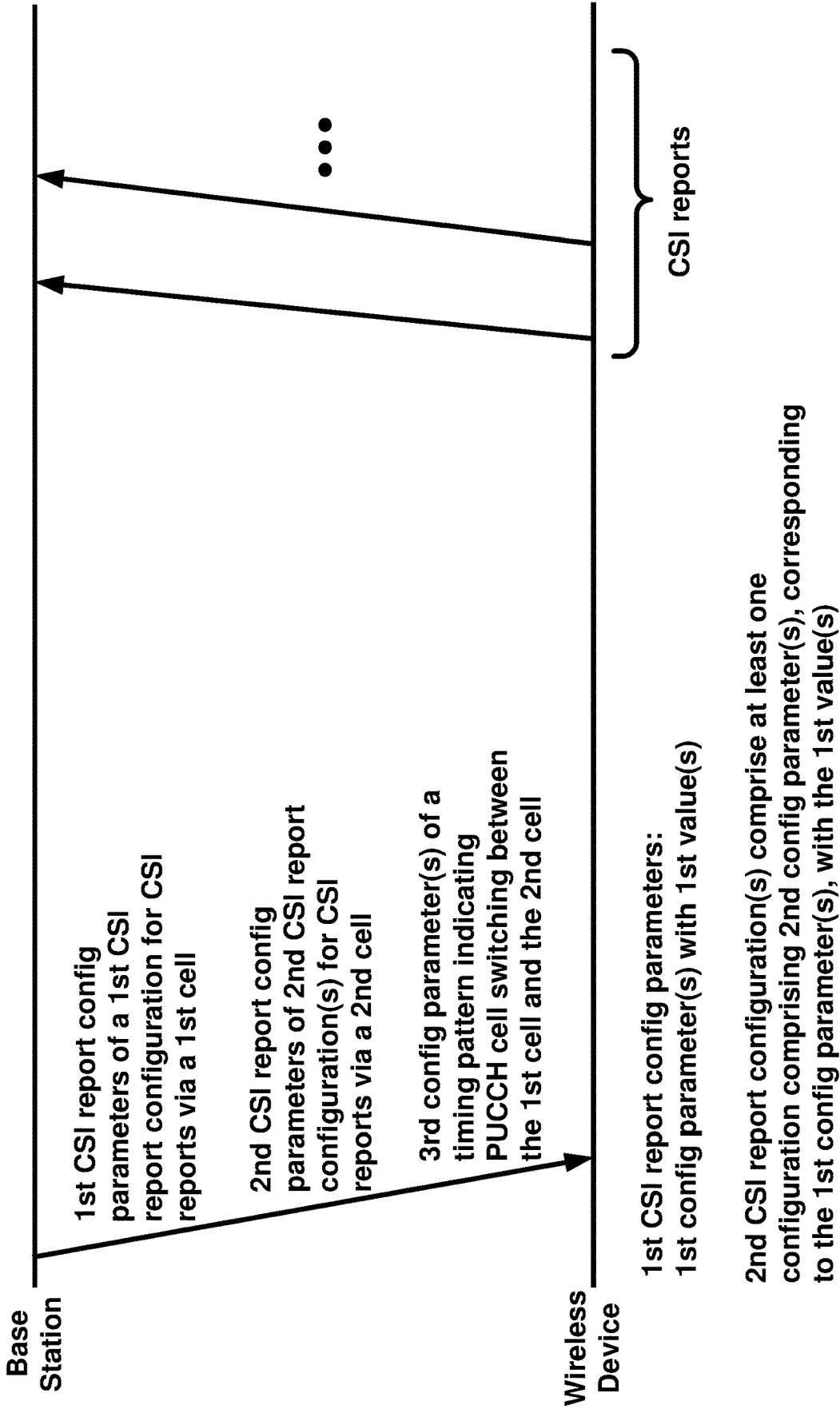


FIG. 23

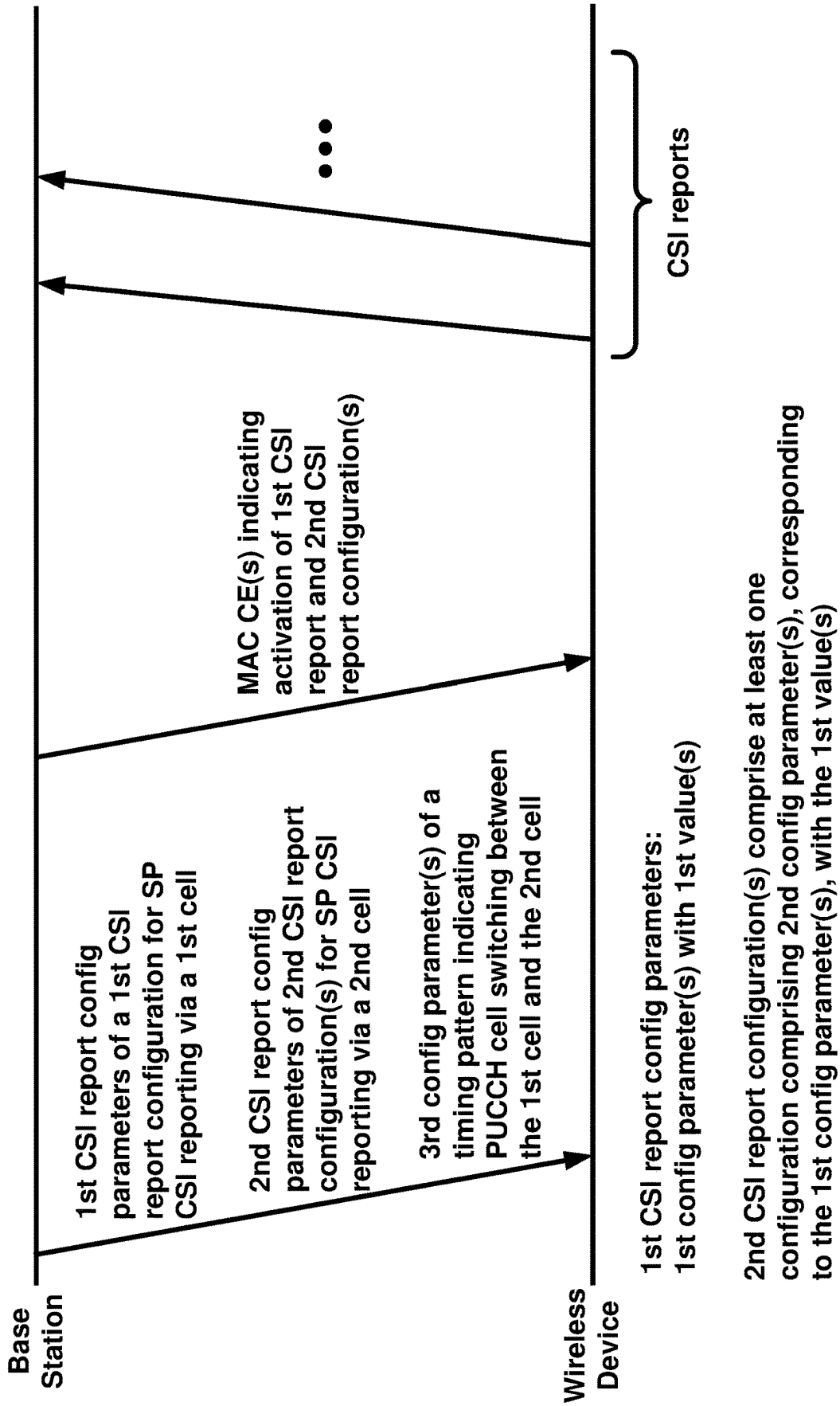


FIG. 24

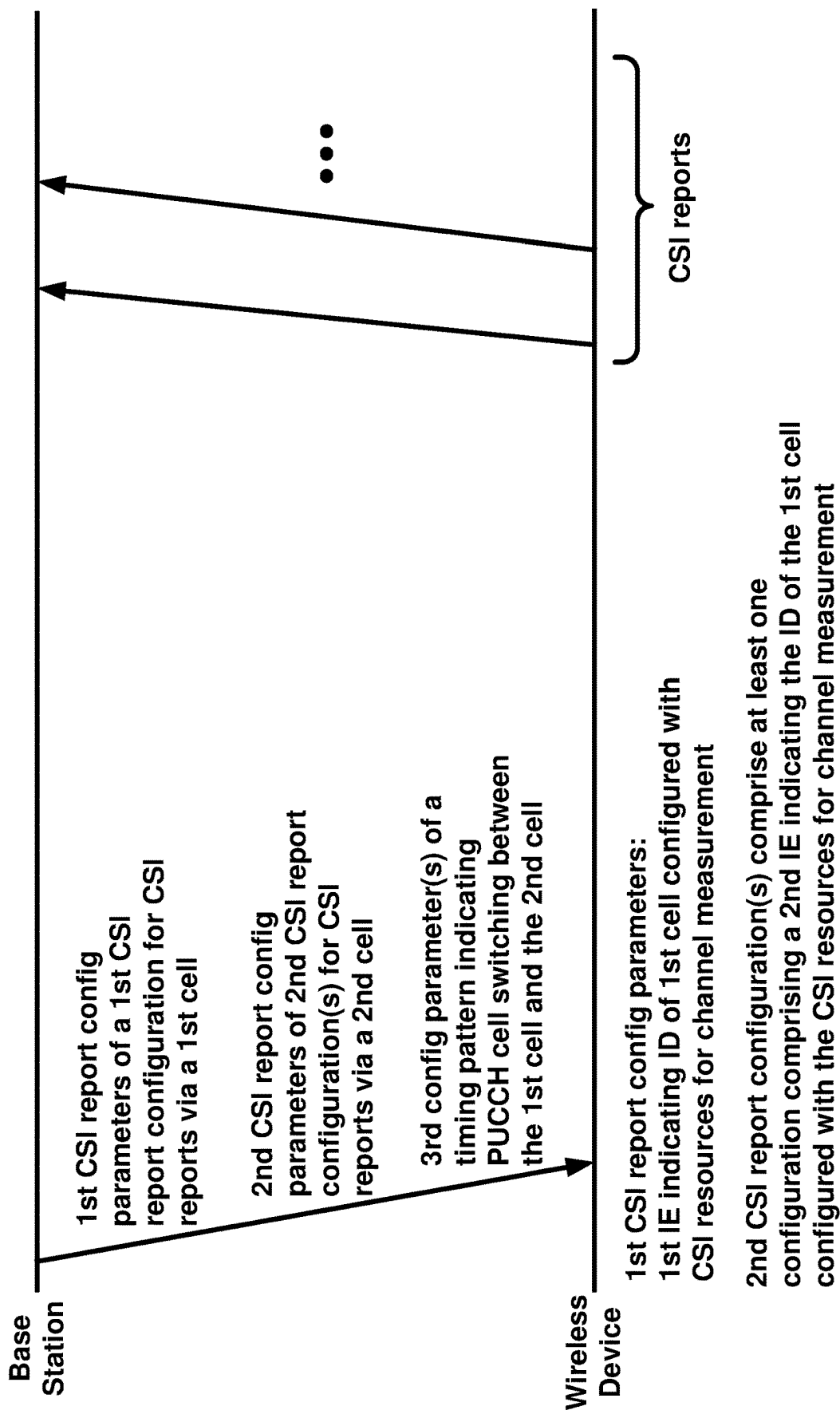


FIG. 25

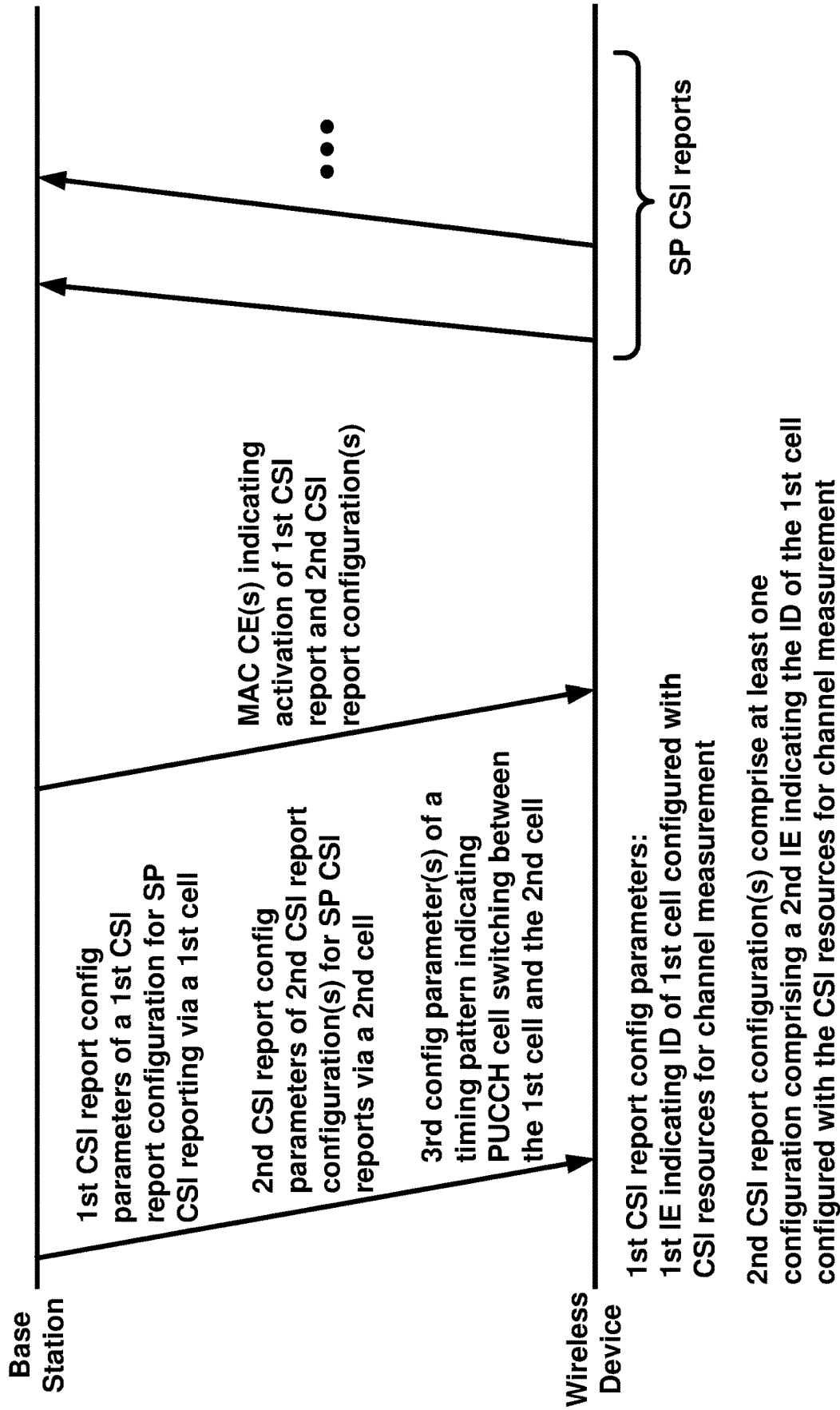


FIG. 26

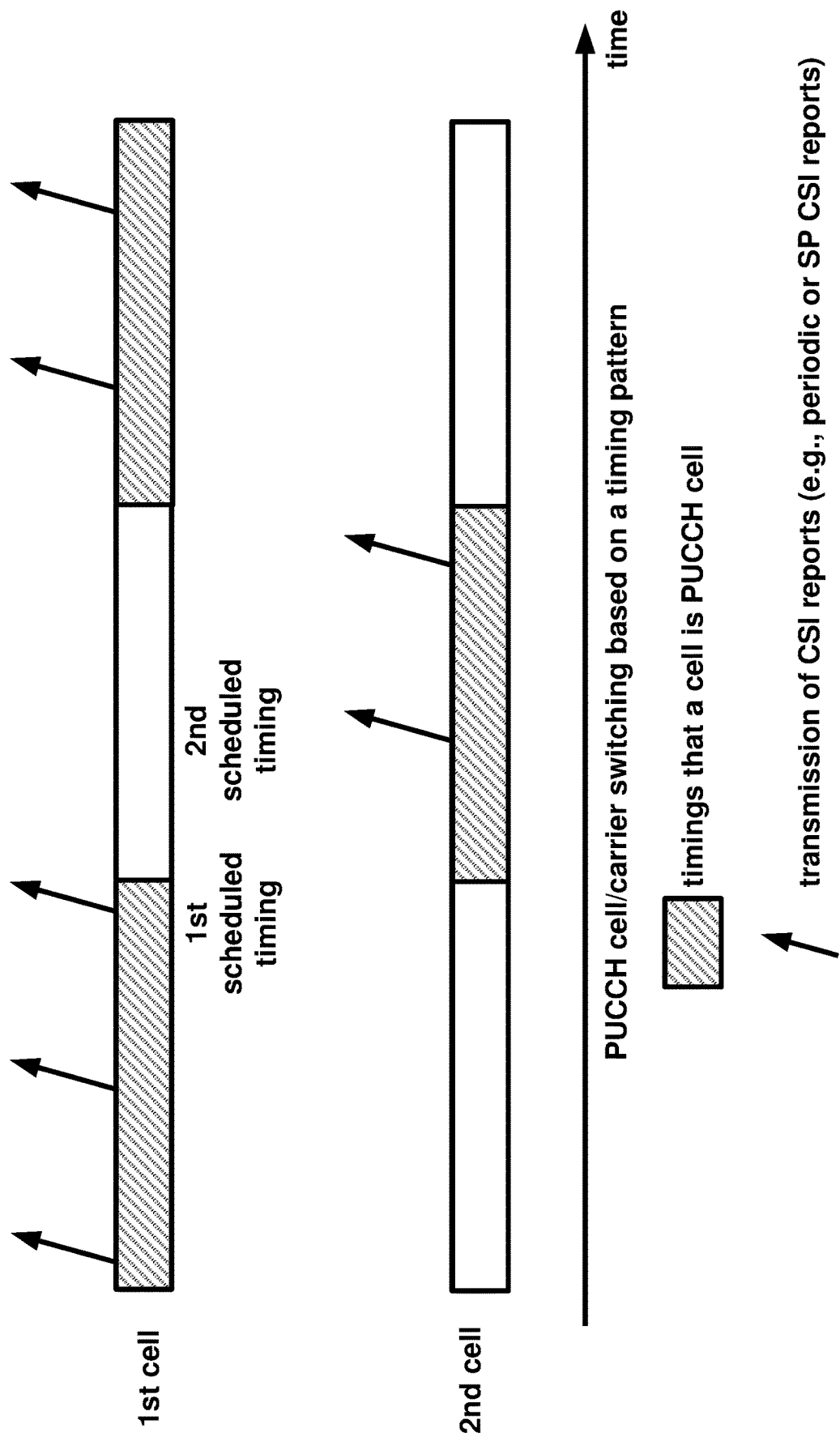


FIG. 27

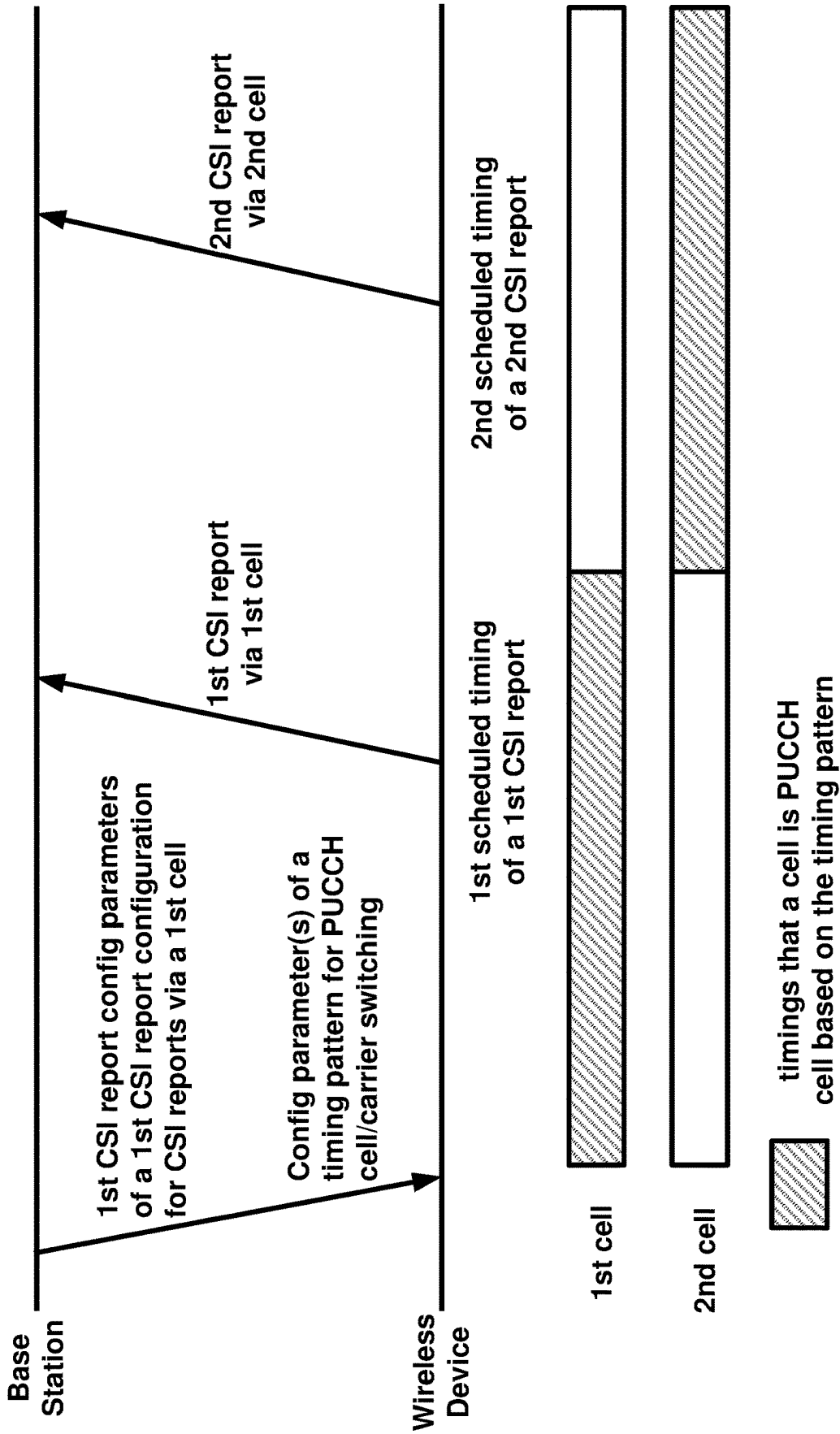


FIG. 28

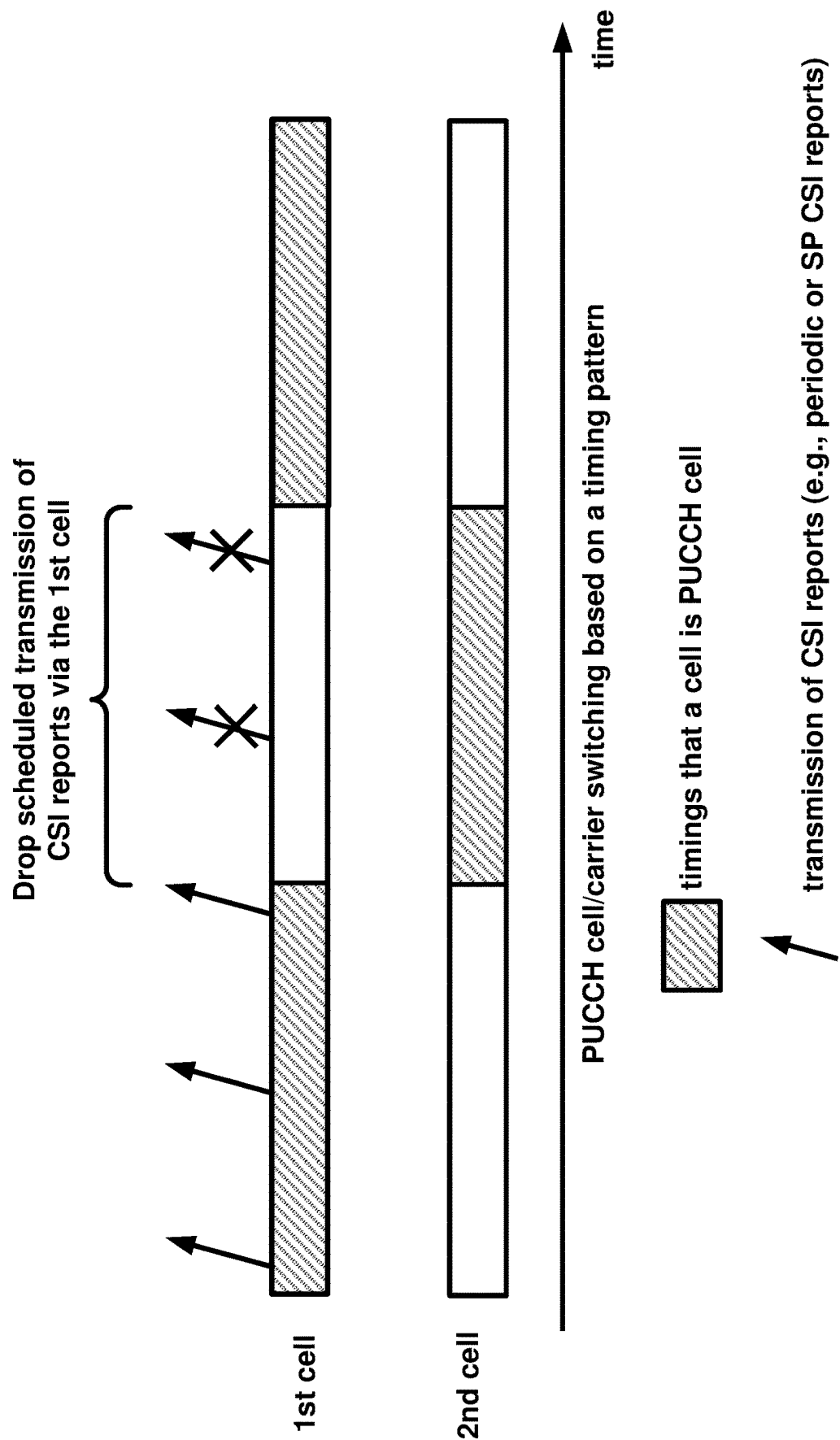


FIG. 29

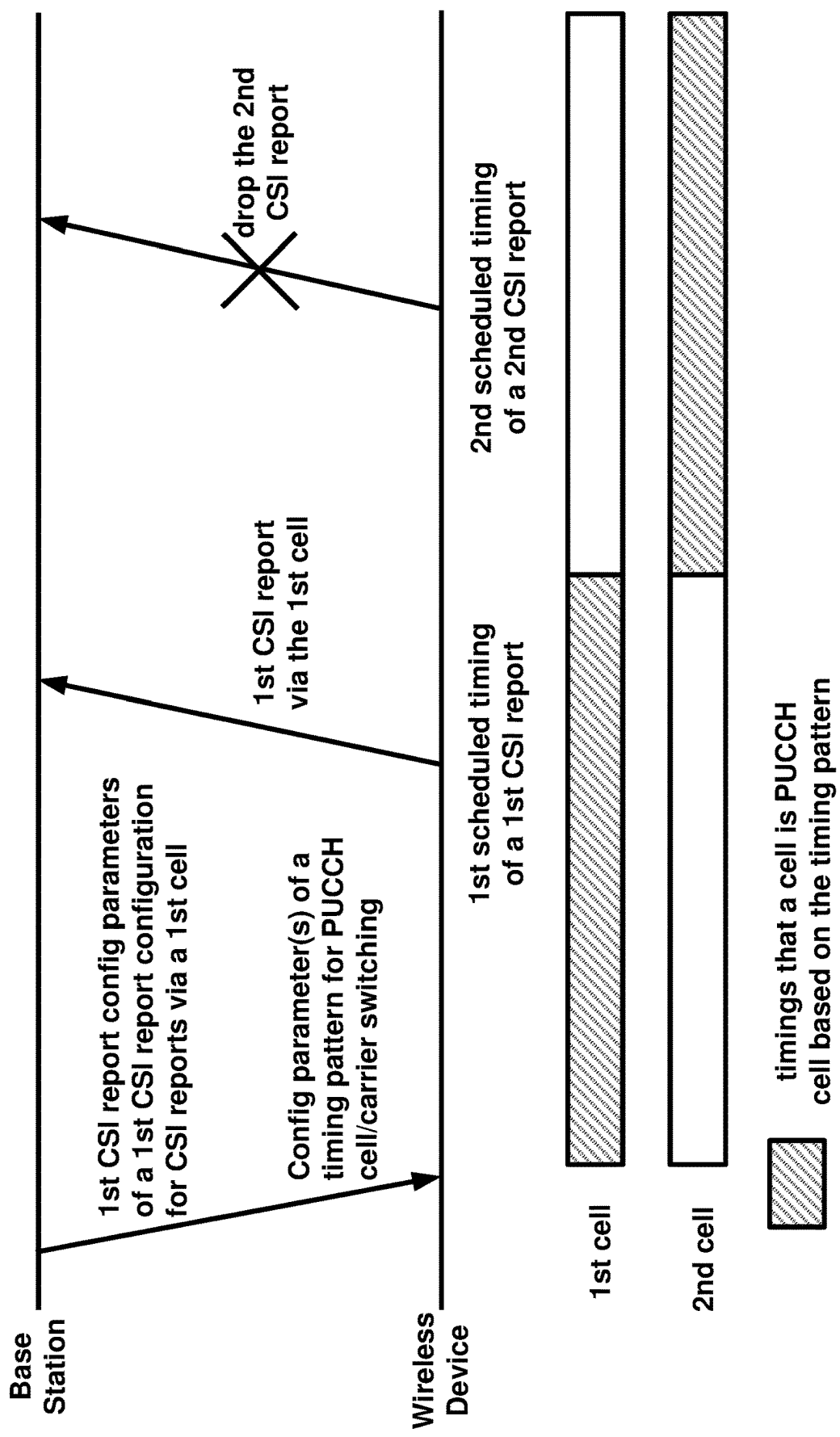


FIG. 30

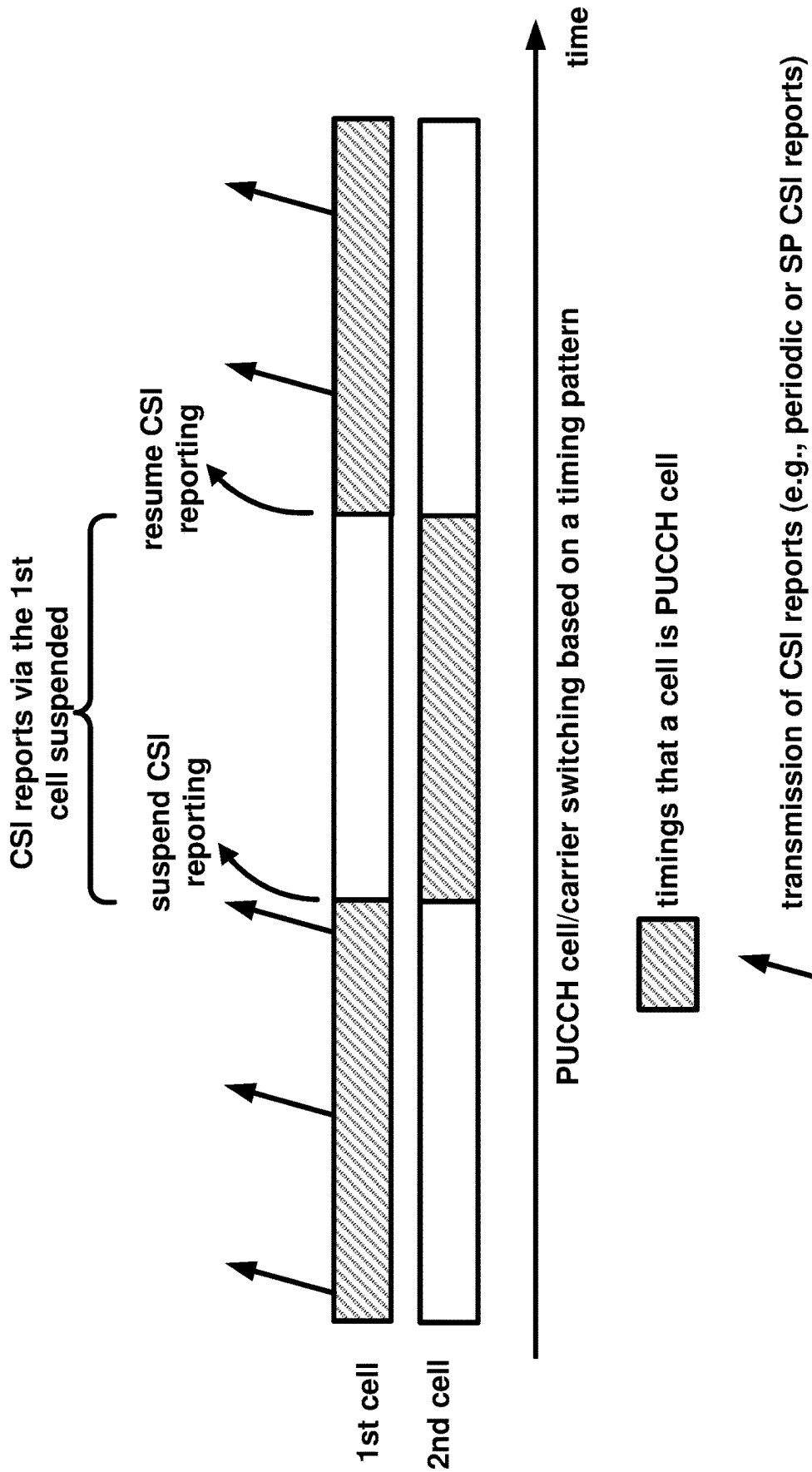


FIG. 31

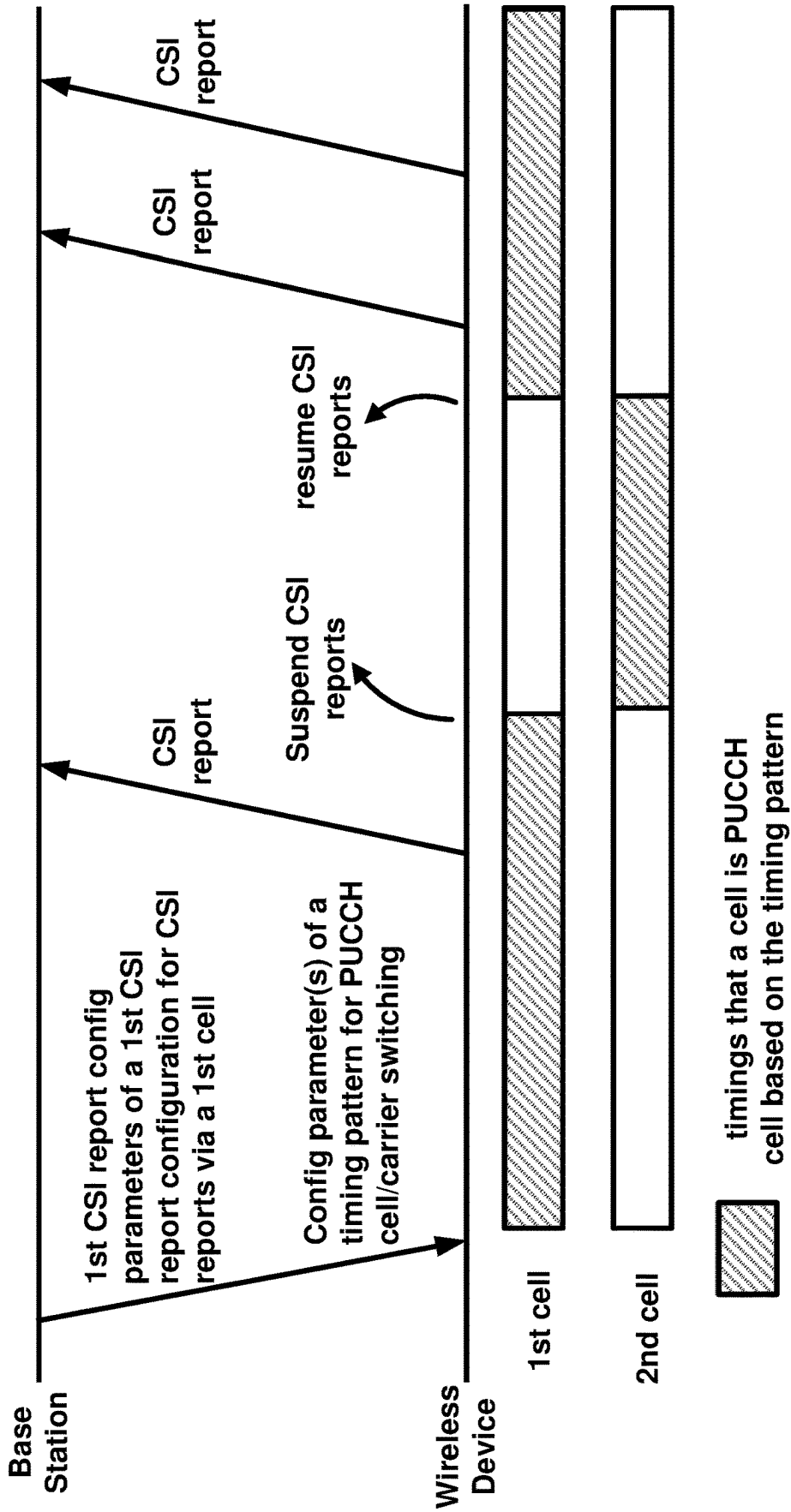


FIG. 32

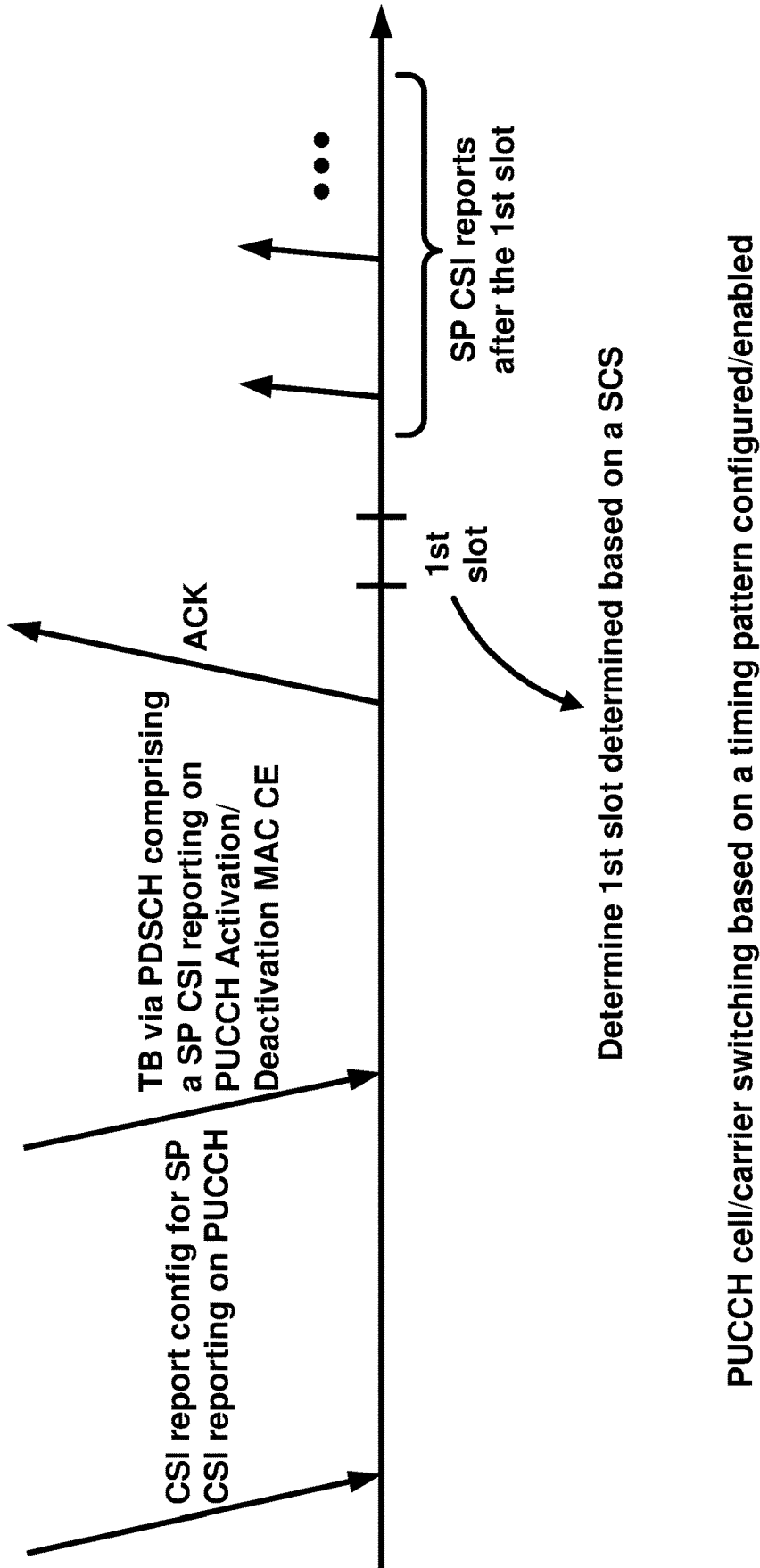


FIG. 33

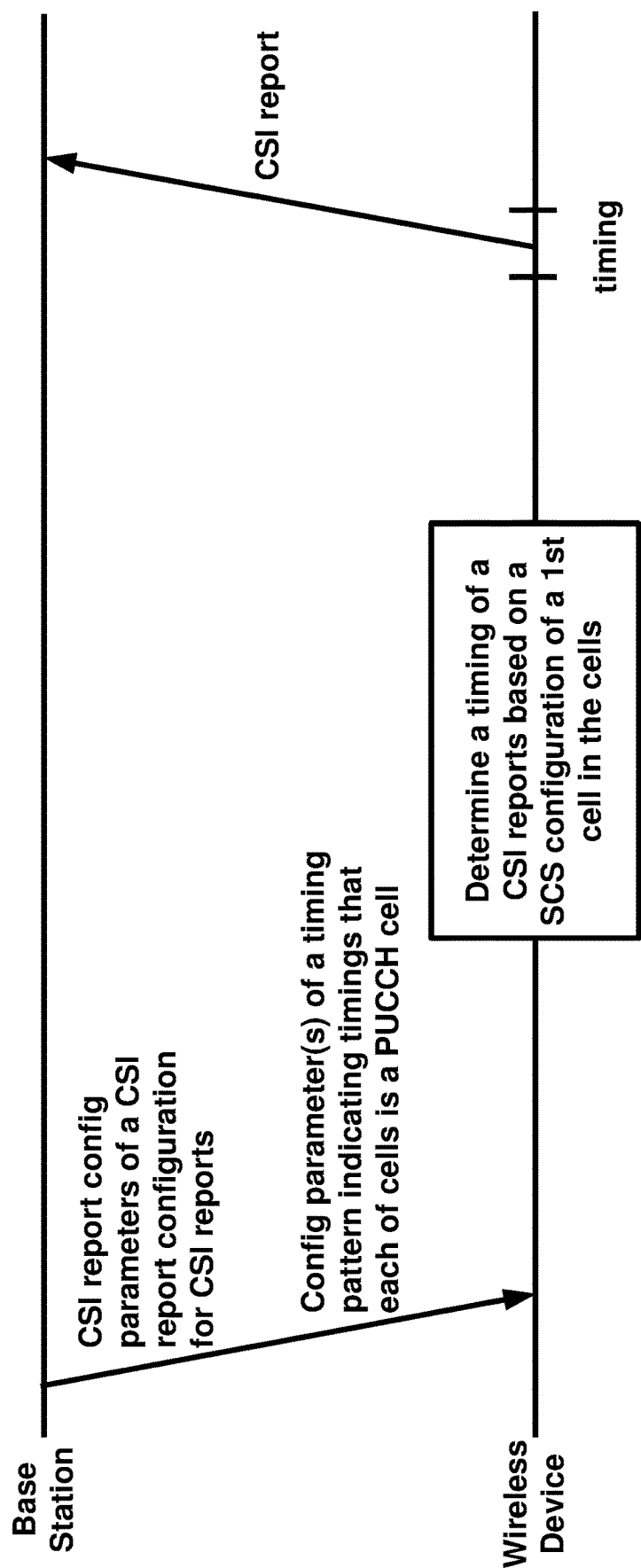


FIG. 34

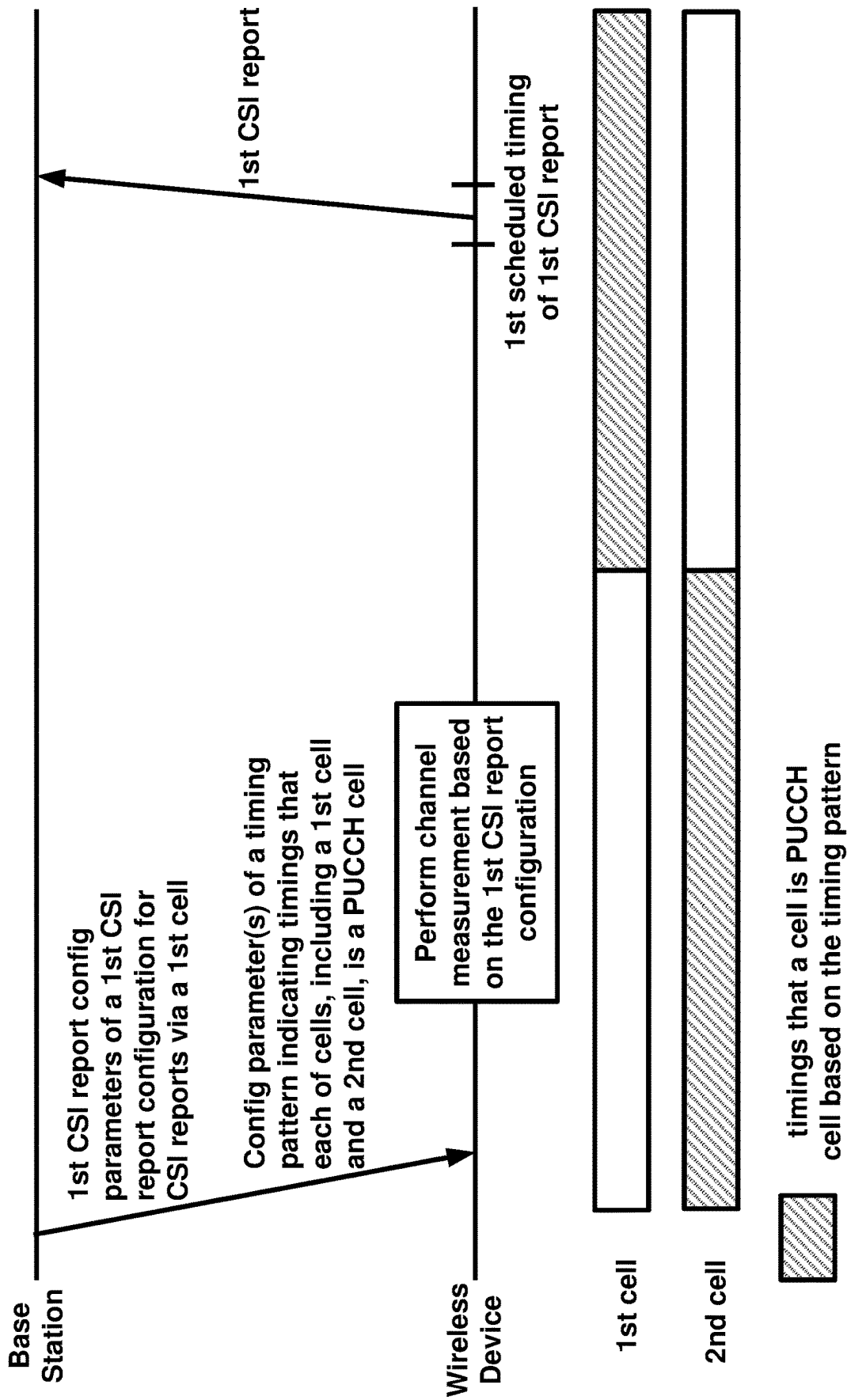


FIG. 35

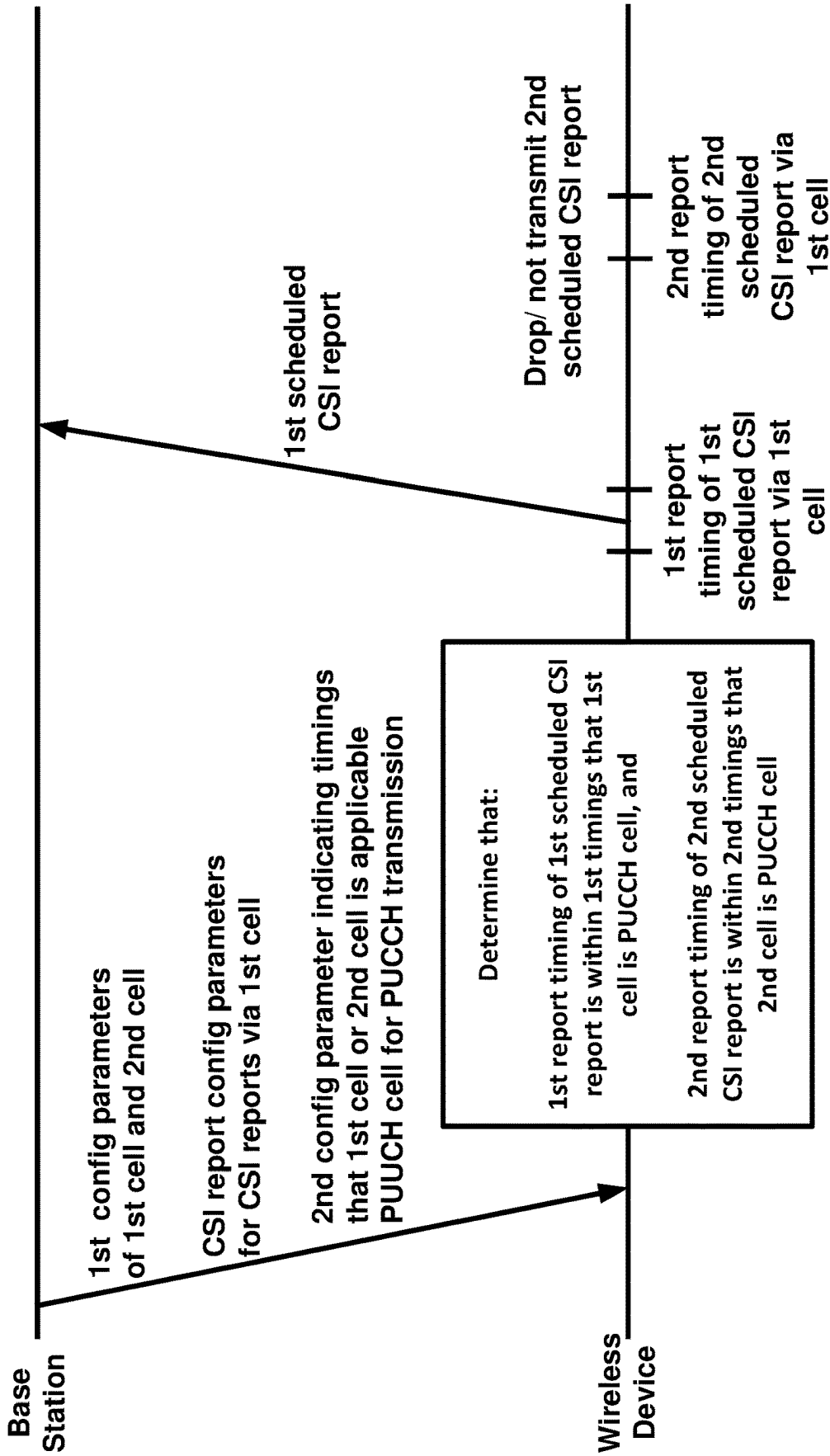


FIG. 36

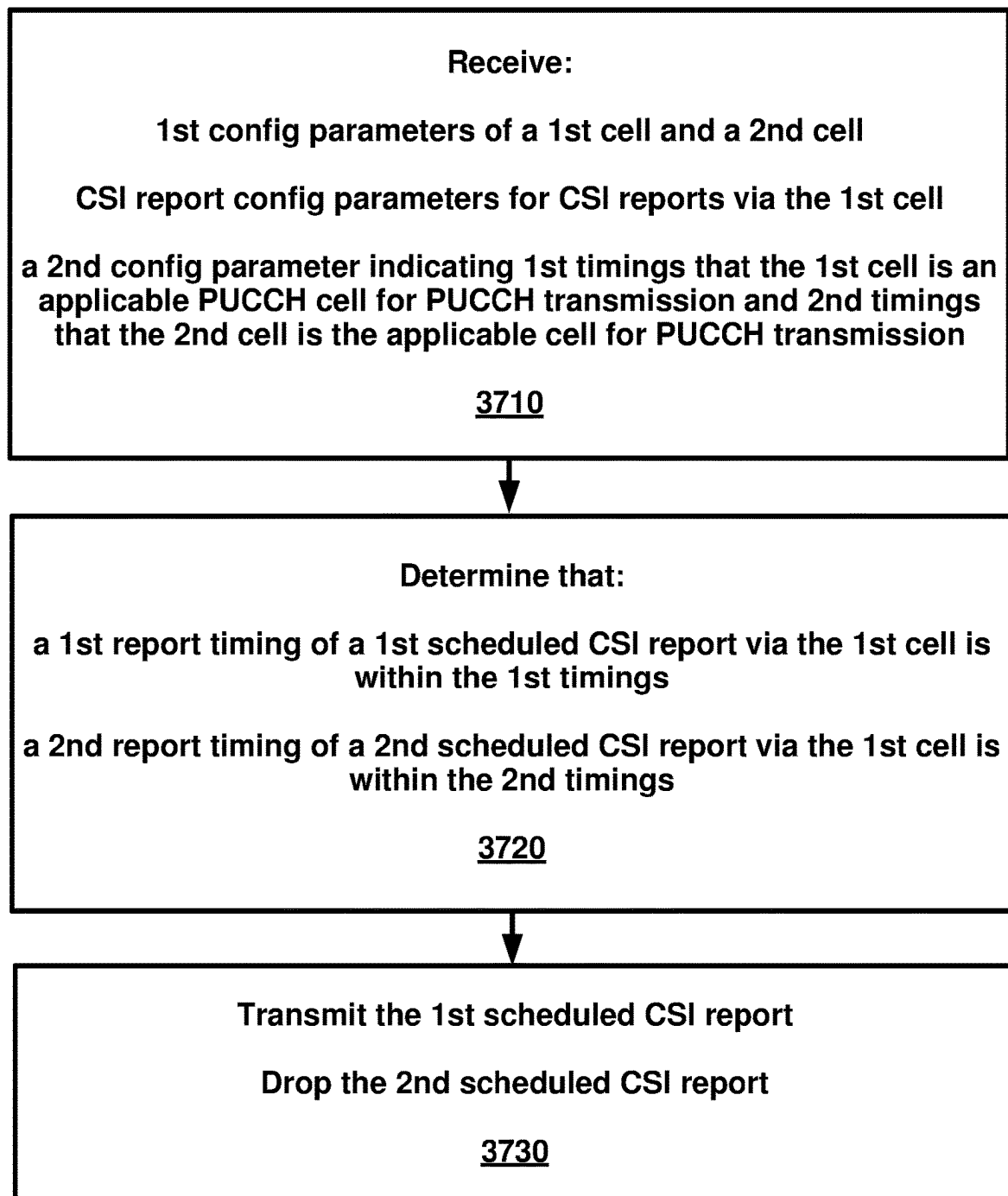
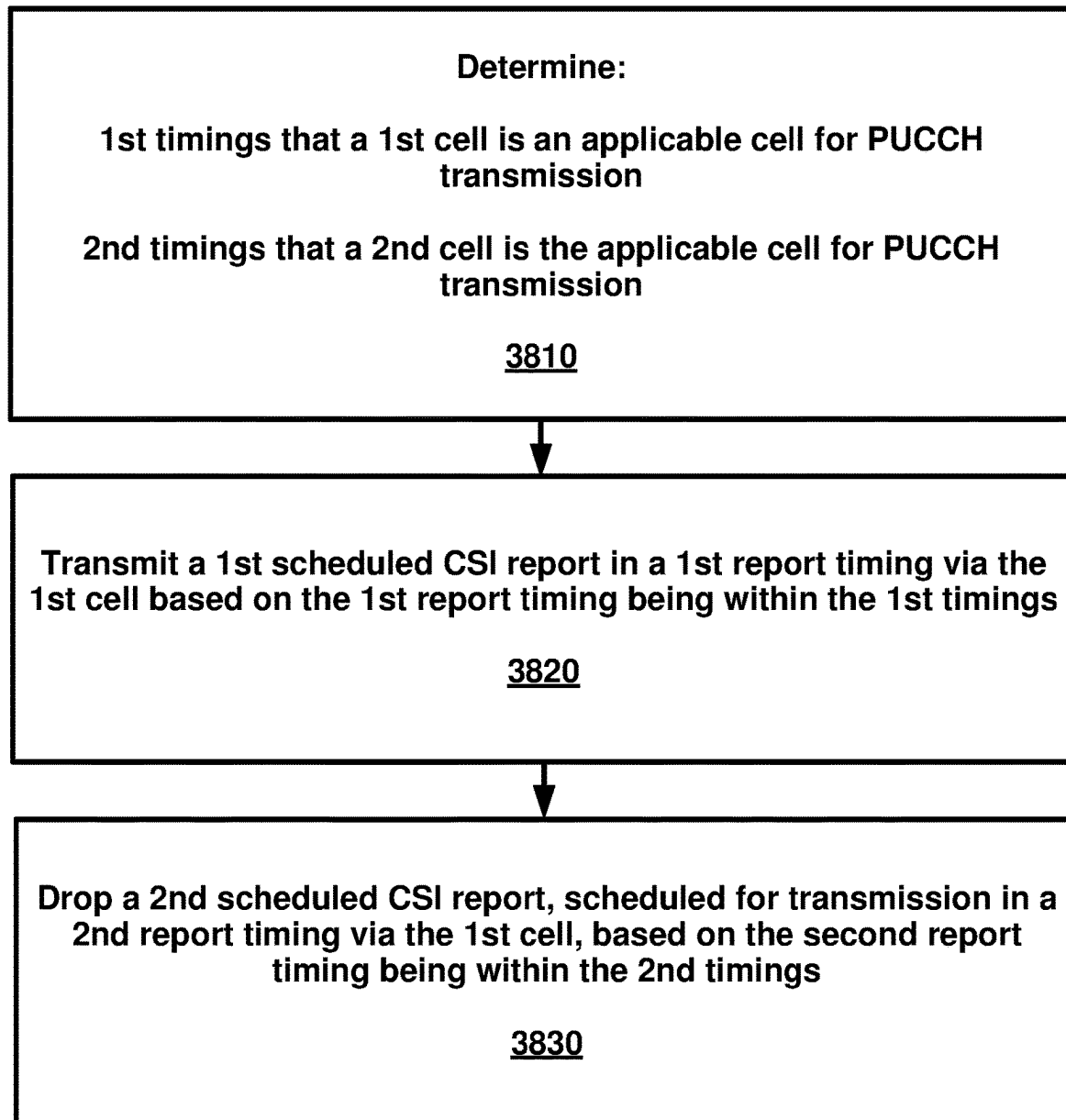


FIG. 37

**FIG. 38**

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PUCCH CELL SWITCHING AND CSI REPORTING

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 18/075,056, filed Dec. 5, 2022, which claims the benefit of U.S. Provisional Application No. 63/287,324, filed Dec. 8, 2021, which is hereby incorporated by reference in its entirety.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A and FIG. 1B show examples of mobile communications systems in accordance with several of various embodiments of the present disclosure.

FIG. 2A and FIG. 2B show examples of user plane and control plane protocol layers in accordance with several of various embodiments of the present disclosure.

FIG. 3 shows example functions and services offered by protocol layers in a user plane protocol stack in accordance with several of various embodiments of the present disclosure.

FIG. 4 shows example flow of packets through the protocol layers in accordance with several of various embodiments of the present disclosure.

FIG. 5A shows example mapping of channels between layers of the protocol stack and different physical signals in downlink in accordance with several of various embodiments of the present disclosure.

FIG. 5B shows example mapping of channels between layers of the protocol stack and different physical signals in uplink in accordance with several of various embodiments of the present disclosure.

FIG. 6 shows example physical layer processes for signal transmission in accordance with several of various embodiments of the present disclosure.

FIG. 7 shows examples of RRC states and RRC state transitions in accordance with several of various embodiments of the present disclosure.

FIG. 8 shows an example time domain transmission structure in NR by grouping OFDM symbols into slots, subframes and frames in accordance with several of various embodiments of the present disclosure.

FIG. 9 shows an example of time-frequency resource grid in accordance with several of various embodiments of the present disclosure.

FIG. 10 shows example adaptation and switching of bandwidth parts in accordance with several of various embodiments of the present disclosure.

FIG. 11A shows example arrangements of carriers in carrier aggregation in accordance with several of various embodiments of the present disclosure.

FIG. 11B shows examples of uplink control channel groups in accordance with several of various embodiments of the present disclosure.

FIG. 12A, FIG. 12B and FIG. 12C show example random access processes in accordance with several of various embodiments of the present disclosure.

FIG. 13A shows example time and frequency structure of SSBs and their associations with beams in accordance with several of various embodiments of the present disclosure.

FIG. 13B shows example time and frequency structure of CSI-RSs and their association with beams in accordance with several of various embodiments of the present disclosure.

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FIG. 14A, FIG. 14B and FIG. 14C show example beam management processes in accordance with several of various embodiments of the present disclosure.

FIG. 15 shows example components of a wireless device and a base station that are in communication via an air interface in accordance with several of various embodiments of the present disclosure.

FIG. 16 shows an example medium access control (MAC) control element (CE) in accordance with several of various embodiments of the present disclosure.

FIG. 17 shows an example time/timing pattern for physical uplink control channel (PUCCH) carrier switching in accordance with several of various embodiments of the present disclosure.

FIG. 18 shows an example time/timing pattern for physical uplink control channel (PUCCH) carrier switching in accordance with several of various embodiments of the present disclosure.

FIG. 19 shows example information elements in accordance with several of various embodiments of the present disclosure.

FIG. 20 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 21 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 22 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 23 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 24 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 25 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 26 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 27 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 28 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 29 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 30 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 31 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 32 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 33 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 34 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 35 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 36 shows an example process in accordance with several of various embodiments of the present disclosure.

FIG. 37 shows an example flow diagram in accordance with several of various embodiments of the present disclosure.

FIG. 38 shows an example flow diagram in accordance with several of various embodiments of the present disclosure.

DETAILED DESCRIPTION

The exemplary embodiments of the disclosed technology enable channel state information (CSI) reporting. The exemplary disclosed embodiments may be implemented in the

technical field of wireless communication systems. More particularly, the embodiments of the disclosed technology may enhance channel state information reporting when the wireless device is configured with physical uplink control channel (PUCCH) cell switching.

The devices and/or nodes of the mobile communications system disclosed herein may be implemented based on various technologies and/or various releases/versions/amendments of a technology. The various technologies include various releases of long-term evolution (LTE) technologies, various releases of 5G new radio (NR) technologies, various wireless local area networks technologies and/or a combination thereof and/or alike. For example, a base station may support a given technology and may communicate with wireless devices with different characteristics. The wireless devices may have different categories that define their capabilities in terms of supporting various features. The wireless device with the same category may have different capabilities. The wireless devices may support various technologies such as various releases of LTE technologies, various releases of 5G NR technologies and/or a combination thereof and/or alike. At least some of the wireless devices in the mobile communications system of the present disclosure may be stationary or almost stationary. In this disclosure, the terms “mobile communications system” and “wireless communications system” may be used interchangeably.

FIG. 1A shows an example of a mobile communications system **100** in accordance with several of various embodiments of the present disclosure. The mobile communications system **100** may be, for example, run by a mobile network operator (MNO) or a mobile virtual network operator (MVNO). The mobile communications system **100** may be a public land mobile network (PLMN) run by a network operator providing a variety of service including voice, data, short messaging service (SMS), multimedia messaging service (MMS), emergency calls, etc. The mobile communications system **100** includes a core network (CN) **106**, a radio access network (RAN) **104** and at least one wireless device **102**.

The CN **106** connects the RAN **104** to one or more external networks (e.g., one or more data networks such as the Internet) and is responsible for functions such as authentication, charging and end-to-end connection establishment. Several radio access technologies (RATs) may be served by the same CN **106**.

The RAN **104** may implement a RAT and may operate between the at least one wireless device **102** and the CN **106**. The RAN **104** may handle radio related functionalities such as scheduling, radio resource control, modulation and coding, multi-antenna transmissions and retransmission protocols. The wireless device and the RAN may share a portion of the radio spectrum by separating transmissions from the wireless device to the RAN and the transmissions from the RAN to the wireless device. The direction of the transmissions from the wireless device to the RAN is known as the uplink and the direction of the transmissions from the RAN to the wireless device is known as the downlink. The separation of uplink and downlink transmissions may be achieved by employing a duplexing technique. Example duplexing techniques include frequency division duplexing (FDD), time division duplexing (TDD) or a combination of FDD and TDD.

In this disclosure, the term wireless device may refer to a device that communicates with a network entity or another device using wireless communication techniques. The wireless device may be a mobile device or a non-mobile (e.g.,

fixed) device. Examples of the wireless device include cellular phone, smart phone, tablet, laptop computer, wearable device (e.g., smart watch, smart shoe, fitness trackers, smart clothing, etc.), wireless sensor, wireless meter, extended reality (XR) devices including augmented reality (AR) and virtual reality (VR) devices, Internet of Things (IoT) device, vehicle to vehicle communications device, road-side units (RSU), automobile, relay node or any combination thereof. In some examples, the wireless device (e.g., a smart phone, tablet, etc.) may have an interface (e.g., a graphical user interface (GUI)) for configuration by an end user. In some examples, the wireless device (e.g., a wireless sensor device, etc.) may not have an interface for configuration by an end user. The wireless device may be referred to as a user equipment (UE), a mobile station (MS), a subscriber unit, a handset, an access terminal, a user terminal, a wireless transmit and receive unit (WTRU) and/or other terminology.

The at least one wireless device may communicate with at least one base station in the RAN **104**. In this disclosure, the term base station may encompass terminologies associated with various RATs. For example, a base station may be referred to as a Node B in a 3G cellular system such as Universal Mobile Telecommunication Systems (UMTS), an evolved Node B (eNB) in a 4G cellular system such as evolved universal terrestrial radio access (E-UTRA), a next generation eNB (ng-eNB), a Next Generation Node B (gNB) in NR and/or a 5G system, an access point (AP) in Wi-Fi and/or other wireless local area networks. A base station may be referred to as a remote radio head (RRH), a baseband unit (BBU) in connection with one or more RRHs, a repeater or relay for coverage extension and/or any combination thereof. In some examples, all protocol layers of a base station may be implemented in one unit. In some examples, some of the protocol layers (e.g., upper layers) of the base station may be implemented in a first unit (e.g., a central unit (CU)) and some other protocol layer (e.g., lower layers) may be implemented in one or more second units (e.g., distributed units (DUs)).

A base station in the RAN **104** includes one or more antennas to communicate with the at least one wireless device. The base station may communicate with the at least one wireless device using radio frequency (RF) transmissions and receptions via RF transceivers. The base station antennas may control one or more cells (or sectors). The size and/or radio coverage area of a cell may depend on the range that transmissions by a wireless device can be successfully received by the base station when the wireless device transmits using the RF frequency of the cell. The base station may be associated with cells of various sizes. At a given location, the wireless device may be in coverage area of a first cell of the base station and may not be in coverage area of a second cell of the base station depending on the sizes of the first cell and the second cell.

A base station in the RAN **104** may have various implementations. For example, a base station may be implemented by connecting a BBU (or a BBU pool) coupled to one or more RRHs and/or one or more relay nodes to extend the cell coverage. The BBU pool may be located at a centralized site like a cloud or data center. The BBU pool may be connected to a plurality of RRHs that control a plurality of cells. The combination of BBU with the one or more RRHs may be referred to as a centralized or cloud RAN (C-RAN) architecture. In some implementations, the BBU functions may be implemented on virtual machines (VMs) on servers at a centralized location. This architecture may be referred to as virtual RAN (vRAN). All, most or a

portion of the protocol layer functions (e.g., all or portions of physical layer, medium access control (MAC) layer and/or higher layers) may be implemented at the BBU pool and the processed data may be transmitted to the RRHs for further processing and/or RF transmission. The links between the BBU pool and the RRHs may be referred to as fronthaul.

In some deployment scenarios, the RAN **104** may include macrocell base stations with high transmission power levels and large coverage areas. In other deployment scenarios, the RAN **104** may include base stations that employ different transmission power levels and/or have cells with different coverage areas. For example, some base station may be macrocell base stations with high transmission powers and/or large coverage areas and other base station may be small cell base stations with comparatively smaller transmission powers and/or coverage areas. In some deployment scenarios, a small cell base station may have coverage that is within or has overlap with coverage area of a macrocell base station. A wireless device may communicate with the macrocell base station while within the coverage area of the macrocell base station. For additional capacity, the wireless device may communicate with both the macrocell base station and the small cell base station while in the overlapped coverage area of the macrocell base station and the small cell base station. Depending on their coverage areas, a small cell base station may be referred to as a microcell base station, a picocell base station, a femtocell base station or a home base station.

Different standard development organizations (SDOs) have specified, or may specify in future, mobile communications systems that have similar characteristics as the mobile communications system **100** of FIG. **1A**. For example, the Third-Generation Partnership Project (3GPP) is a group of SDOs that provides specifications that define 3GPP technologies for mobile communications systems that are akin to the mobile communications system **100**. The 3GPP has developed specifications for third generation (3G) mobile networks, fourth generation (4G) mobile networks and fifth generation (5G) mobile networks. The 3G, 4G and 5G networks are also known as Universal Mobile Telecommunications System (UMTS), Long Term Evolution (LTE) and 5G system (5GS), respectively. In this disclosure, embodiments are described with respect to the RAN implemented in a 3GPP 5G mobile network that is also referred to as next generation RAN (NG-RAN). The embodiments may also be implemented in other mobile communications systems such as 3G or 4G mobile networks or mobile networks that may be standardized in future such as sixth generation (6G) mobile networks or mobile networks that are implemented by standards bodies other than 3GPP. The NG-RAN may be based on a new RAT known as new radio (NR) and/or other radio access technologies such as LTE and/or non-3GPP RATs.

FIG. **1B** shows an example of a mobile communications system **110** in accordance with several of various embodiments of the present disclosure. The mobile communications system **110** of FIG. **1B** is an example of a 5G mobile network and includes a 5G CN (5G-CN) **130**, an NG-RAN **120** and UEs (collectively **112** and individually UE **112A** and UE **112B**). The 5G-CN **130**, the NG-RAN **120** and the UEs **112** of FIG. **1B** operate substantially alike the CN **106**, the RAN **104** and the at least one wireless device **102**, respectively, as described for FIG. **1A**.

The 5G-CN **130** of FIG. **1B** connects the NG-RAN **120** to one or more external networks (e.g., one or more data networks such as the Internet) and is responsible for func-

tions such as authentication, charging and end-to-end connection establishment. The 5G-CN has new enhancements compared to previous generations of CNs (e.g., evolved packet core (EPC) in the 4G networks) including service-based architecture, support for network slicing and control plane/user plane split. The service-based architecture of the 5G-CN provides a modular framework based on service and functionalities provided by the core network wherein a set of network functions are connected via service-based interfaces. The network slicing enables multiplexing of independent logical networks (e.g., network slices) on the same physical network infrastructure. For example, a network slice may be for mobile broadband applications with full mobility support and a different network slice may be for non-mobile latency-critical applications such as industry automation. The control plane/user plane split enables independent scaling of the control plane and the user plane. For example, the control plane capacity may be increased without affecting the user plane of the network.

The 5G-CN **130** of FIG. **1B** includes an access and mobility management function (AMF) **132** and a user plane function (UPF) **134**. The AMF **132** may support termination of non-access stratum (NAS) signaling, NAS signaling security such as ciphering and integrity protection, inter-3GPP access network mobility, registration management, connection management, mobility management, access authentication and authorization and security context management. The NAS is a functional layer between a UE and the CN and the access stratum (AS) is a functional layer between the UE and the RAN. The UPF **134** may serve as an interconnect point between the NG-RAN and an external data network. The UPF may support packet routing and forwarding, packet inspection and Quality of Service (QoS) handling and packet filtering. The UPF may further act as a Protocol Data Unit (PDU) session anchor point for mobility within and between RATs.

The 5G-CN **130** may include additional network functions (not shown in FIG. **1B**) such as one or more Session Management Functions (SMFs), a Policy Control Function (PCF), a Network Exposure Function (NEF), a Unified Data Management (UDM), an Application Function (AF), and/or an Authentication Server Function (AUSF). These network functions along with the AMF **132** and UPF **134** enable a service-based architecture for the 5G-CN.

The NG-RAN **120** may operate between the UEs **112** and the 5G-CN **130** and may implement one or more RATs. The NG-RAN **120** may include one or more gNBs (e.g., gNB **122A** or gNB **122B** or collectively gNBs **122**) and/or one or more ng-eNBs (e.g., ng-eNB **124A** or ng-eNB **124B** or collectively ng-eNBs **124**). The general terminology for gNBs **122** and/or an ng-eNBs **124** is a base station and may be used interchangeably in this disclosure. The gNBs **122** and the ng-eNBs **124** may include one or more antennas to communicate with the UEs **112**. The one or more antennas of the gNBs **122** or ng-eNBs **124** may control one or more cells (or sectors) that provide radio coverage for the UEs **112**.

A gNB and/or an ng-eNB of FIG. **1B** may be connected to the 5G-CN **130** using an NG interface. A gNB and/or an ng-eNB may be connected with other gNBs and/or ng-eNBs using an Xn interface. The NG or the Xn interfaces are logical connections that may be established using an underlying transport network. The interface between a UE and a gNB or between a UE and an ng-eNBs may be referred to as the Uu interface. An interface (e.g., Uu, NG or Xn) may be established by using a protocol stack that enables data and control signaling exchange between entities in the mobile

communications system of FIG. 1B. When a protocol stack is used for transmission of user data, the protocol stack may be referred to as user plane protocol stack. When a protocol stack is used for transmission of control signaling, the protocol stack may be referred to as control plane protocol stack. Some protocol layer may be used in both of the user plane protocol stack and the control plane protocol stack while other protocol layers may be specific to the user plane or control plane.

The NG interface of FIG. 1B may include an NG-User plane (NG-U) interface between a gNB and the UPF 134 (or an ng-eNB and the UPF 134) and an NG-Control plane (NG-C) interface between a gNB and the AMF 132 (or an ng-eNB and the AMF 132). The NG-U interface may provide non-guaranteed delivery of user plane PDUs between a gNB and the UPF or an ng-eNB and the UPF. The NG-C interface may provide services such as NG interface management, UE context management, UE mobility management, transport of NAS messages, paging, PDU session management, configuration transfer and/or warning message transmission.

The UEs 112 and a gNB may be connected using the Uu interface and using the NR user plane and control plane protocol stack. The UEs 112 and an ng-eNB may be connected using the Uu interface using the LTE user plane and control plane protocol stack.

In the example mobile communications system of FIG. 1B, a 5G-CN is connected to a RAN comprised of 4G LTE and/or 5G NR RATs. In other example mobile communications systems, a RAN based on the 5G NR RAT may be connected to a 4G CN (e.g., EPC). For example, earlier releases of 5G standards may support a non-standalone mode of operation where a NR based RAN is connected to the 4G EPC. In an example non-standalone mode, a UE may be connected to both a 5G NR gNB and a 4G LTE eNB (e.g., a ng-eNB) and the control plane functionalities (such as initial access, paging and mobility) may be provided through the 4G LTE eNB. In a standalone of operation, the 5G NR gNB is connected to a 5G-CN and the user plane and the control plane functionalities are provided by the 5G NR gNB.

FIG. 2A shows an example of the protocol stack for the user plan of an NR Uu interface in accordance with several of various embodiments of the present disclosure. The user plane protocol stack comprises five protocol layers that terminate at the UE 200 and the gNB 210. The five protocol layers, as shown in FIG. 2A, include physical (PHY) layer referred to as PHY 201 at the UE 200 and PHY 211 at the gNB 210, medium access control (MAC) layer referred to as MAC 202 at the UE 200 and MAC 212 at the gNB 210, radio link control (RLC) layer referred to as RLC 203 at the UE 200 and RLC 213 at the gNB 210, packet data convergence protocol (PDCP) layer referred to as PDCP 204 at the UE 200 and PDCP 214 at the gNB 210, and service data application protocol (SDAP) layer referred to as SDAP 205 at the UE 200 and SDAP 215 at the gNB 210. The PHY layer, also known as layer 1 (L1), offers transport services to higher layers. The other four layers of the protocol stack (MAC, RLC, PDCP and SDAP) are collectively known as layer 2 (L2).

FIG. 2B shows an example of the protocol stack for the control plan of an NR Uu interface in accordance with several of various embodiments of the present disclosure. Some of the protocol layers (PHY, MAC, RLC and PDCP) are common between the user plane protocol stack shown in FIG. 2A and the control plan protocol stack. The control plane protocol stack also includes the RRC layer, referred to

RRC 206 at the UE 200 and RRC 216 at the gNB 210, that also terminates at the UE 200 and the gNB 210. In addition, the control plane protocol stack includes the NAS layer that terminates at the UE 200 and the AMF 220. In FIG. 2B, the NAS layer is referred to as NAS 207 at the UE 200 and NAS 227 at the AMF 220.

FIG. 3 shows example functions and services offered to other layers by a layer in the NR user plane protocol stack of FIG. 2A in accordance with several of various embodiments of the present disclosure. For example, the SDAP layer of FIG. 3 (shown in FIG. 2A as SDAP 205 at the UE side and SDAP 215 at the gNB side) may perform mapping and de-mapping of QoS flows to data radio bearers. The mapping and de-mapping may be based on QoS (e.g., delay, throughput, jitter, error rate, etc.) associated with a QoS flow. A QoS flow may be a QoS differentiation granularity for a PDU session which is a logical connection between a UE 200 and a data network. A PDU session may contain one or more QoS flows. The functions and services of the SDAP layer include mapping and de-mapping between one or more QoS flows and one or more data radio bearers. The SDAP layer may also mark the uplink and/or downlink packets with a QoS flow ID (QFI).

The PDCP layer of FIG. 3 (shown in FIG. 2A as PDCP 204 at the UE side and PDCP 214 at the gNB side) may perform header compression and decompression (e.g., using Robust Header Compression (ROHC) protocol) to reduce the protocol header overhead, ciphering and deciphering and integrity protection and verification to enhance the security over the air interface, reordering and in-order delivery of packets and discarding of duplicate packets. A UE may be configured with one PDCP entity per bearer.

In an example scenario not shown in FIG. 3, a UE may be configured with dual connectivity and may connect to two different cell groups provided by two different base stations. For example, a base station of the two base stations may be referred to as a master base station and a cell group provided by the master base station may be referred to as a master cell group (MCG). The other base station of the two base stations may be referred to as a secondary base station and the cell group provided by the secondary base station may be referred to as a secondary cell group (SCG). A bearer may be configured for the UE as a split bearer that may be handled by the two different cell groups. The PDCP layer may perform routing of packets corresponding to a split bearer to and/or from RLC channels associated with the cell groups.

In an example scenario not shown in FIG. 3, a bearer of the UE may be configured (e.g., with control signaling) with PDCP packet duplication. A bearer configured with PDCP duplication may be mapped to a plurality of RLC channels each corresponding to different one or more cells. The PDCP layer may duplicate packets of the bearer configured with PDCP duplication and the duplicated packets may be mapped to the different RLC channels. With PDCP packet duplication, the likelihood of correct reception of packets increases thereby enabling higher reliability.

The RLC layer of FIG. 3 (shown in FIG. 2A as RLC 203 at the UE side and RLC 213 at the gNB side) provides service to upper layers in the form of RLC channels. The RLC layer may include three transmission modes: transparent mode (TM), Unacknowledged mode (UM) and Acknowledged mode (AM). The RLC layer may perform error correction through automatic repeat request (ARQ) for the AM transmission mode, segmentation of RLC service data units (SDUs) for the AM and UM transmission modes and re-segmentation of RLC SDUs for AM transmission

mode, duplicate detection for the AM transmission mode, RLC SDU discard for the AM and UM transmission modes, etc. The UE may be configured with one RLC entity per RLC channel.

The MAC layer of FIG. 3 (shown in FIG. 2A as MAC 202 at the UE side and MAC 212 at the gNB side) provides services to the RLC layer in form of logical channels. The MAC layer may perform mapping between logical channels and transport channels, multiplexing/demultiplexing of MAC SDUs belonging to one or more logical channels into/from transport blocks (TBs) delivered to/from the physical layer on transport channels, reporting of scheduling information, error correction through hybrid automatic repeat request (HARQ), priority handling between UEs by means of dynamic scheduling, priority handling between logical channels of one UE by means of logical channel prioritization and/or padding. In case of carrier aggregation, a MAC entity may comprise one HARQ entity per cell. A MAC entity may support multiple numerologies, transmission timings and cells. The control signaling may configure logical channels with mapping restrictions. The mapping restrictions in logical channel prioritization may control the numerology(ies), cell(s), and/or transmission timing(s)/duration(s) that a logical channel may use.

The PHY layer of FIG. 3 (shown in FIG. 2A as PHY 201 at the UE side and PHY 211 at the gNB side) provides transport services to the MAC layer in form of transport channels. The physical layer may handle coding/decoding, HARQ soft combining, rate matching of a coded transport channel to physical channels, mapping of coded transport channels to physical channels, modulation and demodulation of physical channels, frequency and time synchronization, radio characteristics measurements and indication to higher layers, RF processing, and mapping to antennas and radio resources.

FIG. 4 shows example processing of packets at different protocol layers in accordance with several of various embodiments of the present disclosure. In this example, three Internet Protocol (IP) packets that are processed by the different layers of the NR protocol stack. The term SDU shown in FIG. 4 is the data unit that is entered from/to a higher layer. In contrast, a protocol data unit (PDU) is the data unit that is entered to/from a lower layer. The flow of packets in FIG. 4 is for downlink. An uplink data flow through layers of the NR protocol stack is similar to FIG. 4. In this example, the two leftmost IP packets are mapped by the SDAP layer (shown as SDAP 205 and SDAP 215 in FIG. 2A) to radio bearer 402 and the rightmost packet is mapped by the SDAP layer to the radio bearer 404. The SDAP layer adds SDAP headers to the IP packets which are entered into the PDCP layer as PDCP SDUs. The PDCP layer is shown as PDCP 204 and PDCP 214 in FIG. 2A. The PDCP layer adds the PDCP headers to the PDCP SDUs which are entered into the RLC layer as RLC SDUs. The RLC layer is shown as RLC 203 and RLC 213 in FIG. 2A. An RLC SDU may be segmented at the RLC layer. The RLC layer adds RLC headers to the RLC SDUs after segmentation (if segmented) which are entered into the MAC layer as MAC SDUs. The MAC layer adds the MAC headers to the MAC SDUs and multiplexes one or more MAC SDUs to form a PHY SDU (also referred to as a transport block (TB) or a MAC PDU).

In FIG. 4, the MAC SDUs are multiplexed to form a transport block. The MAC layer may multiplex one or more MAC control elements (MAC CEs) with zero or more MAC SDUs to form a transport block. The MAC CEs may also be referred to as MAC commands or MAC layer control

signaling and may be used for in-band control signaling. The MAC CEs may be transmitted by a base station to a UE (e.g., downlink MAC CEs) or by a UE to a base station (e.g., uplink MAC CEs). The MAC CEs may be used for transmission of information useful by a gNB for scheduling (e.g., buffer status report (BSR) or power headroom report (PHR)), activation/deactivation of one or more cells, activation/deactivation of configured radio resources for one or more processes, activation/deactivation of one or more processes, indication of parameters used in one or more processes, etc.

FIG. 5A and FIG. 5B show example mapping between logical channels, transport channels and physical channels for downlink and uplink, respectively in accordance with several of various embodiments of the present disclosure. As discussed before, the MAC layer provides services to higher layer in the form of logical channels. A logical channel may be classified as a control channel, if used for transmission of control and/or configuration information, or a traffic channel if used for transmission of user data. Example logical channels in NR include Broadcast Control Channel (BCCH) used for transmission of broadcast system control information, Paging Control Channel (PCCH) used for carrying paging messages for wireless devices with unknown locations, Common Control Channel (CCCH) used for transmission of control information between UEs and network and for UEs that have no RRC connection with the network, Dedicated Control Channel (DCCH) which is a point-to-point bi-directional channel for transmission of dedicated control information between a UE that has an RRC connection and the network and Dedicated Traffic Channel (DTCH) which is point-to-point channel, dedicated to one UE, for the transfer of user information and may exist in both uplink and downlink.

As discussed before, the PHY layer provides services to the MAC layer and higher layers in the form of transport channels. Example transport channels in NR include Broadcast Channel (BCH) used for transmission of part of the BCCH referred to as master information block (MIB), Downlink Shared Channel (DL-SCH) used for transmission of data (e.g., from DTCH in downlink) and various control information (e.g., from DCCH and CCCH in downlink and part of the BCCH that is not mapped to the BCH), Uplink Shared Channel (UL-SCH) used for transmission of uplink data (e.g., from DTCH in uplink) and control information (e.g., from CCCH and DCCH in uplink) and Paging Channel (PCH) used for transmission of paging information from the PCCH. In addition, Random Access Channel (RACH) is a transport channel used for transmission of random access preambles. The RACH does not carry a transport block. Data on a transport channel (except RACH) may be organized in transport blocks, wherein one or more transport blocks may be transmitted in a transmission time interval (TTI).

The PHY layer may map the transport channels to physical channels. A physical channel may correspond to time-frequency resources that are used for transmission of information from one or more transport channels. In addition to mapping transport channels to physical channels, the physical layer may generate control information (e.g., downlink control information (DCI) or uplink control information (UCI)) that may be carried by the physical channels. Example DCI include scheduling information (e.g., downlink assignments and uplink grants), request for channel state information report, power control command, etc. Example UCI include HARQ feedback indicating correct or incorrect reception of downlink transport blocks, channel state information report, scheduling request, etc. Example

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physical channels in NR include a Physical Broadcast Channel (PBCH) for carrying information from the BCH, a Physical Downlink Shared Channel (PDSCH) for carrying information from the PCH and the DL-SCH, a Physical Downlink Control Channel (PDCCH) for carrying DCI, a Physical Uplink Shared Channel (PUSCH) for carrying information from the UL-SCH and/or UCI, a Physical Uplink Control Channel (PUCCH) for carrying UCI and Physical Random Access Channel (PRACH) for transmission of RACH (e.g., random access preamble).

The PHY layer may also generate physical signals that are not originated from higher layers. As shown in FIG. 5A, example downlink physical signals include Demodulation Reference Signal (DM-RS), Phase Tracking Reference Signal (PT-RS), Channel State Information Reference Signal (CSI-RS), Primary Synchronization Signal (PSS) and Secondary Synchronization Signal (SSS). As shown in FIG. 5B, example uplink physical signals include DM-RS, PT-RS and sounding reference signal (SRS).

As indicated earlier, some of the protocol layers (PHY, MAC, RLC and PDCP) of the control plane of an NR Uu interface, are common between the user plane protocol stack (as shown in FIG. 2A) and the control plane protocol stack (as shown in FIG. 2B). In addition to PHY, MAC, RLC and PDCP, the control plane protocol stack includes the RRC protocol layer and the NAS protocol layer.

The NAS layer, as shown in FIG. 2B, terminates at the UE 200 and the AMF 220 entity of the 5G-C 130. The NAS layer is used for core network related functions and signaling including registration, authentication, location update and session management. The NAS layer uses services from the AS of the Uu interface to transmit the NAS messages.

The RRC layer, as shown in FIG. 2B, operates between the UE 200 and the gNB 210 (more generally NG-RAN 120) and may provide services and functions such as broadcast of system information (SI) related to AS and NAS as well as paging initiated by the 5G-C 130 or NG-RAN 120. In addition, the RRC layer is responsible for establishment, maintenance and release of an RRC connection between the UE 200 and the NG-RAN 120, carrier aggregation configuration (e.g., addition, modification and release), dual connectivity configuration (e.g., addition, modification and release), security related functions, radio bearer configuration/maintenance and release, mobility management (e.g., maintenance and context transfer), UE cell selection and reselection, inter-RAT mobility, QoS management functions, UE measurement reporting and control, radio link failure (RLF) detection and NAS message transfer. The RRC layer uses services from PHY, MAC, RLC and PDCP layers to transmit RRC messages using signaling radio bearers (SRBs). The SRBs are mapped to CCCH logical channel during connection establishment and to DCCH logical channel after connection establishment.

FIG. 6 shows example physical layer processes for signal transmission in accordance with several of various embodiments of the present disclosure. Data and/or control streams from MAC layer may be encoded/decoded to offer transport and control services over the radio transmission link. For example, one or more (e.g., two as shown in FIG. 6) transport blocks may be received from the MAC layer for transmission via a physical channel (e.g., a physical downlink shared channel or a physical uplink shared channel). A cyclic redundancy check (CRC) may be calculated and attached to a transport block in the physical layer. The CRC calculation may be based on one or more cyclic generator polynomials. The CRC may be used by the receiver for error detection. Following the transport block CRC attachment, a

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low-density parity check (LDPC) base graph selection may be performed. In example embodiments, two LDPC base graphs may be used wherein a first LDPC base graph may be optimized for small transport blocks and a second LDPC base graph may be optimized for comparatively larger transport blocks.

The transport block may be segmented into code blocks and code block CRC may be calculated and attached to a code block. A code block may be LDPC coded and the LDPC coded blocks may be individually rate matched. The code blocks may be concatenated to create one or more codewords. The contents of a codeword may be scrambled and modulated to generate a block of complex-valued modulation symbols. The modulation symbols may be mapped to a plurality of transmission layers (e.g., multiple-input multiple-output (MIMO) layers) and the transmission layers may be subject to transform precoding and/or precoding. The precoded complex-valued symbols may be mapped to radio resources (e.g., resource elements). The signal generator block may create a baseband signal and up-convert the baseband signal to a carrier frequency for transmission via antenna ports. The signal generator block may employ mixers, filters and/or other radio frequency (RF) components prior to transmission via the antennas. The functions and blocks in FIG. 6 are illustrated as examples and other mechanisms may be implemented in various embodiments.

FIG. 7 shows examples of RRC states and RRC state transitions at a UE in accordance with several of various embodiments of the present disclosure. A UE may be in one of three RRC states: RRC_IDLE 702, RRC_INACTIVE 704 and RRC_CONNECTED 706. In RRC_IDLE 702 state, no RRC context (e.g., parameters needed for communications between the UE and the network) may be established for the UE in the RAN. In RRC_IDLE 702 state, no data transfer between the UE and the network may take place and uplink synchronization is not maintained. The wireless device may sleep most of the time and may wake up periodically to receive paging messages. The uplink transmission of the UE may be based on a random access process and to enable transition to the RRC_CONNECTED 706 state. The mobility in RRC_IDLE 702 state is through a cell reselection procedure where the UE camps on a cell based on one or more criteria including signal strength that is determined based on the UE measurements.

In RRC_CONNECTED 706 state, the RRC context is established and both the UE and the RAN have necessary parameters to enable communications between the UE and the network. In the RRC_CONNECTED 706 state, the UE is configured with an identity known as a Cell Radio Network Temporary Identifier (C-RNTI) that is used for signaling purposes (e.g., uplink and downlink scheduling, etc.) between the UE and the RAN. The wireless device mobility in the RRC_CONNECTED 706 state is managed by the RAN. The wireless device provides neighboring cells and/or current serving cell measurements to the network and the network may make hand over decisions. Based on the wireless device measurements, the current serving base station may send a handover request message to a neighboring base station and may send a handover command to the wireless device to handover to a cell of the neighboring base station. The transition of the wireless device from the RRC_IDLE 702 state to the RRC_CONNECTED 706 state or from the RRC_CONNECTED 706 state to the RRC_IDLE 702 state may be based on connection establishment and connection release procedures (shown collectively as connection establishment/release 710 in FIG. 7).

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To enable a faster transition to the RRC_CONNECTED 706 state (e.g., compared to transition from RRC_IDLE 702 state to RRC_CONNECTED 706 state), an RRC_INACTIVE 704 state is used for an NR UE wherein, the RRC context is kept at the UE and the RAN. The transition from the RRC_INACTIVE 704 state to the RRC_CONNECTED 706 state is handled by RAN without CN signaling. Similar to the RRC_IDLE 702 state, the mobility in RRC_INACTIVE 704 state is based on a cell reselection procedure without involvement from the network. The transition of the wireless device from the RRC_INACTIVE 704 state to the RRC_CONNECTED 706 state or from the RRC_CONNECTED 706 state to the RRC_INACTIVE 704 state may be based on connection resume and connection inactivation procedures (shown collectively as connection resume/inactivation 712 in FIG. 7). The transition of the wireless device from the RRC_INACTIVE 704 state to the RRC_IDLE 702 state may be based on a connection release 714 procedure as shown in FIG. 7.

In NR, Orthogonal Frequency Division Multiplexing (OFDM), also called cyclic prefix OFDM (CP-OFDM), is the baseline transmission scheme in both downlink and uplink of NR and the Discrete Fourier Transform (DFT) spread OFDM (DFT-s-OFDM) is a complementary uplink transmission in addition to the baseline OFDM scheme. OFDM is multi-carrier transmission scheme wherein the transmission bandwidth may be composed of several narrowband sub-carriers. The subcarriers are modulated by the complex valued OFDM modulation symbols resulting in an OFDM signal. The complex valued OFDM modulation symbols are obtained by mapping, by a modulation mapper, the input data (e.g., binary digits) to different points of a modulation constellation diagram. The modulation constellation diagram depends on the modulation scheme. NR may use different types of modulation schemes including Binary Phase Shift Keying (BPSK), $\pi/2$ -BPSK, Quadrature Phase Shift Keying (QPSK), 16 Quadrature Amplitude Modulation (16QAM), 64QAM and 256QAM. Different and/or higher order modulation schemes (e.g., M-QAM in general) may be used. An OFDM signal with N subcarriers may be generated by processing N subcarriers in parallel for example by using Inverse Fast Fourier Transform (IFFT) processing. The OFDM receiver may use FFT processing to recover the transmitted OFDM modulation symbols. The subcarrier spacing of subcarriers in an OFDM signal is inversely proportional to an OFDM modulation symbol duration. For example, for a 15 KHz subcarrier spacing, duration of an OFDM signal is nearly 66.7 μ s. To enhance the robustness of OFDM transmission in time dispersive channels, a cyclic prefix (CP) may be inserted at the beginning of an OFDM symbol. For example, the last part of an OFDM symbol may be copied and inserted at the beginning of an OFDM symbol. The CP insertion enhanced the OFDM transmission scheme by preserving subcarrier orthogonality in time dispersive channels.

In NR, different numerologies may be used for OFDM transmission. A numerology of OFDM transmission may indicate a subcarrier spacing and a CP duration for the OFDM transmission. For example, a subcarrier spacing in NR may generally be a multiple of 15 KHz and expressed as $\Delta f = 2^{\mu} \cdot 15$ KHz ($\mu=0, 1, 2, \dots$). Example subcarrier spacings used in NR include 15 KHz ($\mu=0$), 30 KHz ($\mu=1$), 60 KHz ($\mu=2$), 120 KHz ($\mu=3$) and 240 KHz ($\mu=4$). As discussed before, a duration of OFDM symbol is inversely proportional to the subcarrier spacing and therefore OFDM symbol duration may depend on the numerology (e.g., the μ value).

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FIG. 8 shows an example time domain transmission structure in NR wherein OFDM symbols are grouped into slots, subframes and frames in accordance with several of various embodiments of the present disclosure. A slot is a group of $N_{\text{symbol}}^{\text{slot}}$ OFDM symbols, wherein the $N_{\text{symbol}}^{\text{slot}}$ may have a constant value (e.g., 14). Since different numerologies results in different OFDM symbol durations, duration of a slot may also depend on the numerology and may be variable. A subframe may have a duration of 1 ms and may be composed of one or more slots, the number of which may depend on the slot duration. The number of slots per subframe is therefore a function of μ and may generally expressed as $N_{\text{symbol}}^{\text{subframe}, \mu}$ and the number of symbols per subframe may be expressed as $N_{\text{symbol}}^{\text{subframe}, \mu} = N_{\text{symbol}}^{\text{slot}} N_{\text{symbol}}^{\text{subframe}, \mu}$. A frame may have a duration of 10 ms and may consist of 10 subframes. The number of slots per frame may depend on the numerology and therefore may be variable. The number of slots per frame may generally be expressed as $N_{\text{symbol}}^{\text{frame}, \mu}$.

An antenna port may be defined as a logical entity such that channel characteristics over which a symbol on the antenna port is conveyed may be inferred from the channel characteristics over which another symbol on the same antenna port is conveyed. For example, for DM-RS associated with a PDSCH, the channel over which a PDSCH symbol on an antenna port is conveyed may be inferred from the channel over which a DM-RS symbol on the same antenna port is conveyed, for example, if the two symbols are within the same resource as the scheduled PDSCH and/or in the same slot and/or in the same precoding resource block group (PRG). For example, for DM-RS associated with a PDCCH, the channel over which a PDCCH symbol on an antenna port is conveyed may be inferred from the channel over which a DM-RS symbol on the same antenna port is conveyed if, for example, the two symbols are within resources for which the UE may assume the same precoding being used. For example, for DM-RS associated with a PBCH, the channel over which a PBCH symbol on one antenna port is conveyed may be inferred from the channel over which a DM-RS symbol on the same antenna port is conveyed if, for example, the two symbols are within a SS/PBCH block transmitted within the same slot, and with the same block index. The antenna port may be different from a physical antenna. An antenna port may be associated with an antenna port number and different physical channels may correspond to different ranges of antenna port numbers.

FIG. 9 shows an example of time-frequency resource grid in accordance with several of various embodiments of the present disclosure. The number of subcarriers in a carrier bandwidth may be based on the numerology of OFDM transmissions in the carrier. A resource element, corresponding to one symbol duration and one subcarrier, may be the smallest physical resource in the time-frequency grid. A resource element (RE) for antenna port p and subcarrier spacing configuration μ may be uniquely identified by $(k, l)_{p, \mu}$, where k is the index of a subcarrier in the frequency domain and l may refer to the symbol position in the time domain relative to some reference point. A resource block may be defined as $N_{\text{SC}}^{\text{RB}} = 12$ subcarriers. Since subcarrier spacing depends on the numerology of OFDM transmission, the frequency domain span of a resource block may be variable and may depend on the numerology. For example, for a subcarrier spacing of 15 KHz (e.g., $\mu=0$), a resource block may be 180 KHz and for a subcarrier spacing of 30 KHz (e.g., $\mu=1$), a resource block may be 360 KHz.

With large carrier bandwidths defined in NR and due to limited capabilities for some UEs (e.g., due to hardware limitations), a UE may not support an entire carrier bandwidth. Receiving on the full carrier bandwidth may imply high energy consumption. For example, transmitting downlink control channels on the full downlink carrier bandwidth may result in high power consumption for wide carrier bandwidths. NR may use a bandwidth adaptation procedure to dynamically adapt the transmit and receive bandwidths. The transmit and receive bandwidth of a UE on a cell may be smaller than the bandwidth of the cell and may be adjusted. For example, the width of the transmit and/or receive bandwidth may change (e.g., shrink during period of low activity to save power); the location of the transmit and/or receive bandwidth may move in the frequency domain (e.g., to increase scheduling flexibility); and the subcarrier spacing of the transmit or receive bandwidth may change (e.g., to allow different services). A subset of the cell bandwidth may be referred to as a Bandwidth Part (BWP) and bandwidth adaptation may be achieved by configuring the UE with one or more BWPs. The base station may configure a UE with a set of downlink BWPs and a set of uplink BWPs. A BWP may be characterized by a numerology (e.g., subcarrier spacing and cyclic prefix) and a set of consecutive resource blocks in the numerology of the BWP. One or more first BWPs of the one or more BWPs of the cell may be active at a time. An active BWP may be an active downlink BWP or an active uplink BWP.

FIG. 10 shows an example of bandwidth part adaptation and switching. In this example, three BWPs (BWP₁ **1004**, BWP₂ **1006** and BWP₃ **1008**) are configured for a UE on a carrier bandwidth. The BWP₁ is configured with a bandwidth of 40 MHz and a numerology with subcarrier spacing of 15 KHz, the BWP₂ is configured with a bandwidth of 10 MHz and a numerology with subcarrier spacing of 15 KHz and the BWP₃ is configured with a bandwidth of 20 MHz and a subcarrier spacing of 60 KHz. The wireless device may switch from a first BWP (e.g., BWP₁) to a second BWP (e.g., BWP₂). An active BWP of the cell may change from the first BWP to the second BWP in response to the BWP switching.

The BWP switching (e.g., BWP switching **1010**, BWP switching **1012**, BWP switching **1014**, or BWP switching **1016** in FIG. 10) may be based on a command from the base station. The command may be a DCI comprising scheduling information for the UE in the second BWP. In case of uplink BWP switching, the first BWP and the second BWP may be uplink BWPs and the scheduling information may be an uplink grant for uplink transmission via the second BWP. In case of downlink BWP switching, the first BWP and the second BWP may be downlink BWPs and the scheduling information may be a downlink assignment for downlink reception via the second BWP.

The BWP switching (e.g., BWP switching **1010**, BWP switching **1012**, BWP switching **1014**, or BWP switching **1016** in FIG. 10) may be based on an expiry of a timer. The base station may configure a wireless device with a BWP inactivity timer and the wireless device may switch to a default BWP (e.g., default downlink BWP) based on the expiry of the BWP inactivity timer. The expiry of the BWP inactivity timer may be an indication of low activity on the current active downlink BWP. The base station may configure the wireless device with the default downlink BWP. If the base station does not configure the wireless device with the default BWP, the default BWP may be an initial downlink BWP. The initial active BWP may be the BWP that the

wireless device receives scheduling information for remaining system information upon transition to an RRC_CONNECTED state.

A wireless device may monitor a downlink control channel of a downlink BWP. For example, the UE may monitor a set of PDCCH candidates in configured monitoring occasions in one or more configured COntrol REsource SETs (CORESETs) according to the corresponding search space configurations. A search space configuration may define how/where to search for PDCCH candidates. For example, the search space configuration parameters may comprise a monitoring periodicity and offset parameter indicating the slots for monitoring the PDCCH candidates. The search space configuration parameters may further comprise a parameter indicating a first symbol with a slot within the slots determined for monitoring PDCCH candidates. A search space may be associated with one or more CORESETs and the search space configuration may indicate one or more identifiers of the one or more CORESETs. The search space configuration parameters may further indicate that whether the search space is a common search space or a UE-specific search space. A common search space may be monitored by a plurality of wireless devices and a UE-specific search space may be dedicated to a specific UE.

FIG. 11A shows example arrangements of carriers in carrier aggregation in accordance with several of various embodiments of the present disclosure. With carrier aggregation, multiple NR component carriers (CCs) may be aggregated. Downlink transmissions to a wireless device may take place simultaneously on the aggregated downlink CCs resulting in higher downlink data rates. Uplink transmissions from a wireless device may take place simultaneously on the aggregated uplink CCs resulting in higher uplink data rates. The component carriers in carrier aggregation may be on the same frequency band (e.g., intra-band carrier aggregation) or on different frequency bands (e.g., inter-band carrier aggregation). The component carriers may also be contiguous or non-contiguous. This results in three possible carrier aggregation scenarios, intra-band contiguous CA **1102**, intra-band non-contiguous CA **1104** and inter-band CA **1106** as shown in FIG. 11A. Depending on the UE capability for carrier aggregation, a UE may transmit and/or receive on multiple carriers or for a UE that is not capable of carrier aggregation, the UE may transmit and/or receive on one component carrier at a time. In this disclosure, the carrier aggregation is described using the term cell and a carrier aggregation capable UE may transmit and/or receive via multiple cells.

In carrier aggregation, a UE may be configured with multiple cells. A cell of the multiple cells configured for the UE may be referred to as a Primary Cell (PCell). The PCell may be the first cell that the UE is initially connected to. One or more other cells configured for the UE may be referred to as Secondary Cells (SCells). The base station may configure a UE with multiple SCells. The configured SCells may be deactivated upon configuration and the base station may dynamically activate or deactivate one or more of the configured SCells based on traffic and/or channel conditions. The base station may activate or deactivate configured SCells using a SCell Activation/Deactivation MAC CE. The SCell Activation/Deactivation MAC CE may comprise a bitmap, wherein each bit in the bitmap may correspond to a SCell and the value of the bit indicates an activation status or deactivation status of the SCell.

An SCell may also be deactivated in response to expiry of a SCell deactivation timer of the SCell. The expiry of an SCell deactivation timer of an SCell may be an indication of

low activity (e.g., low transmission or reception activity) on the SCell. The base station may configure the SCell with an SCell deactivation timer. The base station may not configure an SCell deactivation timer for an SCell that is configured with PUCCH (also referred to as a PUCCH SCell). The configuration of the SCell deactivation timer may be per configured SCell and different SCells may be configured with different SCell deactivation timer values. The SCell deactivation timer may be restarted based on one or more criteria including reception of downlink control information on the SCell indicating uplink grant or downlink assignment for the SCell or reception of downlink control information on a scheduling cell indicating uplink grant or downlink assignment for the SCell or transmission of a MAC PDU based on a configured uplink grant or reception of a configured downlink assignment.

A PCell for a UE may be an SCell for another UE and a SCell for a UE may be PCell for another UE. The configuration of PCell may be UE-specific. One or more SCells of the multiple SCells configured for a UE may be configured as downlink-only SCells, e.g., may only be used for downlink reception and may not be used for uplink transmission. In case of self-scheduling, the base station may transmit signaling for uplink grants and/or downlink assignments on the same cell that the corresponding uplink or downlink transmission takes place. In case of cross-carrier scheduling, the base station may transmit signaling for uplink grants and/or downlink assignments on a cell different from the cell that the corresponding uplink or downlink transmission takes place.

FIG. 11B shows examples of uplink control channel groups in accordance with several of various embodiments of the present disclosure. A base station may configure a UE with multiple PUCCH groups wherein a PUCCH group comprises one or more cells. For example, as shown in FIG. 11B, the base station may configure a UE with a primary PUCCH group 1114 and a secondary PUCCH group 1116. The primary PUCCH group may comprise the PCell 1110 and one or more first SCells. First UCI corresponding to the PCell and the one or more first SCells of the primary PUCCH group may be transmitted by the PUCCH of the PCell. The first UCI may be, for example, HARQ feedback for downlink transmissions via downlink CCs of the PCell and the one or more first SCells. The secondary PUCCH group may comprise a PUCCH SCell and one or more second SCells. Second UCI corresponding to the PUCCH SCell and the one or more second SCells of the secondary PUCCH group may be transmitted by the PUCCH of the PUCCH SCell. The second UCI may be, for example, HARQ feedback for downlink transmissions via downlink CCs of the PUCCH SCell and the one or more second SCells.

FIG. 12A, FIG. 12B and FIG. 12C show example random access processes in accordance with several of various embodiments of the present disclosure. FIG. 12A shows an example of four step contention-based random access (CBRA) procedure. The four-step CBRA procedure includes exchanging four messages between a UE and a base station. Msg1 may be for transmission (or retransmission) of a random access preamble by the wireless device to the base station. Msg2 may be the random access response (RAR) by the base station to the wireless device. Msg3 is the scheduled transmission based on an uplink grant indicated in Msg2 and Msg4 may be for contention resolution.

The base station may transmit one or more RRC messages comprising configuration parameters of the random access parameters. The random access parameters may indicate

radio resources (e.g., time-frequency resources) for transmission of the random access preamble (e.g., Msg1), configuration index, one or more parameters for determining the power of the random access preamble (e.g., a power ramping parameter, a preamble received target power, etc.), a parameter indicating maximum number of preamble transmission, RAR window for monitoring RAR, cell-specific random access parameters and UE specific random access parameters. The UE-specific random access parameters may indicate one or more PRACH occasions for random access preamble (e.g., Msg1) transmissions. The random access parameters may indicate association between the PRACH occasions and one or more reference signals (e.g., SSB or CSI-RS). The random access parameters may further indicate association between the random access preambles and one or more reference signals (e.g., SSB or CSI-RS). The UE may use one or more reference signals (e.g., SSB(s) or CSI-RS(s)) and may determine a random access preamble to use for Msg1 transmission based on the association between the random access preambles and the one or more reference signals. The UE may use one or more reference signals (e.g., SSB(s) or CSI-RS(s)) and may determine the PRACH occasion to use for Msg1 transmission based on the association between the PRACH occasions and the reference signals. The UE may perform a retransmission of the random access preamble if no response is received with the RAR window following the transmission of the preamble. UE may use a higher transmission power for retransmission of the preamble. UE may determine the higher transmission power of the preamble based on the power ramping parameter.

Msg2 is for transmission of RAR by the base station. Msg2 may comprise a plurality of RARs corresponding to a plurality of random access preambles transmitted by a plurality of UEs. Msg2 may be associated with a random access temporary radio identifier (RA-RNTI) and may be received in a common search space of the UE. The RA-RNTI may be based on the PRACH occasion (e.g., time and frequency resources of a PRACH) in which a random access preamble is transmitted. RAR may comprise a timing advance command for uplink timing adjustment at the UE, an uplink grant for transmission of Msg3 and a temporary C-RNTI. In response to the successful reception of Msg2, the UE may transmit the Msg3. Msg3 and Msg4 may enable contention resolution in case of CBRA. In a CBRA, a plurality of UEs may transmit the same random access preamble and may consider the same RAR as being corresponding to them. UE may include a device identifier in Msg3 (e.g., a C-RNTI, temporary C-RNTI or other UE identity). Base station may transmit the Msg4 with a PDSCH and UE may assume that the contention resolution is successful in response to the PDSCH used for transmission of Msg4 being associated with the UE identifier included in Msg3.

FIG. 12B shows an example of a contention-free random access (CFRA) process. Msg 1 (random access preamble) and Msg 2 (random access response) in FIG. 12B for CFRA may be analogous to Msg 1 and Msg 2 in FIG. 12A for CBRA. In an example, the CFRA procedure may be initiated in response to a PDCCH order from a base station. The PDCCH order for initiating the CFRA procedure by the wireless device may be based on a DCI having a first format (e.g., format 1_0). The DCI for the PDCCH order may comprise a random access preamble index, an UL/SUL indicator indicating an uplink carrier of a cell (e.g., normal uplink carrier or supplementary uplink carrier) for transmission of the random access preamble, a SS/PBCH index

indicating the SS/PBCH that may be used to determine a RACH occasion for PRACH transmission, a PRACH mask index indicating the RACH occasion associated with the SS/PBCH indicated by the SS/PBCH index for PRACH transmission, etc. In an example, the CFRA process may be started in response to a beam failure recovery process. The wireless device may start the CFRA for the beam failure recovery without a command (e.g., PDCCH order) from the base station and by using the wireless device dedicated resources.

FIG. 12C shows an example of a two-step random access process comprising two messages exchanged between a wireless device and a base station. Msg A may be transmitted by the wireless device to the base station and may comprise one or more transmissions of a preamble and/or one or more transmissions of a transport block. The transport block in Msg A and Msg 3 in FIG. 12A may have similar and/or equivalent contents. The transport block of Msg A may comprise data and control information (e.g., SR, HARQ feedback, etc.). In response to the transmission of Msg A, the wireless device may receive Msg B from the base station. Msg B in FIG. 12C and Msg 2 (e.g., RAR) illustrated in FIGS. 12A and 12B may have similar and/or equivalent content.

The base station may periodically transmit synchronization signals (SSs), e.g., primary SS (PSS) and secondary SS (SSS) along with PBCH on each NR cell. The PSS/SSS together with PBCH is jointly referred to as a SS/PBCH block. The SS/PBCH block enables a wireless device to find a cell when entering to the mobile communications network or find new cells when moving within the network. The SS/PBCH block spans four OFDM symbols in time domain. The PSS is transmitted in the first symbol and occupies 127 subcarriers in frequency domain. The SSS is transmitted in the third OFDM symbol and occupies the same 127 subcarriers as the PSS. There are eight and nine empty subcarriers on each side of the SSS. The PBCH is transmitted on the second OFDM symbol occupying 240 subcarriers, the third OFDM symbol occupying 48 subcarriers on each side of the SSS, and on the fourth OFDM symbol occupying 240 subcarriers. Some of the PBCH resources indicated above may be used for transmission of the demodulation reference signal (DMRS) for coherent demodulation of the PBCH. The SS/PBCH block is transmitted periodically with a period ranging from 5 ms to 160 ms. For initial cell search or for cell search during inactive/idle state, a wireless device may assume that the SS/PBCH block is repeated at least every 20 ms.

In NR, transmissions using of antenna arrays, with many antenna elements, and beamforming plays an important role specially in higher frequency bands. Beamforming enables higher capacity by increasing the signal strength (e.g., by focusing the signal energy in a specific direction) and by lowering the amount interference received at the wireless devices. The beamforming techniques may generally be divided to analog beamforming and digital beamforming techniques. With digital beamforming, signal processing for beamforming is carried out in the digital domain before digital-to-analog conversion and detailed control of both amplitude and phase of different antenna elements may be possible. With analog beamforming, the signal processing for beamforming is carried out in the analog domain and after the digital to analog conversion. The beamformed transmissions may be in one direction at a time. For example, the wireless devices that are in different directions relative to the base station may receive their downlink

transmissions at different times. For analog receiver-side beamforming, the receiver may focus its receiver beam in one direction at a time.

In NR, the base station may use beam sweeping for transmission of SS/PBCH blocks. The SS/PBCH blocks may be transmitted in different beams using time multiplexing. The set of SS/PBCH blocks that are transmitted in one beam sweep may be referred to as a SS/PBCH block set. The period of PBCH/SSB block transmission may be a time duration between a SS/PBCH block transmission in a beam and the next SS/PBCH block transmission in the same beam. The period of SS/PBCH block is, therefore, also the period of the SS/PBCH block set.

FIG. 13A shows example time and frequency structure of SS/PBCH blocks and their associations with beams in accordance with several of various embodiments of the present disclosure. In this example, a SS/PBCH block (also referred to as SSB) set comprise L SSBs wherein an SSB in the SSB set is associated with (e.g., transmitted in) one of L beams of a cell. The transmission of SSBs of an SSB set may be confined within a 5 ms interval, either in a first half-frame or a second half-frame of a 10 ms frame. The number of SSBs in an SSB set may depend on the frequency band of operation. For example, the number of SSBs in a SSB set may be up to four SSBs in frequency bands below 3 GHz enabling beam sweeping of up to four beams, up to eight SSBs in frequency bands between 3 GHz and 6 GHz enabling beam sweeping of up to eight beams, and up to sixty four SSBs in higher frequency bands enabling beam sweeping of up to sixty four beams. The SSs of an SSB may depend on a physical cell identity (PCI) of the cell and may be independent of which beam of the cell is used for transmission of the SSB. The PBCH of an SSB may indicate a time index parameter and the wireless device may determine the relative position of the SSB within the SSB set using the time index parameter. The wireless device may use the relative position of the SSB within an SSB set for determining the frame timing and/or determining RACH occasions for a random access process.

A wireless device entering the mobile communications network may first search for the PSS. After detecting the PSS, the wireless device may determine the synchronization up to the periodicity of the PSS. By detecting the PSS, the wireless device may determine the transmission timing of the SSS. The wireless device may determine the PCI of the cell after detecting the SSS. The PBCH of a SS/PBCH block is a downlink physical channel that carries the MIB. The MIB may be used by the wireless device to obtain remaining system information (RMSI) that is broadcast by the network. The RMSI may include System Information Block 1 (SIB1) that contains information required for the wireless device to access the cell.

As discussed earlier, the wireless device may determine a time index parameter from the SSB. The PBCH comprises a half-frame parameter indicating whether the SSB is in the first 5 ms half or the second 5 ms half of a 10 ms frame. The wireless device may determine the frame boundary using the time index parameter and the half-frame parameter. In addition, the PBCH may comprise a parameter indicating the system frame number (SFN) of the cell.

The base station may transmit CSI-RS and a UE may measure the CSI-RS to obtain channel state information (CSI). The base station may configure the CSI-RS in a UE-specific manner. In some scenarios, same set of CSI-RS resources may be configured for multiple UEs and one or more resource elements of a CSI-RS resource may be shared among multiple UEs. A CSI-RS resource may be configured

such that it does not collide with a CORESET configured for the wireless device and/or with a DMRS of a PDSCH scheduled for the wireless device and/or transmitted SSBs. The UE may measure one or more CSI-RSs configured for the UE and may generate a CSI report based on the CSI-RS measurements and may transmit the CSI report to the base station for scheduling, link adaptation and/or other purposes.

NR supports flexible CSI-RS configurations. A CSI-RS resource may be configured with single or multiple antenna ports and with configurable density. Based on the number of configured antenna ports, a CSI-RS resource may span different number of OFDM symbols (e.g., 1, 2, and 4 symbols). The CSI-RS may be configured for a downlink BWP and may use the numerology of the downlink BWP. The CSI-RS may be configured to cover the full bandwidth of the downlink BWP or a portion of the downlink BWP. In some case, the CSI-RS may be repeated in every resource block of the CSI-RS bandwidth, referred to as CSI-RS with density equal to one. In some cases, the CSI-RS may be configured to be repeated in every other resource block of the CSI-RS bandwidth. CSI-RS may be non-zero power (NZP) CSI-RS or zero-power (ZP) CSI-RS.

The base station may configure a wireless device with one or more sets of NZP CSI-RS resources. The base station may configure the wireless device with a NZP CSI-RS resource set using an RRC information element (IE) NZP-CSI-RS-ResourceSet indicating a NZP CSI-RS resource set identifier (ID) and parameters specific to the NZP CSI-RS resource set. An NZP CSI-RS resource set may comprise one or more CSI-RS resources. An NZP CSI-RS resource set may be configured as part of the CSI measurement configuration.

The CSI-RS may be configured for periodic, semi-persistent or aperiodic transmission. In case of the periodic and semi-persistent CSI-RS configurations, the wireless device may be configured with a CSI resource periodicity and offset parameter that indicate a periodicity and corresponding offset in terms of number of slots. The wireless device may determine the slots that the CSI-RSs are transmitted. For semi-persistent CSI-RS, the CSI-RS resources for CSI-RS transmissions may be activated and deactivated by using a semi-persistent (SP) CSI-CSI Resource Set Activation/Deactivation MAC CE. In response to receiving a MAC CE indicating activation of semi-persistent CSI-RS resources, the wireless device may assume that the CSI-RS transmissions will continue until the CSI-RS resources for CSI-RS transmissions are activated.

As discussed before, CSI-RS may be configured for a wireless device as NZP CSI-RS or ZP CSI-RS. The configuration of the ZP CSI-RS may be similar to the NZP CSI-RS with the difference that the wireless device may not carry out measurements for the ZP CSI-RS. By configuring ZP CSI-RS, the wireless device may assume that a scheduled PDSCH that includes resource elements from the ZP CSI-RS is rate matched around those ZP CSI-RS resources. For example, a ZP CSI-RS resource configured for a wireless device may be an NZP CSI-RS resource for another wireless device. For example, by configuring ZP CSI-RS resources for the wireless device, the base station may indicate to the wireless device that the PDSCH scheduled for the wireless device is rate matched around the ZP CSI-RS resources.

A base station may configure a wireless device with channel state information interference measurement (CSI-IM) resources. Similar to the CSI-RS configuration, configuration of locations and density of CSI-IM resources may be flexible. The CSI-IM resources may be periodic (configured with a periodicity), semi-persistent (configured with a

periodicity and activated and deactivated by MAC CE) or aperiodic (triggered by a DCI).

Tracking reference signals (TRSs) may be configured for a wireless device as a set of sparse reference signals to assist the wireless in time and frequency tracking and compensating time and frequency variations in its local oscillator. The wireless device may further use the TRSs for estimating channel characteristics such as delay spread or doppler frequency. The base station may use a CSI-RS configuration for configuring TRS for the wireless device. The TRS may be configured as a resource set comprising multiple periodic NZP CSI-RS resources.

A base station may configure a UE and the UE may transmit sounding reference signals (SRSs) to enable uplink channel sounding/estimation at the base station. The SRS may support up to four antenna ports and may be designed with low cubic metric enabling efficient operation of the wireless device amplifier. The SRS may span one or more (e.g., one, two or four) consecutive OFDM symbols in time domain and may be located within the last n (e.g., six) symbols of a slot. In the frequency domain, the SRS may have a structure that is referred to as a comb structure and may be transmitted on every N th subcarrier. Different SRS transmissions from different wireless devices may have different comb structures and may be multiplexed in frequency domain.

A base station may configure a wireless device with one or more SRS resource sets and an SRS resource set may comprise one or more SRS resources. The SRS resources in an SRS resources set may be configured for periodic, semi-persistent or aperiodic transmission. The periodic SRS and the semi-persistent SRS resources may be configured with periodicity and offset parameters. The Semi-persistent SRS resources of a configured semi-persistent SRS resource set may be activated or deactivated by a semi-persistent (SP) SRS Activation/Deactivation MAC CE. The set of SRS resources included in an aperiodic SRS resource set may be activated by a DCI. A value of a field (e.g., an SRS request field) in the DCI may indicate activation of resources in an aperiodic SRS resource set from a plurality of SRS resource sets configured for the wireless device.

An antenna port may be associated with one or more reference signals. The receiver may assume that the one or more reference signals, associated with the antenna port, may be used for estimating channel corresponding to the antenna port. The reference signals may be used to derive channel state information related to the antenna port. Two antenna ports may be referred to as quasi co-located if characteristics (e.g., large-scale properties) of the channel over which a symbol is conveyed on one antenna port may be inferred from the channel over which a symbol is conveyed from another antenna port. For example, a UE may assume that radio channels corresponding to two different antenna ports have the same large-scale properties if the antenna ports are specified as quasi co-located. In some cases, the UE may assume that two antenna ports are quasi co-located based on signaling received from the base station. Spatial quasi-colocation (QCL) between two signals may be, for example, due to the two signals being transmitted from the same location and in the same beam. If a receive beam is good for a signal in a group of signals that are spatially quasi co-located, it may be assumed also be good for the other signals in the group of signals.

The CSI-RS in the downlink and the SRS in uplink may serve as quasi-location (QCL) reference for other physical downlink channels and physical uplink channels, respectively. For example, a downlink physical channel (e.g.,

PDSCH or PDCCH) may be spatially quasi co-located with a downlink reference signal (e.g., CSI-RS or SSB). The wireless device may determine a receive beam based on measurement on the downlink reference signal and may assume that the determined received beam is also good for reception of the physical channels (e.g., PDSCH or PDCCH) that are spatially quasi co-located with the downlink reference signal. Similarly, an uplink physical channel (e.g., PUSCH or PUCCH) may be spatially quasi co-located with an uplink reference signal (e.g., SRS). The base station may determine a receive beam based on measurement on the uplink reference signal and may assume that the determined received beam is also good for reception of the physical channels (e.g., PUSCH or PUCCH) that are spatially quasi co-located with the uplink reference signal.

The Demodulation Reference Signals (DM-RSs) enables channel estimation for coherent demodulation of downlink physical channels (e.g., PDSCH, PDCCH and PBH) and uplink physical channels (e.g., PUSCH and PUCCH). The DM-RS may be located early in the transmission (e.g., front-loaded DM-RS) and may enable the receiver to obtain the channel estimate early and reduce the latency. The time-domain structure of the DM-RS (e.g., symbols wherein the DM-RS are located in a slot) may be based on different mapping types.

The Phase Tracking Reference Signals (PT-RSs) enables tracking and compensation of phase variations across the transmission duration. The phase variations may be, for example, due to oscillator phase noise. The oscillator phase noise may become more severe in higher frequencies (e.g., mmWave frequency bands). The base station may configure the PT-RS for uplink and/or downlink. The PT-RS configuration parameters may indicate frequency and time density of PT-RS, maximum number of ports (e.g., uplink ports), resource element offset, configuration of uplink PT-RS without transform precoder (e.g., CP-OFDM) or with transform precoder (e.g., DFT-s-OFDM), etc. The subcarrier number and/or resource blocks used for PT-RS transmission may be based on the C-RNTI of the wireless device to reduce risk of collisions between PT-RSs of wireless devices scheduled on overlapping frequency domain resources.

FIG. 13B shows example time and frequency structure of CSI-RSs and their association with beams in accordance with several of various embodiments of the present disclosure. A beam of the L beams shown in FIG. 13B may be associated with a corresponding CSI-RS resource. The base station may transmit the CSI-RSs using the configured CSI-RS resources and a UE may measure the CSI-RSs (e.g., received signal received power (RSRP) of the CSI-RSs) and report the CSI-RS measurements to the base station based on a reporting configuration. For example, the base station may determine one or more transmission configuration indication (TCI) states and may indicate the one or more TCI states to the UE (e.g., using RRC signaling, a MAC CE and/or a DCI). Based on the one or more TCI states indicated to the UE, the UE may determine a downlink receive beam and receive downlink transmissions using the receive beam. In case of a beam correspondence, the UE may determine a spatial domain filter of a transmit beam based on spatial domain filter of a corresponding receive beam. Otherwise, the UE may perform an uplink beam selection procedure to determine the spatial domain filter of the transmit beam. The UE may transmit one or more SRSs using the SRS resources configured for the UE and the base station may measure the SRSs and determine/select the transmit beam for the UE based on the SRS measurements. The base station may indicate the selected beam to the UE. The CSI-RS resources shown

in FIG. 13B may be for one UE. The base station may configure different CSI-RS resources associated with a given beam for different UEs by using frequency division multiplexing.

A base station and a wireless device may perform beam management procedures to establish beam pairs (e.g., transmit and receive beams) that jointly provide good connectivity. For example, in the downlink direction, the UE may perform measurements for a beam pair and estimate channel quality for a transmit beam by the base station (or a transmission reception point (TRP) more generally) and the receive beam by the UE. The UE may transmit a report indicating beam pair quality parameters. The report may comprise one or more parameters indicating one or more beams (e.g., a beam index, an identifier of reference signal associated with a beam, etc.), one or more measurement parameters (e.g., RSRP), a precoding matrix indicator (PMI), a channel quality indicator (CQI), and/or a rank indicator (RI).

FIG. 14A, FIG. 14B and FIG. 14C show example beam management processes (referred to as P1, P2 and P3, respectively) in accordance with several of various embodiments of the present disclosure. The P1 process shown in FIG. 14A may enable, based on UE measurements, selection of a base station (or TRP more generally) transmit beam and/or a wireless device receive beam. The TRP may perform a beam sweeping procedure where the TRP may sequentially transmit reference signals (e.g., SSB and/or CSI-RS) on a set of beams and the UE may select a beam from the set of beams and may report the selected beam to the TRP. The P2 procedure as shown in FIG. 14B may be a beam refinement procedure. The selection of the TRP transmit beam and the UE receive beam may be regularly reevaluated due to movements and/or rotations of the UE or movement of other objects. In an example, the base station may perform the beam sweeping procedure over a smaller set of beams and the UE may select the best beam over the smaller set. In an example, the beam shape may be narrower compared to beam selected based on the P1 procedure. Using the P3 procedure as shown in FIG. 14C, the TRP may fix its transmit beam and the UE may refine its receive beam.

A wireless device may receive one or more messages from a base station. The one or more messages may comprise one or more RRC messages. The one or more messages may comprise configuration parameters of a plurality of cells for the wireless device. The plurality of cells may comprise a primary cell and one or more secondary cells. For example, the plurality of cells may be provided by a base station and the wireless device may communicate with the base station using the plurality of cells. For example, the plurality of cells may be provided by multiple base station (e.g., in case of dual and/or multi-connectivity). The wireless device may communicate with a first base station, of the multiple base stations, using one or more first cells of the plurality of cells. The wireless device may communicate with a second base station of the multiple base stations using one or more second cells of the plurality of cells.

The one or more messages may comprise configuration parameters used for processes in physical, MAC, RLC, PCDP, SDAP, and/or RRC layers of the wireless device. For example, the configuration parameters may include values of timers used in physical, MAC, RLC, PCDP, SDAP, and/or RRC layers. For example, the configuration parameters may include parameters for configuring different channels (e.g., physical layer channel, logical channels, RLC channels, etc.) and/or signals (e.g., CSI-RS, SRS, etc.).

Upon starting a timer, the timer may start running until the timer is stopped or until the timer expires. A timer may be restarted if it is running. A timer may be started if it is not running (e.g., after the timer is stopped or after the timer expires). A timer may be configured with or may be associated with a value (e.g., an initial value). The timer may be started or restarted with the value of the timer. The value of the timer may indicate a time duration that the timer may be running upon being started or restarted and until the timer expires. The duration of a timer may not be updated until the timer is stopped or expires (e.g., due to BWP switching). This specification may disclose a process that includes one or more timers. The one or more timers may be implemented in multiple ways. For example, a timer may be used by the wireless device and/or base station to determine a time window [t1, t2], wherein the timer is started at time t1 and expires at time t2 and the wireless device and/or the base station may be interested in and/or monitor the time window [t1, t2], for example to receive a specific signaling. Other examples of implementation of a timer may be provided.

FIG. 15 shows example components of a wireless device and a base station that are in communication via an air interface in accordance with several of various embodiments of the present disclosure. The wireless device 1502 may communicate with the base station 1542 over the air interface 1532. The wireless device 1502 may include a plurality of antennas. The base station 1542 may include a plurality of antennas. The plurality of antennas at the wireless device 1502 and/or the base station 1542 enables different types of multiple antenna techniques such as beamforming, single-user and/or multi-user MIMO, etc.

The wireless device 1502 and the base station 1542 may have one or more of a plurality of modules/blocks, for example RF front end (e.g., RF front end 1530 at the wireless device 1502 and RF front end 1570 at the base station 1542), Data Processing System (e.g., Data Processing System 1524 at the wireless device 1502 and Data Processing System 1564 at the base station 1542), Memory (e.g., Memory 1512 at the wireless device 1502 and Memory 1542 at the base station 1542). Additionally, the wireless device 1502 and the base station 1542 may have other modules/blocks such as GPS (e.g., GPS 1514 at the wireless device 1502 and GPS 1554 at the base station 1542).

An RF front end module/block may include circuitry between antennas and a Data Processing System for proper conversion of signals between these two modules/blocks. An RF front end may include one or more filters (e.g., Filter(s) 1526 at RF front end 1530 or Filter(s) 1566 at the RF front end 1570), one or more amplifiers (e.g., Amplifier(s) 1528 at the RF front end 1530 and Amplifier(s) 1568 at the RF front end 1570). The Amplifier(s) may comprise power amplifier (s) for transmission and low-noise amplifier(s) (LNA(s)) for reception.

The Data Processing System 1524 and the Data Processing System 1564 may process the data to be transmitted or the received signals by implementing functions at different layers of the protocol stack such as PHY, MAC, RLC, etc. Example PHY layer functions that may be implemented by the Data Processing System 1524 and/or 1564 may include forward error correction, interleaving, rate matching, modulation, precoding, resource mapping, MIMO processing, etc. Similarly, one or more functions of the MAC layer, RLC layer and/or other layers may be implemented by the Data Processing System 1524 and/or the Data Processing System 1564. One or more processes described in the present disclosure may be implemented by the Data Processing System 1524 and/or the Data Processing System 1564. A

Data Processing System may include an RF module (RF module 1516 at the Data Processing System 1524 and RF module 1556 at the Data Processing System 1564) and/or a TX/RX processor (e.g., TX/RX processor 1518 at the Data Processing System 1524 and TX/RX processor 1558 at the Data Processing System 1566) and/or a central processing unit (CPU) (e.g., CPU 1520 at the Data Processing System 1524 and CPU 1560 at the Data Processing System 1564) and/or a graphical processing unit (GPU) (e.g., GPU 1522 at the Data Processing System 1524 and GPU 1562 at the Data Processing System 1564).

The Memory 1512 may have interfaces with the Data Processing System 1524 and the Memory 1552 may have interfaces with Data Processing System 1564, respectively. The Memory 1512 or the Memory 1552 may include non-transitory computer readable mediums (e.g., Storage Medium 1510 at the Memory 1512 and Storage Medium 1550 at the Memory 1552) that may store software code or instructions that may be executed by the Data Processing System 1524 and Data Processing System 1564, respectively, to implement the processes described in the present disclosure. The Memory 1512 or the Memory 1552 may include random-access memory (RAM) (e.g., RAM 1506 at the Memory 1512 or RAM 1546 at the Memory 1552) or read-only memory (ROM) (e.g., ROM 1508 at the Memory 1512 or ROM 1548 at the Memory 1552) to store data and/or software codes.

The Data Processing System 1524 and/or the Data Processing System 1564 may be connected to other components such as a GPS module 1514 and a GPS module 1554, respectively, wherein the GPS module 1514 and a GPS module 1554 may enable delivery of location information of the wireless device 1502 to the Data Processing System 1524 and location information of the base station 1542 to the Data Processing System 1564. One or more other peripheral components (e.g., Peripheral Component(s) 1504 or Peripheral Component(s) 1544) may be configured and connected to the data Processing System 1524 and data Processing System 1564, respectively.

In example embodiments, a wireless device may be configured with parameters and/or configuration arrangements. For example, the configuration of the wireless device with parameters and/or configuration arrangements may be based on one or more control messages that may be used to configure the wireless device to implement processes and/or actions. The wireless device may be configured with the parameters and/or the configuration arrangements regardless of the wireless device being in operation or not in operation. For example, software, firmware, memory, hardware and/or a combination thereof and/or alike may be configured in a wireless device regardless of the wireless device being in operation or not operation. The configured parameters and/or settings may influence the actions and/or processes performed by the wireless device when in operation.

In example embodiments, a wireless device may receive one or more messages comprising configuration parameters. For example, the one or more messages may comprise radio resource control (RRC) messages. A parameter of the configuration parameters may be in at least one of the one or more messages. The one or more messages may comprise information element (IEs). An information element may be a structural element that includes single or multiple fields. The fields in an IE may be individual contents of the IE. The terms configuration parameter, IE and field may be used equally in this disclosure. The IEs may be implemented using a nested structure, wherein an IE may include one or more other IEs and an IE of the one or more other IEs may

include one or more additional IEs. With this structure, a parent IE contains all the offspring IEs as well. For example, a first IE containing a second IE, the second IE containing a third IE, and the third IE containing a fourth IE may imply that the first IE contains the third IE and the fourth IE.

In an example, in PUCCH carrier switching (e.g., for HARQ-ACK), PUCCH carrier switching for different cells operated may be considered for cells that are part of the active UL carrier aggregation (CA) configuration.

In an example, PUCCH carrier switching may be based on dynamic indication in DCI.

In an example, PUCCH carrier switching may be based on one or more rules (e.g., semi-static rules).

In an example, PUCCH carrier switching may be based on an RRC configured PUCCH cell timing pattern of applicable/candidate PUCCH cells.

In an example, PUCCH carrier switching may be based on dynamic indication in a DCI scheduling a PUCCH and/or based on semi-static configuration. In an example, dynamic indication and/or semi-static configuration may be subject to separate wireless device capabilities (e.g., separate capability information elements may indicate whether the wireless device is capable of PUCCH carrier switching based on dynamic indication and whether the wireless device is capable of PUCCH carrier switching based on semi-static configuration). In an example, the semi-static PUCCH carrier switching configuration operation may be based on RRC configured PUCCH cell timing pattern of applicable/candidate PUCCH cells. In an example, PUCCH carrier switching may be across cells with different numerologies.

In an example, a maximum number of candidate PUCCH cells/carriers for PUCCH carrier switching may be configured or may be pre-configured. In an example, the maximum number of candidate PUCCH cells/carriers for PUCCH carrier switching may be based on wireless device capability. In an example, the wireless device may transmit one or more capability messages comprising one or more information elements indicating the wireless device capability in terms of the maximum number of candidate PUCCH cells/carriers for PUCCH carrier switching. The wireless device may receive configuration parameters of the candidate PUCCH cells/carriers for PUCCH carrier switching based on the wireless device capability.

In an example, for PUCCH carrier switching, the PUCCH resource configuration may be per UL BWP (e.g., per candidate cell and UL BWP of that specific candidate cell).

In an example, for PUCCH carrier switching, the PUCCH resource configuration (e.g., pucch-Config/PUCCH-ConfigurationList) may be per UL BWP (e.g., per candidate cell and UL BWP of that specific candidate cell).

In an example, for PUCCH carrier switching, PUCCH carrier switching may be among different TDD cells with PUCCH configured on the normal uplink (NUL) carrier.

In an example, for semi-static and dynamic indication of PUCCH cell switching, the PUCCH repetition factor may be determined based on the PUCCH format or PUCCH resource on the target PUCCH cell for the first repetition.

In an example, PUCCH cell switching may be between two cells.

In an example, PUCCH cell switching may be between more than two cells.

In an example, semi-static PUCCH carrier switching may be applicable to different uplink control information (UCI) types including HARQ feedback SR and CSI.

In an example, for semi-static PUCCH cell switching, PCell/PSCell/PUCCH-SCell may be a reference cell. In an example, the time domain pattern configurations may be

based on the numerology of the reference cell. In an example, the PDSCH to HARQ-ACK offset k_1 may be interpreted based on the numerology and PUCCH configuration of a reference cell to be able to apply the time-domain PUCCH cell switching pattern.

In an example for PUCCH carrier switching based on dynamic indication in DCI scheduling a PUCCH, the PDSCH to HARQ feedback timing/offset (e.g., k_1 value) may be based on the numerology of the dynamically indicated target PUCCH cell.

In an example, a wireless device may not expect overlapping PUCCH slots with dynamic PUCCH cell indication on more than one carrier. The gNB may dynamically indicate a single PUCCH cell for a final PUCCH slot.

In an example, for semi-static PUCCH carrier switching, the time-domain pattern configuration may be based on the following properties: a single time-domain pattern may be configured per PUCCH cell group; the granularity of the time-domain pattern may be one slot of the PCell/PSCell/PUCCH-SCell. In an example, the time-domain pattern may be applied periodically. In an example, the period may be one frame (10 ms). In an example, the period may be RRC configured (e.g., the wireless device may receive one or more configuration parameters indicating the period). In an example, the timing pattern may define, for each slot of the PCell/PSCell/PUCCH-SCell, at least the applicable PUCCH cell.

In an example, for semi-static PUCCH carrier switching, the PDSCH to HARQ feedback timing/offset (e.g., k_1 value) may be interpreted based on the numerology and PUCCH configuration of a reference cell to be able to apply the time-domain PUCCH carrier switching pattern.

In an example, for semi-static PUCCH carrier switching, the PUCCH resource indicator (PRI) of a may be interpreted based on the PUCCH configuration of determined target PUCCH cell.

In an example, the periodicity/length of the time-domain pattern for semi-static PUCCH cell switching may be directly determined by the RRC configuration of the time domain pattern (e.g., based on a pucchCellPattern IE). In an example, pucchCellPattern may have a variable length (e.g., a variable length of $1 \dots \maxNrofSlots$).

In an example, for PUCCH carrier switching based on semi-static operation, for the case the PCell slot to be longer than the target PUCCH cell slot or sub-slot (e.g., multiple target PUCCH cell slots overlapping with a single PCell slot), the first target PUCCH slot overlapping with the PCell slot may be used for UCI transmission.

In an example, for PUCCH carrier switching based on semi-static operation, for the case the PCell slot to be longer than the target PUCCH cell slot or sub-slot (e.g., multiple target PUCCH cell slots overlapping with a single PCell slot), a relative slot-offset within the reference cell slot may be used, the relative slot offset may be configured in the time domain pattern (e.g., time domain pattern may contain 'cell index' and 'slot_offset' for each reference cell slot).

In an example, for PUCCH carrier switching based on semi-static operation, for the case the PCell slot to be shorter than the target PUCCH cell slot, the UE may not expect the same UCI type (e.g., HARQ-ACK, SR or CSI) from more than one PCell PUCCH slot to be overlapping with a single dynamically indicated PUCCH cell slot. In an example, there may be HARQ-ACK only present in either of the overlapping slots, but not in more than one overlapping slot.

In an example, for PUCCH carrier switching based on semi-static operation, for the case the PCell slot to be shorter than the target PUCCH cell slot, the UE may not expect a

semi-static PUCCH cell configuration, where a single target PUCCH slot/sub-slot may be overlapping with more than one PCell slot/sub-slot.

In an example, for PUCCH carrier switching based on dynamic indication in DCI scheduling a PUCCH, the PDSCH to HARQ-ACK offset k_1 is interpreted based on the numerology of the dynamically indicated target PUCCH cell.

In an example, in addition to HARQ feedback of PDSCH dynamically scheduled by a DCI indicating a PUCCH carrier, the dynamic target carrier indication also applies to: HARQ feedback corresponding to the first SPS PDSCH activated by activation DCI based on the indication in the activation DCI; HARQ feedback corresponding to the SPS Release DCI based on the indication in the release DCI; triggered PUCCH for Type 3 codebook (CB) and one-shot triggering for HARQ feedback retransmission based on the indication in the triggering DCI.

In an example, UE may not expect overlapping PUCCH slots with dynamic PUCCH cell indication on more than one cell, e.g., gNB may only dynamically indicate a single PUCCH cell for a final PUCCH slot.

In an example, for PUCCH cell switching based on dynamic indication in the DCI, a dedicated DCI field for the DCI scheduling PDSCH may be used to indicate the target PUCCH cell.

In an example, the dynamic target PUCCH cell indication may apply to HARQ-ACK corresponding to SCell dormancy indication without scheduling PDSCH.

In an example, PUCCH cell switching may be based on dynamic indication in the DCI using DCI format 1_2 for a UE supporting DCI format 1_2. In an example, the presence of the 'PUCCH carrier switching' bitfield in DCI format 1_2 may be RRC configured.

In an example, for semi-static PUCCH carrier switching, the time-domain pattern periodicity may be RRC configured, using candidate values of applicable periodicities from dl-UL-TransmissionPeriodicity and dl-UL-TransmissionPeriodicity-v1530 (e.g., {ms0p5, ms0p625, ms1, ms1p25, ms2, ms2p5, ms3, ms4, ms5, ms10}).

In an example, for PUCCH carrier switching, the PUCCH configuration (e.g., pucch-Config/PUCCH-Configuration-List) may be per UL BWP (e.g., per candidate cell and UL BWP of that specific candidate cell).

In an example, semi-static PUCCH carrier switching may be applicable to UCI types including HARQ feedback, scheduling request (SR) and channel state information (CSI).

In an example, SPS HARQ-ACK deferral and PUCCH cell switching based on the semi-static time domain pattern may be simultaneously configured. For the target slot determination of SPS HARQ-ACK deferral, the UE may first determine a next PUCCH slot on the cell for PUCCH transmission using the semi-static time-domain PUCCH cell pattern and the related rules for semi-static PUCCH cell switching. The UE may determine, based on the SPS HARQ-ACK deferral rules, if this PUCCH slot on the PUCCH cell for transmission is the target PUCCH slot or not.

In an example, one-shot HARQ-ACK re-transmission and semi-static PUCCH cell switching may be simultaneously configured.

In an example, for PUCCH cell switching based on semi-static operation, for the case the PCell slot to be longer than the target PUCCH cell slot or sub-slot (e.g., multiple target PUCCH cell slots overlapping with a single PCell slot,

the first target PUCCH slot overlapping with the PCell slot may be used for UCI transmission.

In an example, the time-domain pattern for semi-static PUCCH cell switching may be separately configurable for the primary and secondary PUCCH cell group.

In an example, the time-domain pattern for semi-static PUCCH cell switching may be based on the reference SCS configuration provided by tdd-UL-DL-ConfigurationCommon and may be common to every configured UL BWP (e.g., of PCell/SPCell/PUCCH SCell).

In an example, for PUCCH cell switching based on semi-static operation, the UE may not expect a semi-static PUCCH cell configuration, where a single target PUCCH slot/sub-slot would be overlapping with more than one PCell slot/sub-slot.

In an example, for semi-static PUCCH cell switching, if the alternative PUCCH cell (e.g., PUCCH SCell) is deactivated or the alternative PUCCH Cell is dormant, the UE may not apply time-domain pattern and the UCI may be transmitted on PCell/SPCell/PUCCH SCell.

In an example, for dynamic PUCCH cell switching, the UE may not expect a PUCCH slot with UCI on PCell/SPCell/PUCCH SCell to overlap with a PUCCH slot with HARQ-ACK on the dynamically indicated alternative PUCCH cell. In an example, the UCI on PCell/SPCell/PUCCH SCell dropped due to collision with semi-static DL symbols, SSB, and symbols indicated by pdccch-ConfigSIB1 in MIB for a CORESET for Type0-PDCCH CSS set may be exempted and may not be multiplexed on the PUCCH on the alternative PUCCH cell.

In an example, an IE ServCellIndex may concern a short identity, used to uniquely identify a serving cell (e.g., the PCell, the PSCell or an SCell) across the cell groups. Value 0 may apply for the PCell, while the SCellIndex that may have previously been assigned may apply for SCells. A value of ServCellIndex may be between 0 and maxNrofServingCells-1.

In an example, an IE SCellIndex may concern a short identity, used to identify an SCell. The value range may be shared across the Cell Groups. A value of SCellIndex may be between 1 and 31.

In an example, an IE ServingCellConfig may be used to configure (e.g., add and/or modify) the UE with a serving cell, which may be the SpCell or an SCell of a master cell group (MCG) or secondary cell group (SCG). The parameters herein may be mostly UE specific but may partly also be cell specific (e.g., in additionally configured bandwidth parts). Reconfiguration between a PUCCH and PUCCHless SCell may be supported using an SCell release and add. A field/parameter csi-RS-ValidationWithDCI may indicate how the UE may perform periodic and semi-persistent CSI-RS reception in a slot. The presence of this field may indicate that the UE uses DCI detection to validate whether to receive CSI-RS.

In an example, an IE CSI-MeasConfig may be used to configure CSI-RS (reference signals) belonging to the serving cell in which CSI-MeasConfig is included, channel state information reports to be transmitted on PUCCH on the serving cell in which CSI-MeasConfig is included and channel state information reports on PUSCH triggered by DCI received on the serving cell in which CSI-MeasConfig is included. A field aperiodicTriggerStateList may contain trigger states for dynamically selecting one or more aperiodic and semi-persistent reporting configurations and/or triggering one or more aperiodic CSI-RS resource sets for channel and/or interference measurement. A field/parameter csi-ReportConfigToAddModList may indicate configured

CSI report settings. A field/parameter `csi-ResourceConfig-ToAddModList` may indicate configured CSI resource settings. A field/parameter `csi-SSB-ResourceSetToAddModList` may indicate pool of CSI-SSB-ResourceSet which may be referred to from CSI-ResourceConfig. A field/parameter `nzp-CSI-RS-ResourceSetToAddModList` may indicate pool of NZP-CSI-RS-ResourceSet which may be referred to from CSI-ResourceConfig or from MAC CEs. A field/parameter `nzp-CSI-RS-ResourceToAddModList` may indicate pool of NZP-CSI-RS-Resource which may be referred to from NZP-CSI-RS-ResourceSet.

In an example, `csi-ReportConfigToAddModList` may indicate configured CSI report settings.

In an example, an IE `CSI-ReportConfig` may be used to configure a periodic or semi-persistent report sent on PUCCH in the cell in which the `CSI-ReportConfig` is included, or to configure a semi-persistent or aperiodic report sent on PUSCH triggered by DCI received on the cell in which the `CSI-ReportConfig` is included (in this case, the cell on which the report is sent may be determined by the received DCI). A field/parameter `carrier` may indicate in which serving cell the `CSI-ResourceConfig` indicated below are to be found. If the field is absent, the resources may be on the same serving cell as this report configuration. A field/parameter `cqi-FormatIndicator` may indicate whether the UE may report a single (wideband) or multiple (subband) CQI. A field/parameter `cqi-Table` may indicate which CQI table to use for CQI calculation. A field/parameter `csi-IM-ResourcesForInterference` may indicate CSI IM resources for interference measurement. `csi-ResourceConfigId` of a `CSI-ResourceConfig` included in the configuration of the serving cell indicated with the field “carrier” above. The `CSI-ResourceConfig` indicated here may contain CSI-IM resources. The `bwp-Id` in that `CSI-ResourceConfig` may be the same value as the `bwp-Id` in the `CSI-ResourceConfig` indicated by `resourcesForChannelMeasurement`. A field/parameter `csi-ReportingBand` may indicate a contiguous or non-contiguous subset of subbands in the bandwidth part which CSI may be reported for. Each bit in the bit-string may represent one subband. The right-most bit in the bit string may represent the lowest subband in the BWP. The choice may determine the number of subbands (subbands3 for 3 subbands, subbands4 for 4 subbands, and so on). A field/parameter `nrofReportedRS` may indicate the number (N) of measured RS resources to be reported per report setting in a non-group-based report. $N \leq N_{max}$, where N_{max} may be 2 or 4 depending on UE capability. A field/parameter `nzp-CSI-RS-ResourcesForInterference` may indicate NZP CSI RS resources for interference measurement. `csi-ResourceConfigId` of a `CSI-ResourceConfig` included in the configuration of the serving cell indicated with the field “carrier” above. The `CSI-ResourceConfig` indicated here may contain NZP-CSI-RS resources. The `bwp-Id` in that `CSI-ResourceConfig` may be the same value as the `bwp-Id` in the `CSI-ResourceConfig` indicated by `resourcesForChannelMeasurement`. A field/parameter `pmi-FormatIndicator` may indicate whether the UE may report a single (wideband) or multiple (subband) PMI. A field/parameter `pucch-CSI-ResourceList` may indicate which PUCCH resource to use for reporting on PUCCH. A field/parameter `reportConfigType` may indicate time domain behavior of reporting configuration. A field/parameter `reportFreqConfiguration` may indicate reporting configuration in the frequency domain. A field/parameter `reportQuantity` may indicate the CSI related quantities to report. A field/parameter `reportSlotConfig` may indicate periodicity and slot offset. A field/parameter `subbandSize` may indicate one out of two possible

BWP-dependent values for the subband size. A field/parameter `timeRestrictionForChannelMeasurements` may indicate time domain measurement restriction for the channel (signal) measurements. A field/parameter `timeRestrictionForInterferenceMeasurements` may indicate time domain measurement restriction for interference measurements. A field/parameter `pucch-Resource` may indicate PUCCH resource for the associated uplink BWP.

In an example, an IE `CSI-ReportConfigId` may be used to identify one `CSI-ReportConfig`.

In an example, an IE `CSI-ResourceConfig` may define a group of one or more NZP-CSI-RS-ResourceSet, CSI-IM-ResourceSet and/or CSI-SSB-ResourceSet. A field/parameter `bwp-Id` may indicate the DL BWP which the CSI-RS associated with this `CSI-ResourceConfig` are located in. A field/parameter `csi-IM-ResourceSetList` may indicate list of references to CSI-IM resources used for CSI measurement and reporting in a CSI-RS resource set. It may contain up to `maxNrofCSI-IM-ResourceSetsPerConfig` resource sets if `resourceType` is ‘aperiodic’ and 1 otherwise. A field/parameter `csi-ResourceConfigId` may be used in `CSI-ReportConfig` to refer to an instance of `CSI-ResourceConfig`. A field/parameter `csi-SSB-ResourceSetList` may indicate list of references to SSB resources used for CSI measurement and reporting in a CSI-RS resource set. A field/parameter `nzp-CSI-RS-ResourceSetList` may indicate list of references to NZP CSI-RS resources used for beam measurement and reporting in a CSI-RS resource set. It may contain up to `maxNrofNZP-CSI-RS-ResourceSetsPerConfig` resource sets if `resourceType` is ‘aperiodic’ and 1 otherwise. A field/parameter `resourceType` may indicate time domain behavior of resource configuration. It may not apply to resources provided in the `csi-SSB-ResourceSetList`.

In an example, an IE `CSI-ResourceConfigId` may be used to identify a `CSI-ResourceConfig`. It may take a value between 0 and `maxNrofCSI-ResourceConfigurations-1`.

In an example, an IE `CSI-ResourcePeriodicityAndOffset` may be used to configure a periodicity and a corresponding offset for periodic and semi-persistent CSI resources, and for periodic and semi-persistent reporting on PUCCH. The periodicity and the offset may be given in number of slots. The periodicity value `slots4` may correspond to 4 slots, value `slots5` may correspond to 5 slots, and so on.

In an example, an IE `CSI-RS-ResourceMapping` may be used to configure the resource element mapping of a CSI-RS resource in time- and frequency domain. A field/parameter `cdm-Type` may indicate code division multiplexing (CDM) type. A field/parameter `density` may indicate density of CSI-RS resource measured in RE/port/PRB. A field/parameter `firstOFDMSymbolInTimeDomain2` may indicate time domain allocation within a physical resource block. A field/parameter `firstOFDMSymbolInTimeDomain` may indicate time domain allocation within a physical resource block. The field may indicate the first OFDM symbol in the PRB used for CSI-RS. A field/parameter `freqBand` may indicate wideband or partial band CSI-RS. A field/parameter `frequencyDomainAllocation` may indicate frequency domain allocation within a physical resource block.

In an example, an IE `CSI-FrequencyOccupation` may be used to configure the frequency domain occupation of a channel state information measurement resource (e.g., NZP-CSI-RS-Resource, CSI-IM-Resource). A field/parameter `nrofRBs` may indicate number of PRBs across which this CSI resource spans. If the configured value is larger than the width of the corresponding BWP, the UE may assume that the actual CSI-RS bandwidth is equal to the width of the BWP. A field/parameter `startingRB` may indicate PRB where

this CSI resource starts in relation to common resource block #0 (CRB #0) on the common resource block grid.

In an example, an IE PUCCH-Config may be used to configure UE specific PUCCH parameters (per BWP).

In an example, an IE PUCCH-ConfigCommon is used to configure the cell specific PUCCH parameters.

In an example, a CSI Resource Setting CSI-ResourceConfig may contain a configuration of a list of $S \geq 1$ CSI Resource Sets (given by higher layer parameter `csi-RS-ResourceSetList`), where the list may comprise references to either or both of NZP CSI-RS resource set(s) and SS/PBCH block set(s) or the list may comprise of references to CSI-IM resource set(s). A CSI Resource Setting may be located in the DL BWP identified by the higher layer parameter `BWP-id`, and CSI Resource Settings linked to a CSI Report Setting may have the same DL BWP.

In an example, the time domain behavior of the CSI-RS resources within a CSI Resource Setting may be indicated by the higher layer parameter `resourceType` and may be set to aperiodic, periodic, or semi-persistent. For periodic and semi-persistent CSI Resource Settings, the number of CSI-RS Resource Sets configured may be limited to $S=1$. For periodic and semi-persistent CSI Resource Settings, the configured periodicity and slot offset may be given in the numerology of its associated DL BWP, as given by `BWP-id`. When a UE is configured with multiple CSI-ResourceConfigs comprising the same NZP CSI-RS resource ID, the same time domain behavior may be configured for the CSI-ResourceConfigs. When a UE is configured with multiple CSI-ResourceConfigs comprising the same CSI-IM resource ID, the same time-domain behavior may be configured for the CSI-ResourceConfigs. CSI Resource Settings linked to a CSI Report Setting may have the same time domain behavior.

In an example, the following may be configured via higher layer signaling for one or more CSI Resource Settings for channel and interference measurement: CSI-IM resource for interference measurement; NZP CSI-RS resource for interference measurement; NZP CSI-RS resource for channel measurement.

In an example, the UE may assume that the NZP CSI-RS resource(s) for channel measurement and the CSI-IM resource(s) for interference measurement configured for one CSI reporting may be resource-wise QCLed with respect to 'typeD'. When NZP CSI-RS resource(s) is used for interference measurement, the UE may assume that the NZP CSI-RS resource for channel measurement and the CSI-IM resource or NZP CSI-RS resource(s) for interference measurement configured for one CSI reporting are QCLed with respect to 'typeD'.

In an example, the Reporting configuration for CSI may be aperiodic (using PUSCH), periodic (using PUCCH) or semi-persistent (using PUCCH, and DCI activated PUSCH). The CSI-RS Resources may be periodic, semi-persistent, or aperiodic. Example supported combinations of CSI Reporting configurations and CSI-RS Resource configurations and how the CSI Reporting is triggered for each CSI-RS Resource configuration are shown below. Periodic CSI-RS may be configured by higher layers. Semi-persistent CSI-RS may be activated and deactivated. Aperiodic CSI-RS may be configured and triggered/activated.

For Periodic CSI-RS configuration: periodic CSI reporting may not be based on dynamic triggering/activation; for semi-persistent CSI reporting on PUCCH, the UE may receive an activation command and for semi-persistent CSI reporting on PUSCH, the UE may receive triggering on

DCI; aperiodic CSI reporting may be triggered by DCI and subselection indication may be possible.

For semi-persistent CSI-RS configuration: periodic CSI reporting may not be supported; for semi-persistent CSI reporting on PUCCH, the UE may receive an activation command, for semi-persistent reporting on PUSCH, the UE may receive triggering on DCI; aperiodic CSI reporting may be triggered by DCI and subselection indication may be possible.

For aperiodic CSI-RS configuration: periodic CSI reporting may not be supported; semi-persistent CSI reporting may not be supported; aperiodic CSI reporting may be triggered by DCI and subselection indication may be possible.

In an example, for a periodic or semi-persistent CSI report on PUCCH, the periodicity T_{CSI} (measured in slots) and the slot offset T_{offset} may be configured by the higher layer parameter `reportSlotConfig`. The UE may transmit the CSI report in frames with SFN f and slot number within the frame $n_{s,f}^{\mu}$ satisfying $(N_{symb}^{frame,\mu} n_{s,f}^{\mu} - T_{offset}) \bmod T_{CSI} = 0$, where μ is the SCS configuration of the UL BWP the CSI report is transmitted on.

In an example, for semi-persistent or periodic CSI, a CSI-ReportConfig may be linked to periodic or semi-persistent Resource Setting(s). When one Resource Setting (given by higher layer parameter `resourcesForChannelMeasurement`) is configured, the Resource Setting may be for channel measurement for L1-RSRP or for channel and interference measurement for L1-SINR computation. When two Resource Settings are configured, the first Resource Setting (given by higher layer parameter `resourcesForChannelMeasurement`) may be for channel measurement and the second Resource Setting (given by higher layer parameter `csi-IM-ResourcesForInterference`) may be used for interference measurement performed on CSI-IM. For L1-SINR computation, the second Resource Setting (given by higher layer parameter `csi-IM-ResourcesForInterference` or higher layer parameter `nzp-CSI-RS-ResourceForInterference`) may be used for interference measurement performed on CSI-IM or on NZP CSI-RS.

In an example, a UE may be configured with a CSI-ReportConfig with the higher layer parameter `reportQuantity` set to either 'none', 'cri-RI-PMI-CQI', 'cri-RI-i1', 'cri-RI-i1-CQI', 'cri-RI-CQI', 'cri-RSRP', 'cri-SINR', 'ssb-Index-RSRP', 'ssb-Index-SINR' or 'cri-RI-LI-PMI-CQI'.

In an example, for semi-persistent reporting on PUCCH, the PUCCH resource used for transmitting the CSI report may be configured by `reportConfigType`. Semi-persistent reporting on PUCCH may be activated by an activation command, which selects one of the semi-persistent Reporting Settings for use by the UE on the PUCCH. When the UE would transmit a PUCCH with HARQ-ACK information in slot n corresponding to the PDSCH carrying the activation command, the indicated semi-persistent Reporting Setting may be applied starting from the first slot that is after slot $n + 3N_{symb}^{subframe,\mu}$ where μ is the SCS configuration for the PUCCH.

In an example, a UE may be semi-statically configured by higher layers to perform periodic CSI Reporting on the PUCCH. A UE may be configured by higher layers for multiple periodic CSI Reports corresponding to multiple higher layer configured CSI Reporting Settings, where the associated CSI Resource Settings are higher layer configured. Periodic CSI reporting on PUCCH formats 2, 3, 4 may support Type I CSI with wideband granularity.

In an example, a UE may perform semi-persistent CSI reporting on the PUCCH applied starting from the first slot

that is after slot $n+3N_{\text{subframe},\mu}$ when the UE would transmit a g slot PUCCH with HARQ-ACK information in slot n corresponding to the PDSCH carrying the activation command where Q is the SCS configuration for the PUCCH. The activation command may contain one or more Reporting Settings where the associated CSI Resource Settings may be configured. Semi-persistent CSI reporting on the PUCCH may support Type I CSI. Semi-persistent CSI reporting on the PUCCH format 2 may support Type I CSI with wideband frequency granularity. Semi-persistent CSI reporting on PUCCH formats 3 or 4 may support Type I CSI with wideband and sub-band frequency granularities and Type II CSI Part 1.

In an example, when the UE is configured with CSI Reporting on PUCCH formats 2, 3 or 4, each PUCCH resource may be configured for each candidate UL BWP.

In an example, if the UE is in an active semi-persistent CSI reporting configuration on PUCCH and has not received a deactivation command, the CSI reporting may take place when the BWP in which the reporting is configured to take place is the active BWP, otherwise the CSI reporting may be suspended.

In an example, the network may activate and deactivate the configured Semi-persistent CSI reporting on PUCCH of a Serving Cell by sending the SP CSI reporting on PUCCH Activation/Deactivation MAC CE. The configured Semi-persistent CSI reporting on PUCCH may be initially deactivated upon configuration and after a handover.

In an example, if the MAC entity receives an SP CSI reporting on PUCCH Activation/Deactivation MAC CE on a Serving Cell, the MAC entity may indicate to lower layers the information regarding the SP CSI reporting on PUCCH Activation/Deactivation MAC CE.

In an example, the SP CSI reporting on PUCCH Activation/Deactivation MAC CE may be identified by a MAC subheader with a corresponding LCID. An example SP CSI reporting on PUCCH Activation/Deactivation MAC CE is shown in FIG. 16. It may have a fixed size of 16 bits. A Serving Cell ID field may indicate the identity of the Serving Cell for which the MAC CE applies. The length of the field may be 5 bits. A BWP ID field may indicate a UL BWP for which the MAC CE applies as the codepoint of the DCI bandwidth part indicator field. The length of the BWP ID field may be 2 bits. An S_i field may indicate the activation/deactivation status of the Semi-Persistent CSI report configuration within $\text{csi-ReportConfigToAddModList}$. So may refer to the report configuration which includes PUCCH resources for SP CSI reporting in the indicated BWP and has the lowest CSI-ReportConfigId within the list with type set to $\text{semiPersistentOnPUCCH}$, S_1 to the report configuration which includes PUCCH resources for SP CSI reporting in the indicated BWP and has the second lowest CSI-ReportConfigId and so on. If the number of report configurations within the list with type set to $\text{semiPersistentOnPUCCH}$ in the indicated BWP is less than $i+1$, MAC entity may ignore the S_i field. The S_i field may be set to 1 to indicate that the corresponding Semi-Persistent CSI report configuration may be activated. The S_i field may be set to 0 to indicate that the corresponding Semi-Persistent CSI report configuration i may be deactivated. The R field may be Reserved bit, set to 0.

A wireless device may be configured with PUCCH carrier switching for transmission of uplink control information including channel state information (CSI) report. The wireless device may receive configuration parameter(s) indicating a semi-static time/timing pattern. The time/timing pattern may indicate timings that a candidate PUCCH cell/

carrier, in a plurality of candidate PUCCH cells/carriers, is configured as a PUCCH cell. Existing CSI measurement and/or reporting processes may lead to inefficient wireless device and wireless network performance, e.g., in terms of throughput and/or latency, when the wireless device is configured with semi-static PUCCH cell/carrier switching. There is a need to enhance the existing CSI measurement and/or reporting processes when the wireless device is configured with semi-static PUCCH cell/carrier switching. Example embodiments enhance the existing CSI measurement and/or reporting processes when the wireless device is configured with semi-static PUCCH cell/carrier switching.

In an example embodiment, a wireless device may receive one or more messages (e.g., one or more RRC messages) comprising configuration parameters. The configuration parameters may comprise first configuration parameters of a plurality of cells. The plurality of cells may be associated with (e.g., provided by) one or more base stations. In an example, a first plurality of cells (e.g., grouped into a first cell group, e.g., a master cell group (MCG)) of the plurality of cells may be provided by a first base station (e.g., a master base station) and a second plurality of cells (e.g., grouped into a second cell group, e.g., a secondary cell group (SCG)) of the plurality of cells may be provided by a second base station (e.g., a secondary base station).

In an example, the plurality of cells may be grouped into one or more PUCCH groups. Each PUCCH group, in the one or more PUCCH groups, may comprise one or more cells configured with PUCCH resources. Uplink control information (e.g., HARQ feedback, SR, CSI reports, etc.) associated with a PUCCH group (e.g., associated with the cells of the PUCCH group, e.g., HARQ feedbacks associated with downlink TBs received via cells of the PUCCH group, CSI reports for CSI channel measurement reports for the cells of the PUCCH group, etc.) may be transmitted via PUCCH resources configured for the one or more cells.

In an example, the one or more PUCCH groups may comprise a first PUCCH group and a second PUCCH group. The first PUCCH group may comprise one or more first cells configured with PUCCH and the second PUCCH group may comprise one or more second cells configured PUCCH. The one or more first cells, in the first PUCCH group, configured with PUCCH may comprise a SpCell (e.g., primary cell (PCell) or a primary secondary cell (PSCell)). The first PUCCH group may be referred to as a primary PUCCH group. The second PUCCH group may be referred to as a secondary PUCCH group. A first secondary cell in one or more second cells of the second PUCCH group (e.g., secondary PUCCH group), configured with PUCCH, may be referred to a PUCCH SCell. In an example, the SpCell (e.g., PCell or PSCell) of the primary PUCCH group may be an anchor cell for transmission of uplink control associated with the primary PUCCH group. In an example, the PUCCH SCell of the secondary PUCCH group may be an anchor cell for transmission of uplink control associated with the secondary PUCCH group.

In an example as shown in FIG. 17, a PUCCH group may comprise a plurality of cells/carriers configured with PUCCH. The plurality of cells/carriers configured with PUCCH may be referred to as candidate PUCCH cells/carriers. The wireless device may determine timings that a candidate PUCCH cell/carrier, in the candidate PUCCH cells/carriers, is configured as PUCCH cell based on a time/timing pattern. The wireless device may receive one or more configuration parameters (e.g., RRC parameters) indicating the time/timing pattern. The time/timing pattern may indicate/define timings that a candidate PUCCH cell, in the

candidate PUCCH cells, is configured as PUCCH cell. In an example, a wireless device capability parameter may indicate a transitioning time (e.g., a minimum required transitioning time) from a first candidate PUCCH cell to a second candidate PUCCH cell. In an example, the timing/time pattern configured for the wireless device may be based on the wireless device capability. The wireless device may transmit uplink control information (e.g., HARQ feedback, SR, CSI report, etc.) based on radio resources of a PUCCH cell. The timing pattern may be periodically repeated based on a periodicity. In an example, the periodicity may be in a first number of slots. In an example, the periodicity may be based on a first number of slots associated with a numerology of a reference cell/carrier. In an example, the periodicity may be pre-configured and/or fixed (e.g., one frame/10 ms) or may be configurable (e.g., based on receiving an RRC configuration parameter indicating the periodicity).

In an example, a cell in the PUCCH group may be referred to the reference cell. For example, the reference cell may be a SpCell (e.g., PCell or SPCell) for a primary PUCCH group. In an example, the reference cell may be a PUCCH SCell for a secondary PUCCH group. In an example as shown in FIG. 18, the time/timing pattern may indicate that a corresponding candidate PUCCH cell, in the candidate PUCCH cells of a PUCCH group, is a configured PUCCH cell for each slot of the reference cell. The time/timing pattern of a PUCCH group may be based on a numerology (e.g., a slot duration) of the reference cell. Each slot of the reference cell/carrier may be associated with a candidate PUCCH cell (e.g., a candidate PUCCH cell in a PUCCH group) and the wireless may determine the number of slots that a candidate PUCCH cell is configured as a PUCCH cell based on the slot duration of the reference cell/carrier and the slot duration of the candidate PUCCH cell/carrier. In the example shown in FIG. 18, a slot duration of a reference cell/carrier may be two times the slot duration of cell/carrier 1 (e.g., based on a numerology of the reference cell/carrier and a numerology of the cell/carrier 1), may be the same as the slot duration of the cell/carrier 2 (e.g., based on the numerology of the reference cell/carrier and a numerology of the cell/carrier 2), and may be four times the slot duration of cell/carrier 3 (e.g., based on the numerology of the reference cell/carrier and a numerology of the cell/carrier 3). In the example shown in FIG. 18, cell/carrier 1 is the configured PUCCH cell for two slots of cell/carrier 1, cell/carrier 2 is the configured PUCCH cell for one slot of cell/carrier 2, cell/carrier 3 is the configured PUCCH cell for four slots of cell/carrier 3, and so on.

In an example as shown in FIG. 19, configuration parameters of a serving cell (e.g., received via a ServingCellConfig IE), in a plurality of cells configured for a wireless device, may comprise a csi-MeasConfig IE indicating setting up a CSI-MeasConfig configuration. The CSI-MeasConfig IE may be used to configure CSI-RS (reference signals) and CSI reports to be transmitted on PUCCH on the serving cell. The CSI-MeasConfig may comprise a csi-ReportConfig-ToAddModList IE indicating a list of one or more CSI report configurations. The one or more CSI report configurations in the list of the one or more CSI report configurations may comprise a first CSI report configuration and a second CSI report configuration.

The first CSI report configuration may comprise a first identifier of the first CSI report configuration, e.g., may comprise a first reportConfigId indicating the first identifier of the first CSI report configuration. The first CSI report configuration may comprise a first carrier IE indicating a first cell identifier of a serving cell for which a CSI-

ResourceConfig, associated with the first CSI report configuration and comprising configuration parameters of CSI resources (e.g., reference signals (e.g., CSI-RS reference signals)) for CSI channel measurements associated with the first CSI report configuration, is configured and may be found. The first CSI report configuration may comprise a resourcesForChannelMeasurement IE indicating a CSI-ResourceConfigId of a CSI resource configuration comprising the configuration parameters of the CSI resources (e.g., reference signals (e.g., CSI-RS reference signals)) for CSI channel measurements associated with the first CSI report configuration. The first CSI report configuration may comprise a first reportConfigType parameter indicating one of a periodic, semiPersistentOnPUCCH, semiPersistentOnPUSCH, and aperiodic CSI reporting types. In an example, the first reportConfigType parameter may indicate the semi-PersistentOnPUCCH CSI reporting type. The semi-PersistentOnPUCCH IE may comprise a reportSlotConfig parameter indicating a periodicity and offset for semi-persistent CSI reports on PUCCH and a pucch-CSI-ResourceList parameter indicating which PUCCH resource to use for reporting on PUCCH. In an example, the first reportConfigType parameter may indicate the periodic reporting type. The periodic IE may comprise a reportSlotConfig parameter indicating a periodicity and offset for periodic CSI reports on PUCCH and a pucch-CSI-ResourceList parameter indicating which PUCCH resource to use for reporting on PUCCH.

The second CSI report configuration may comprise a second identifier of the second CSI report configuration, e.g., may comprise a second reportConfigId indicating the second identifier of the second CSI report configuration. The second CSI report configuration may comprise a second carrier IE indicating a second cell identifier of a serving cell for which a CSI-ResourceConfig, associated with the second CSI report configuration and comprising configuration parameters of CSI resources (e.g., reference signals (e.g., CSI-RS reference signals)) for CSI channel measurements associated with the second CSI report configuration, is configured and may be found. The second CSI report configuration may comprise a resourcesForChannelMeasurement IE indicating a CSI-ResourceConfigId of a CSI resource configuration comprising the configuration parameters of the CSI resources (e.g., reference signals (e.g., CSI-RS reference signals)) for CSI channel measurements associated with the second CSI report configuration. The second CSI report configuration may comprise a second reportConfigType parameter indicating one of a periodic, semiPersistentOnPUCCH, semiPersistentOnPUSCH, and aperiodic CSI reporting types. In an example, the second reportConfigType parameter may indicate the semi-PersistentOnPUCCH CSI reporting type. The semi-PersistentOnPUCCH IE may comprise a reportSlotConfig parameter indicating a periodicity and offset for semi-persistent CSI reports on PUCCH and a pucch-CSI-ResourceList parameter indicating which PUCCH resource to use for reporting on PUCCH. In an example, the first reportConfigType parameter may indicate the periodic reporting type. The periodic IE may comprise a reportSlotConfig parameter indicating a periodicity and offset for periodic CSI reports on PUCCH and a pucch-CSI-ResourceList parameter indicating which PUCCH resource to use for reporting on PUCCH.

In an example as shown in FIG. 20, the configuration parameters of a serving cell may comprise a CSI-MeasConfig IE comprising a csi-ReportConfigToAddModList IE indicating a list of one or more CSI report configurations. The one or more CSI report configurations in the list of the one or more CSI report configurations may comprise a first

CSI report configuration (e.g., CSI-ReportConfig1). The first CSI report configuration may comprise a reportConfig-Type parameter indicating a semi-persistent CSI reporting type. The wireless device may receive a SP CSI reporting on PUCCH Activation/Deactivation MAC CE indicating activation of the first CSI report configuration. An example format of the SP CSI reporting on PUCCH Activation/Deactivation MAC CE is shown in FIG. 16. In response to receiving SP CSI reporting on PUCCH Activation/Deactivation MAC CE indicating activation of the first CSI report configuration and/or in response to transmitting an acknowledgement indicating successful reception of a transport block (received via PDSCH) comprising the MAC CE, the wireless device may transmit SP CSI reports based on the first CSI report configuration.

In an example, the configuration parameters of a serving cell may comprise a CSI-MeasConfig IE comprising a csi-ReportConfigToAddModList IE indicating a list of one or more CSI report configurations. The one or more CSI report configurations in the list of the one or more CSI report configurations may comprise a first CSI report configuration (e.g., CSI-ReportConfig1). The first CSI report configuration may comprise a reportConfigType parameter indicating a periodic CSI reporting type. In response to receiving the configuration parameters of the first CSI report configuration, the wireless device may transmit periodic CSI reports based on the first CSI report configuration.

In example embodiments as shown in FIG. 21, FIG. 23 and FIG. 24, a wireless device may receive one or more messages (e.g., one or more RRC messages) comprising configuration parameters of a plurality of cells comprising a first cell and a second cell. The configuration parameters (e.g., first configuration parameters of the first cell) may comprise first CSI report configuration parameters of a first CSI report configuration for transmission of CSI report via the first cell (e.g., via PUCCH resources of the first cell). The first CSI report configuration may be one of one or more first CSI report configurations (in a first list of CSI report configurations) that are in a CSI-MeasConfig IE included in configuration parameters of the first cell (e.g., in a serving-CellConfig of the first cell). The configuration parameters (e.g., second configuration parameters of the second cell) may comprise second CSI report configuration parameters of one or more second CSI report configurations for transmission of CSI report via the second cell (e.g., via PUCCH resources of the second cell). The one or more second CSI report configurations may be among a second list of one or more second CSI report configurations that are in a CSI-MeasConfig IE included in configuration parameters of the second cell (e.g., in a servingCellConfig of the second cell). The first cell and the second cell may be in a cell group (e.g., a primary PUCCH group or a second PUCCH group). The first cell and the second cell may be among a plurality of candidate PUCCH cells configured with PUCCH resources wherein a candidate PUCCH cell, in the candidate PUCCH cells, may be a PUCCH cell based on a time/timing pattern (e.g., a time/timing pattern configured using a configuration (e.g., RRC) parameter). An example of the time/timing pattern indicating timings that the first cell is the PUCCH cell and timings that the second cell is the PUCCH cell is shown in FIG. 21. The configuration parameters may comprise at least one third configuration parameter of the time/timing pattern. The time/timing pattern may indicate first timings/durations that the first cell is a PUCCH cell and second timings/durations that the second cell is a PUCCH cell. The reception of the at least one third configuration parameter may indicate that PUCCH cell/carrier switching is

configured for the wireless device. The at least one third configuration parameter may indicate that PUCCH cell/carrier switching between a plurality of cells/carriers, comprising the first cell/carrier and the second cell/carrier, is enabled/configured.

In examples as shown in FIG. 23 and FIG. 24, the first CSI report configuration parameters of the first CSI report configuration may comprise a first configuration parameter that may have a first value. Based on receiving the at least one third configuration parameter and/or based on the PUCCH cell/carrier switching being enabled/configured, e.g., based on PUCCH cell/carrier switching between a plurality of cells/carriers comprising the first cell/carrier and the second cell/carrier being configured/enabled, the one or more second CSI report configurations may comprise at least one second CSI report configuration comprising a second configuration parameter, corresponding to the first configuration parameter, with the same value as the first value of the first configuration parameter in the first CSI report configuration. The first configuration parameter and the second configuration parameter may be for configuring the same parameter for CSI reporting. For example, the first configuration parameter of the first CSI report configuration may be a first carrier IE indicating a serving cell identifier of a serving cell for CSI channel measurements and the second configuration parameter of the second CSI report configuration may be a second carrier IE indicating the same serving cell identifier. For example, the first configuration parameters of the first CSI report configuration may be a first resourcesForChannelMeasurement IE indicating a first CSI-ResourceConfigId of a CSI resource configuration (e.g., CSI resource configuration included in the configuration of the serving cell indicated by the first carrier IE) indicating resources for CSI channel measurement. The second configuration parameter of the second CSI report configuration may be a second resourcesForChannelMeasurement IE indicating the same first CSI-ResourceConfigId of the same CSI resource configuration (e.g., same CSI resource configuration included in the configuration of the same serving cell). For example, based on receiving the at least one third configuration parameter and/or based on the PUCCH cell/carrier switching being enabled/configured, e.g., based on PUCCH cell/carrier switching between a plurality of cells/carriers comprising the first cell/carrier and the second cell/carrier being configured/enabled, the first CSI report configuration parameters may comprise a first carrier IE, indicating a first serving cell identifier, and a first resourceFoeChannelMeasurement IE indicating a first CSI-ResourceConfigId and the one or more second CSI report configurations may comprise at least one second CSI report configuration comprising a second carrier IE indicating the same first serving cell identifier and a second resourceFoeChannelMeasurement IE indicate the same first CSI-ResourceConfigId.

In an example, the first CSI report configuration and the second CSI report configuration may have the same report type configuration (e.g., both may be periodic CSI reports or both may be semi-persistent CSI reports).

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports, associated with the first CSI report configuration, are periodic CSI reports. The second CSI report configuration parameters may comprise a second reporting config type parameter indicating that the second CSI reports, associated with the second CSI report configuration, are periodic CSI reports.

In an example, the first CSI report configuration parameters may comprise a first report config type parameter

indicating that the CSI reports, associated with the first CSI report configuration, are semi-persistent CSI reports. The second CSI report configuration parameters may a second reporting config type parameter indicating that the second CSI reports, associated with the second CSI report configuration, are semi-persistent CSI reports. In an example as shown in FIG. 24, the wireless device may receive a first activation command (e.g., first SP CSI reporting on PUCCH activation/deactivation MAC CE) indicating activation of the first CSI report configuration and may receive a second activation command (e.g., a second SP CSI reporting on PUCCH activation/deactivation MAC CE) indicating activation of the second CSI report configuration. The wireless device may activate the first CSI report configuration (e.g., may start transmission of CSI reports) in response to receiving the first activation command and may activate the second CSI report configuration (e.g., may start transmission of CSI reports) in response to receiving the second activation command.

The wireless device may transmit CSI reports based on the first CSI report configuration parameters and the second CSI report configuration parameters. For example, the wireless device may transmit first CSI reports based on the first CSI report configuration parameters and via the first cell (via PUCCH resources of the first cell) and second CSI reports based on second CSI report configuration parameters and via the second cell (e.g., via PUCCH resources of the second cell).

In an example embodiment, PUCCH cell switching between a plurality of cells/carriers (e.g., comprising a first cell/carrier and a second cell/carrier) may be configured/enabled for a wireless device. A first CSI reporting configuration may be configured for CSI reporting (e.g., periodic CSI reporting) via PUCCH resources on the first cell/carrier (e.g., via PUCCH resources of an active UL BWP of the first cell/carrier). The first CSI reporting configuration may comprise a first configuration parameter with a first value. One or more second CSI reporting configurations may be configured for CSI reporting (e.g., periodic CSI reporting) via PUCCH resources on the second cell/carrier (e.g., via PUCCH resources of an active UL BWP of the second cell/carrier). The one or more second CSI reporting configurations may comprise at least one second CSI reporting configuration comprising a second configuration parameter, corresponding to the first configuration parameter, with the first value. Based on the PUCCH cell switching between the first cell/carrier and the second cell/carrier being configured/enabled, the one or more second CSI reporting configurations may comprise at least one second CSI reporting configuration comprising a second configuration parameter, corresponding to the first configuration parameter, with the first value.

In an example embodiment, PUCCH cell/carrier switching between a plurality of cells/carriers (e.g., comprising a first cell/carrier and a second cell/carrier) may be configured/enabled for a wireless device. A first CSI reporting configuration may be configured and activated (e.g., activated based on a MAC CE) for SP CSI reporting via PUCCH resources on the first cell/carrier (e.g., via PUCCH resources of an active UL BWP of the first cell/carrier). The first CSI reporting configuration may comprise a first configuration parameter with a first value. One or more second CSI reporting configurations may be configured and activated (e.g., activated based on one or more MAC CEs) for SP CSI reporting via PUCCH resources on the second cell/carrier (e.g., via PUCCH resources of an active UL BWP of the second cell/carrier). The one or more second CSI reporting

configurations may comprise at least one second CSI reporting configuration comprising a second configuration parameter, corresponding to the first configuration parameter, with the first value. Based on the PUCCH cell switching between the first cell/carrier and the second cell/carrier being configured/enabled, the one or more second CSI reporting configurations may comprise at least one second CSI reporting configuration comprising a second configuration parameter, corresponding to the first configuration parameter, with the first value.

In example embodiments as shown in FIG. 22, FIG. 25 and FIG. 26, a wireless device may receive one or more messages (e.g., one or more RRC messages) comprising configuration parameters of a plurality of cells comprising a first cell and a second cell. The configuration parameters (e.g., first configuration parameters of the first cell) may comprise first CSI report configuration parameters of a first CSI report configuration for transmission of CSI report via the first cell (e.g., via PUCCH resources of the first cell). The first CSI report configuration may be one of one or more first CSI report configurations (e.g., in a first list of CSI report configurations) that are in a CSI-MeasConfig IE included in configuration parameters of the first cell (e.g., in a serving-CellConfig of the first cell). The configuration parameters (e.g., second configuration parameters of the second cell) may comprise second CSI report configuration parameters of one or more second CSI report configurations for transmission of CSI report via the second cell (e.g., via PUCCH resources of the second cell). The one or more second CSI report configurations may be in a second list of CSI report configurations in a CSI-MeasConfig IE included in configuration parameters of the second cell (e.g., in a servingCellConfig of the second cell). The first cell and the second cell may be in a cell group (e.g., a primary PUCCH group or a second PUCCH group). The first cell and the second cell may be among a plurality of candidate PUCCH cells configured with PUCCH resources wherein a candidate PUCCH cell, in the candidate PUCCH cells, may be a PUCCH cell based on a time/timing pattern (e.g., a time/timing pattern configured using a configuration (e.g., RRC) parameter). An example of the time/timing pattern indicating timings that the first cell is the PUCCH cell and timings that the second cell is the PUCCH cell is shown in FIG. 22. The configuration parameters may comprise at least one third configuration parameter of the time/timing pattern. The time/timing pattern may indicate first timings/durations that the first cell is a PUCCH cell and second timings/durations that the second cell is a PUCCH cell. The reception of the at least one third configuration parameter may indicate that PUCCH cell/carrier switching is configured for the wireless device. The at least one third configuration parameter may indicate that PUCCH cell/carrier switching between a plurality of cells, comprising the first cell/carrier and the second cell/carrier, is enabled/configured.

In examples as shown in FIG. 25 and FIG. 26, the first CSI report configuration parameters of the first CSI report configuration may comprise at least one first parameter (e.g., a first carrier IE) indicating an identifier of a first serving cell configured with CSI resources/reference signals for CSI channel measurement. The first CSI report configuration parameters may further comprise a first resourcesForChannelMeasurement IE indicating a first CSI-ResourceConfigId of a CSI resource configuration (e.g., CSI resource configuration included in the configuration of the serving cell indicated by the first carrier IE) indicating resources for CSI channel measurement. Based on receiving the at least one third configuration parameter and/or based on the PUCCH

cell/carrier switching being enabled/configured, e.g., based on PUCCH cell/carrier switching between a plurality of cells/carriers comprising the first cell/carrier and the second cell/carrier being configured/enabled, the one or more second CSI report configurations may comprise at least one second CSI report configuration comprising at least one second parameter (e.g., a second carrier IE) indicating the same identifier of the first serving cell configured with CSI resources/reference signals for CSI channel measurement. In an example, based on receiving the at least one third configuration parameter and/or based on the PUCCH cell/carrier switching being enabled/configured, e.g., based on PUCCH cell/carrier switching between a plurality of cells/carriers comprising the first cell/carrier and the second cell/carrier being configured/enabled, the at least one second CSI report configuration may further comprise a second resourcesForChannelMeasurement IE indicating the same first CSI-ResourceConfigId of the same CSI resource configuration (e.g., same CSI resource configuration included in the configuration of the same serving cell).

In an example, the first CSI report configuration and the second CSI report configuration may have the same report type configuration (e.g., both may be periodic CSI reports or both may be semi-persistent CSI reports).

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports, associated with the first CSI report configuration, are periodic CSI reports. The second CSI report configuration parameters may comprise a second reporting config type parameter indicating that the second CSI reports, associated with the second CSI report configuration, are periodic CSI reports.

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports, associated with the first CSI report configuration, are semi-persistent CSI reports. The second CSI report configuration parameters may comprise a second reporting config type parameter indicating that the second CSI reports, associated with the second CSI report configuration, are semi-persistent CSI reports. In an example as shown in FIG. 26, the wireless device may receive a first activation command (e.g., first SP CSI reporting on PUCCH activation/deactivation MAC CE) indicating activation of the first CSI report configuration and may receive a second activation command (e.g., a second SP CSI reporting on PUCCH activation/deactivation MAC CE) indicating activation of the second CSI report configuration. The wireless device may activate the first CSI report configuration (e.g., may start transmission of CSI reports) in response to receiving the first activation command and may activate the second CSI report configuration (e.g., may start transmission of CSI reports) in response to receiving the second activation command.

The wireless device may transmit CSI reports based on the first CSI report configuration parameters and the second CSI report configuration parameters. For example, the wireless device may transmit first CSI reports based on the first CSI report configuration parameters and via the first cell (via PUCCH resources of the first cell) and second CSI reports based on second CSI report configuration parameters and via the second cell (e.g., via PUCCH resources of the second cell).

In an example embodiment, PUCCH cell/carrier switching between a plurality of cells/carriers (e.g., comprising a first cell/carrier and a second cell/carrier) may be configured/enabled for a wireless device. A first CSI reporting configuration may be configured for CSI reporting (e.g.,

periodic CSI reporting) via PUCCH resources on the first cell/carrier (e.g., via PUCCH resources of an active UL BWP of the first cell). The first CSI reporting configuration may indicate (e.g., may comprise a carrier IE indicating) a first serving cell comprising CSI resources for channel measurement for the CSI reports. The first CSI reporting configuration may indicate a first identifier of a first CSI resource configuration of the first serving cell for channel measurement for the CSI reports. One or more second CSI reporting configurations may be configured for CSI reporting (e.g., periodic CSI reporting) via PUCCH resources on the second cell (e.g., via PUCCH resources of an active UL BWP of the second cell). The one or more second CSI reporting configurations may comprise at least one second CSI reporting configuration indicating (e.g., comprising a carrier IE indicating) the first serving cell and/or indicating the first identifier of the first CSI resource configuration of the first serving cell. Based on the PUCCH cell/carrier switching between the first cell/carrier and the second cell/carrier being configured/enabled, the one or more second CSI reporting configurations may comprise at least one second CSI reporting configuration indicating (e.g., comprising a carrier IE indicating) the first serving cell and/or indicating the first identifier of the first CSI resource configuration of the first serving cell.

In an example embodiment, PUCCH cell/carrier switching between a plurality of cells/carriers (e.g., comprising a first cell/carrier and a second cell/carrier) may be configured/enabled for a wireless device. A first CSI reporting configuration may be configured and activated (e.g., activated based on a MAC CE) for SP CSI reporting via PUCCH resources on the first cell/carrier (e.g., via PUCCH resources of an active UL BWP of the first cell). The first CSI reporting configuration may indicate (e.g., may comprise a carrier IE indicating) a first serving cell comprising CSI resources for channel measurement for the CSI reports. The first CSI reporting configuration may indicate a first identifier of a first CSI resource configuration of the first serving cell for channel measurement for the CSI reports. One or more second CSI reporting configurations may be configured and activated (e.g., activated based on one or more MAC CEs) for SP CSI reporting via PUCCH resources on the second cell (e.g., via PUCCH resources of an active UL BWP of the second cell). The one or more second CSI reporting configurations may comprise at least one second CSI reporting configuration indicating (e.g., comprising a carrier IE indicating) the first serving cell and/or indicating the first identifier of the first CSI resource configuration of the first serving cell for channel measurement for the CSI reports. Based on the PUCCH cell switching between the first cell/carrier and the second cell/carrier being configured/enabled, the one or more second CSI reporting configurations may comprise at least one second CSI reporting configuration indicating (e.g., comprising a carrier IE indicating) the first serving cell and/or indicating the first identifier of the first CSI resource configuration of the first serving cell for channel measurement for the CSI reports.

In example embodiments as shown in FIG. 27 and FIG. 28, a wireless device may receive one or more messages (e.g., one or more RRC messages) comprising configuration parameters of a plurality of cells comprising a first cell and a second cell. The configuration parameters (e.g., first configuration parameters of the first cell) may comprise first CSI report configuration parameters of a first CSI report configuration for transmission of CSI report via the first cell (e.g., via PUCCH resources of the first cell). The first cell and the second cell may be among a plurality of candidate

PUCCH cells configured with PUCCH resources wherein a candidate PUCCH cell, in the candidate PUCCH cells, may be a PUCCH cell based on a time/timing pattern (e.g., a time/timing pattern configured using a configuration (e.g., RRC) parameter). An example of the time/timing pattern indicating timings that the first cell is the PUCCH cell and timings that the second cell is the PUCCH cell is shown in FIG. 27. The configuration parameters may comprise at least one configuration parameter of the time/timing pattern. The time/timing pattern may indicate a first timing/time duration that the first cell is a PUCCH cell and a second timing/time duration that the second cell is a PUCCH cell. The reception of the at least one configuration parameter may indicate that PUCCH cell/carrier switching is configured for the wireless device. The at least one configuration parameter may indicate that PUCCH cell/carrier switching between a plurality of cells, comprising the first cell/carrier and the second cell/carrier, is enabled/configured.

The wireless device may determine that a first scheduled timing of a first CSI report overlaps and/or is within the first timing/time duration that the first cell is the PUCCH cell and may determine that a second scheduled timing of a second CSI report overlaps (e.g., in one or more symbols) with the second timing/time duration that the second cell is the PUCCH cell. Based on the first scheduled timing of the first CSI report overlapping and/or being within the first timing/time duration that the first cell is the PUCCH cell, the wireless device may transmit the first CSI report via the first cell (e.g., via first PUCCH resources of the first cell) and in the first scheduled timing of the first CSI report. Based on the second scheduled timing of the second CSI report overlapping (e.g., in one or more symbols) with the second timing/time duration that the second cell is the PUCCH cell, the wireless device may transmit the second CSI report via the second cell (e.g., in the second scheduled timing or in a third timing that is based on the second scheduled timing and via second PUCCH resources of the second cell).

In an example, the wireless device may determine the first scheduled timing of the first CSI report based on the first CSI report configuration parameters. For example, the first CSI report configuration parameters may comprise a first reportSlotConfig parameter indicating a first periodicity and slot offset. The first scheduled timing of the first CSI report may be based on the first periodicity and slot offset indicated by the first reportSlotConfig parameter. The wireless device may determine the first scheduled timing based on the first periodicity and slot offset indicated by the first reportSlotConfig parameter. In an example, the first scheduled timing of the first CSI report may further be based on a subcarrier spacing (SCS) configuration of an active uplink BWP (e.g., active uplink BWP of the first cell) configured with PUCCH resources to be used for transmission of the first CSI report. In an example, the wireless device may determine the first scheduled timing further based on the SCS configuration of an active uplink BWP (e.g., active uplink BWP of the first cell) configured with PUCCH resources to be used for transmission of the first CSI report. In an example, the second scheduled timing of the second CSI report may be based on the first CSI report configuration parameters. For example, the second scheduled timing may be based on the first periodicity and slot offset indicated by the first reportSlotConfig parameter. In an example, the second scheduled timing may further be based on SCS configuration of an active uplink BWP (e.g., active uplink BWP of the first cell or active uplink BWP of the second cell). The wireless device may determine the second scheduled timing based on

the first periodicity and slot offset indicated by the first reportSlotConfig parameter and/or based on the SCS configuration.

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports, associated with the CSI report configuration, are periodic CSI reports.

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports, associated with the CSI report configuration, are semi-persistent (SP) CSI reports. The wireless device may receive an activation command (e.g., a SP CSI reporting on PUCCH activation/deactivation MAC CE) indicating activation of the first CSI report configuration. Transmission of the CSI reports may be after/in response to receiving the activation command.

In an example, the wireless device may transmit the first CSI report based on the first CSI report configuration parameters.

In an example, the wireless device may receive second CSI report configuration parameters of a second CSI report configuration for transmission of CSI reports via the second cell (e.g., via PUCCH resources of the second cell). In an example, transmitting the second CSI report via the second cell (e.g., in the second scheduled timing of the second CSI report or in a third timing that may be based on the second scheduled timing) may be based on the second CSI report configuration parameters. In an example, the second CSI report configuration parameters may comprise a second reportSlotConfig parameter indicating a second periodicity and slot offset and the wireless device may determine the second scheduled timing or the third timing based on the second periodicity and slot offset. In an example, the wireless device may determine the second scheduled timing or the third timing further based on a SCS configuration of the second cell (e.g., active BWP of the second cell).

In an example, the wireless device may determine a first plurality of timings comprising the first scheduled timing and the second scheduled timing based on the first CSI report configuration parameters (e.g., based on the first reportSlotConfig parameter) and may determine second plurality of timings comprising the third timing based on the second CSI report configuration parameters (e.g., based on the second reportSlotConfig parameter). The third timing may be one of the second plurality of timings that is adjacent to and/or after the second scheduled timing.

In an example, transmitting the second CSI report may further be based on one or more configuration parameters (e.g., a carrier IE indicating the serving cell identifier of a serving cell comprising CSI resources/reference signals for which channel measurement takes place and/or a CSI resource configuration ID of the CSI resources/reference signals) of the first CSI report configuration parameters having the same values as corresponding configuration parameters in the second CSI report configuration parameters.

In an example embodiment, for periodic CSI reporting configured for a first cell, if during a time window comprising a scheduled timing of a CSI report, a PUCCH cell is switched from the first cell to a second cell, the wireless device may transmit the CSI report via the second cell.

In an example embodiment, for semi-persistent CSI reporting on PUCCH configured and activated for the first cell, if during a time window comprising a scheduled timing of a SP CSI report, a PUCCH cell is switched from the first cell to a second cell, the wireless device may transmit the SP CSI report via the second cell.

In example embodiments as shown in FIG. 29 and FIG. 30, a wireless device may receive one or more messages (e.g., one or more RRC messages) comprising configuration parameters of a plurality of cells comprising a first cell and a second cell. The configuration parameters (e.g., first configuration parameters of the first cell) may comprise first CSI report configuration parameters of a first CSI report configuration for transmission of CSI report via the first cell (e.g., via PUCCH resources of the first cell). The first cell and the second cell may be among a plurality of candidate PUCCH cells configured with PUCCH resources wherein a candidate PUCCH cell, in the candidate PUCCH cells, may be a PUCCH cell based on a time/timing pattern (e.g., a time/timing pattern configured using a configuration (e.g., RRC) parameter). An example of the time/timing pattern indicating timings that the first cell is the PUCCH cell and timings that the second cell is the PUCCH cell is shown in FIG. 29. The configuration parameters may comprise at least one configuration parameter of the time/timing pattern. The time/timing pattern may indicate a first timing/time duration that the first cell is a PUCCH cell and a second timing/time duration that the second cell is a PUCCH cell. The reception of the at least one configuration parameter may indicate that PUCCH cell/carrier switching is configured for the wireless device. The at least one configuration parameter may indicate that PUCCH cell/carrier switching between a plurality of cells, comprising the first cell/carrier and the second cell/carrier, is enabled/configured.

The wireless device may determine that a first scheduled timing of a first CSI report overlaps and/or is within the first timing/time duration that the first cell is the PUCCH cell and may determine that a second scheduled timing of a second CSI report overlaps (e.g., in one or more symbols) with the second timing/time duration that the second cell is the PUCCH cell. Based on the first scheduled timing of the first CSI report overlapping and/or being within the first timing/time duration that the first cell is the PUCCH cell, the wireless device may transmit the first CSI report via the first cell (e.g., via first PUCCH resources of the first cell) and in the first scheduled timing of the first CSI report. Based on the second scheduled timing of the second CSI report overlapping (e.g., in one or more symbols) with the second timing/time duration that the second cell is the PUCCH cell, the wireless device may drop the second CSI report.

In an example, the wireless device may determine the first scheduled timing of the first CSI report based on the first CSI report configuration parameters. For example, the first CSI report configuration parameters may comprise a first reportSlotConfig parameter indicating a first periodicity and slot offset. The first scheduled timing of the first CSI report may be based on the first periodicity and slot offset indicated by the first reportSlotConfig parameter. The wireless device may determine the first scheduled timing based on the first periodicity and slot offset indicated by the first reportSlotConfig parameter. In an example, the first scheduled timing of the first CSI report may further be based on a subcarrier spacing (SCS) configuration of an active uplink BWP (e.g., active uplink BWP of the first cell) configured with PUCCH resources to be used for transmission of the first CSI report. In an example, the wireless device may determine the first scheduled timing further based on the SCS configuration of an active uplink BWP (e.g., active uplink BWP of the first cell) configured with PUCCH resources to be used for transmission of the first CSI report. In an example, the second scheduled timing of the second CSI report may be based on the first CSI report configuration parameters. For example, the second scheduled timing may be based on the

first periodicity and slot offset indicated by the first reportSlotConfig parameter. In an example, the second scheduled timing may further be based on SCS configuration of an active uplink BWP (e.g., active uplink BWP of the first cell or active uplink BWP of the second cell). The wireless device may determine the second scheduled timing based on the first periodicity and slot offset indicated by the first reportSlotConfig parameter and/or based on the SCS configuration.

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports, associated with the CSI report configuration, are periodic CSI reports.

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports, associated with the CSI report configuration, are semi-persistent (SP) CSI reports. The wireless device may receive an activation command (e.g., a SP CSI reporting on PUCCH activation/deactivation MAC CE) indicating activation of the first CSI report configuration. Transmission of the CSI reports may be after/in response to receiving the activation command.

In an example, the wireless device may transmit the first CSI report based on the first CSI report configuration parameters.

In an example, the wireless device may receive second CSI report configuration parameters of a second CSI report configuration for transmission of CSI reports via the second cell (e.g., via PUCCH resources of the second cell). In an example, dropping the second CSI report may further be based on one or more configuration parameters (e.g., a carrier IE indicating the serving cell identifier of a serving cell comprising CSI resources/reference signals for which channel measurement takes place and/or a CSI resource configuration ID of the CSI resources/reference signals) of the first CSI report configuration parameters having different values from corresponding configuration parameters in the second CSI report configuration parameters.

In an example embodiment, for periodic CSI reporting configured for a first cell, if during a time window comprising a scheduled timing of a CSI report, a PUCCH cell is switched from the first cell to a second cell, the wireless device may drop the CSI report.

In an example embodiment, for semi-persistent CSI reporting on PUCCH configured and activated for the first cell, if during a time window comprising a scheduled timing of a SP CSI report, a PUCCH cell is switched from the first cell to a second cell, the wireless device may drop the SP CSI report.

In example embodiments as shown in FIG. 31 and FIG. 32, a wireless device may receive one or more messages (e.g., one or more RRC messages) comprising configuration parameters of a plurality of cells comprising a first cell and a second cell. The configuration parameters (e.g., first configuration parameters of the first cell) may comprise first CSI report configuration parameters of a first CSI report configuration for transmission of CSI report via the first cell (e.g., via PUCCH resources of the first cell). The first cell and the second cell may be among a plurality of candidate PUCCH cells configured with PUCCH resources wherein a candidate PUCCH cell, in the candidate PUCCH cells, may be a PUCCH cell based on a time/timing pattern (e.g., a time/timing pattern configured using a configuration (e.g., RRC) parameter). An example of the time/timing pattern indicating timings that the first cell is the PUCCH cell and timings that the second cell is the PUCCH cell is shown in FIG. 31. The configuration parameters may comprise at least

one configuration parameter of the time/timing pattern. The time/timing pattern may indicate a first timing/time duration that the first cell is a PUCCH cell and a second timing/time duration that the second cell is a PUCCH cell. The reception of the at least one configuration parameter may indicate that PUCCH cell/carrier switching is configured for the wireless device. The at least one configuration parameter may indicate that PUCCH cell/carrier switching between a plurality of cells, comprising the first cell/carrier and the second cell/carrier, is enabled/configured.

The wireless device may determine that a first scheduled timing of a first CSI report overlaps and/or is within the first timing/time duration that the first cell is the PUCCH cell. Based on the first scheduled timing of the first CSI report overlapping and/or being within the first timing/time duration that the first cell is the PUCCH cell, the wireless device may transmit the first CSI report via the first cell (e.g., via first PUCCH resources of the first cell) and in the first scheduled timing of the first CSI report. The wireless device may suspend CSI reporting (e.g., CSI reporting associated with the first CSI report configuration) during the second timing/time duration that the second cell is the PUCCH cell.

In an example, the wireless device may determine the first scheduled timing of the first CSI report based on the first CSI report configuration parameters. For example, the first CSI report configuration parameters may comprise a first reportSlotConfig parameter indicating a first periodicity and slot offset. The first scheduled timing of the first CSI report may be based on the first periodicity and slot offset indicated by the first reportSlotConfig parameter. The wireless device may determine the first scheduled timing based on the first periodicity and slot offset indicated by the first reportSlotConfig parameter. In an example, the first scheduled timing of the first CSI report may further be based on a subcarrier spacing (SCS) configuration of an active uplink BWP (e.g., active uplink BWP of the first cell) configured with PUCCH resources to be used for transmission of the first CSI report.

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports, associated with the CSI report configuration, are periodic CSI reports.

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports, associated with the CSI report configuration, are semi-persistent (SP) CSI reports. The wireless device may receive an activation command (e.g., a SP CSI reporting on PUCCH activation/deactivation MAC CE) indicating activation of the first CSI report configuration. Transmission of the CSI reports may be after/in response to receiving the activation command.

In an example, the wireless device may transmit the first CSI report based on the first CSI report configuration parameters.

In an example, the wireless device may receive second CSI report configuration parameters of a second CSI report configuration for transmission of CSI reports via the second cell (e.g., via PUCCH resources of the second cell). In an example, suspending the CSI reporting, during the second timing/time duration that the second cell is the PUCCH cell, may further be based on one or more configuration parameters (e.g., a carrier IE indicating the serving cell identifier of a serving cell comprising CSI resources/reference signals for which channel measurement takes place and/or a CSI resource configuration ID of the CSI resources/reference signals) of the first CSI report configuration parameters

having different values from corresponding configuration parameters in the second CSI report configuration parameters.

In an example embodiment, for periodic CSI reporting configured for a first cell, if PUCCH cell is switched from the first cell to a second cell during a time window, the wireless device may suspend the CSI reports during the time window.

In an example embodiment, for semi-persistent CSI reporting on PUCCH configured and activated for the first cell, if PUCCH cell is switched from the first cell to a second cell during a time window, the wireless device may suspend the SP CSI reports during the time window.

In example embodiments as shown in FIG. 33, a wireless device may receive one or more messages (e.g., one or more RRC messages) comprising configuration parameters. The configuration parameters may comprise first CSI report configuration parameters of a first CSI report configuration for transmission of semi-persistent (SP) CSI report (e.g., via PUCCH resources of a first cell). The first CSI report configuration parameters may comprise a report config type parameter indicating that CSI reports, associated with the first CSI report configuration, are SP CSI reports. The configuration parameters may comprise at least one configuration parameter of a time/timing pattern indicating timings that each candidate PUCCH cell, in a plurality of candidate PUCCH cells, is a PUCCH cell. Each candidate PUCCH cell, in the plurality of candidate PUCCH cells, may be configured with PUCCH resources and the wireless device may determine, based on the time/timing pattern indicated by the at least one configuration parameter, timings that each candidate PUCCH cell, in the plurality of candidate PUCCH cells, is a PUCCH cell. The reception of the at least one configuration parameter may indicate that PUCCH cell/carrier switching is configured for the wireless device. The at least one configuration parameter may indicate that PUCCH cell/carrier switching between the plurality of candidate PUCCH cells is enabled/configured.

The wireless device may receive, via a PDSCH, a transport block comprising a SP CSI reporting on PUCCH Activation/Deactivation MAC CE indicating activation of the first CSI report configuration. The SP CSI reporting on PUCCH activation/deactivation MAC CE may comprise a field with a value indicating that the first SP CSI report configuration is activated. In response to receiving the transport block comprising the SP CSI reporting on PUCCH Activation/Deactivation MAC CE, the wireless device may start transmitting SP CSI reports associated with the CSI report configuration.

In an example embodiment, the wireless device may determine a timing of a first slot based on subcarrier spacing (SCS) configuration for the PUCCH of a cell, in the plurality of cells, that is the PUCCH cell at the time that the transport block comprising the SP CSI reporting on PUCCH Activation/Deactivation MAC CE is received. Transmission of the SP CSI reports may be after the first slot. The wireless device may transmit SP CSI reports after the first slot and based on the first CSI report configuration parameters.

In an example embodiment, a wireless device may perform semi-persistent CSI reporting on the PUCCH applied starting from the first slot that is after slot $n+3N_{\text{slot}}^{\text{subframe},\mu}$ when the wireless would transmit a PUCCH with HARQ-ACK information in slot n corresponding to the PDSCH carrying the activation command (e.g., SP CSI reporting on PUCCH activation MAC CE) where μ is the SCS configuration for the PUCCH of the cell that is the PUCCH cell at

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the time that PDSCH is received in case PUCCH carrier switching is configured/enabled.

In an example embodiment, the wireless device may determine a timing of a first slot based on subcarrier spacing (SCS) configuration for the PUCCH of a cell, in the plurality of cells, that is the PUCCH cell at the time that an acknowledgement associated with the transport block comprising the SP CSI reporting on PUCCH Activation/Deactivation MAC CE is transmitted. Transmission of the SP CSI reports may be after the first slot. The wireless device may transmit SP CSI reports after the first slot and based on the first CSI report configuration parameters.

In an example embodiment, a wireless device may perform semi-persistent CSI reporting on the PUCCH applied starting from the first slot that is after slot $n+3N_{\text{slot}}^{\text{subframe},\mu}$ when the wireless would transmit a PUCCH with HARQ-ACK information in slot n corresponding to the PDSCH carrying the activation command (e.g., SP CSI reporting on PUCCH activation MAC CE) where μ is the SCS configuration for the PUCCH of the cell that is the PUCCH cell at the time that an acknowledgement associated with the activation command is transmitted in case PUCCH carrier switching is configured/enabled.

In an example embodiment, the wireless device may determine a timing of a first slot based on subcarrier spacing (SCS) configuration for the PUCCH of a primary cell (PCell) or a primary secondary cell (PSCell, e.g., in case the transport block is received via a cell in the secondary cell group (SCG)), or a PUCCH SCell (e.g., in case the transport block is received via a cell in a secondary PUCCH group) or a reference cell. Transmission of the SP CSI reports may be after the first slot. The wireless device may transmit SP CSI reports after the first slot and based on the first CSI report configuration parameters.

In an example embodiment, a wireless device may perform semi-persistent CSI reporting on the PUCCH applied starting from the first slot that is after slot $n+3N_{\text{slot}}^{\text{subframe},\mu}$ when the wireless would transmit a PUCCH with HARQ-ACK information in slot n corresponding to the PDSCH carrying the activation command (e.g., SP CSI reporting on PUCCH activation MAC CE) where μ is the SCS configuration for the PUCCH of the PCell/PSCell/PUCCH SCell/a reference cell in case PUCCH carrier switching is configured/enabled].

In an example embodiment as shown in FIG. 34, a wireless device may receive one or more messages (e.g., one or more RRC messages) comprising configuration parameters. The configuration parameters may comprise CSI report configuration parameters of a CSI report configuration for transmission of CSI report (e.g., via PUCCH resources of a first cell). The CSI report configuration parameters may comprise a report config type parameter indicating that CSI reports, associated with the CSI report configuration, are periodic CSI report or indicating that CSI reports, associated with the CSI report configuration, are SP CSI reports. The configuration parameters may comprise at least one configuration parameter of a time/timing pattern indicating timings that each candidate PUCCH cell, in a plurality of candidate PUCCH cells, is a PUCCH cell. Each candidate PUCCH cell, in the plurality of candidate PUCCH cells, may be configured with PUCCH resources and the wireless device may determine, based on the time/timing pattern indicated by the at least one configuration parameter, timings that each candidate PUCCH cell, in the plurality of candidate PUCCH cells, is a PUCCH cell. The reception of the at least one configuration parameter may indicate that PUCCH cell/carrier switching is configured for the wireless

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device. The at least one configuration parameter may indicate that PUCCH cell/carrier switching between the plurality of candidate PUCCH cells is enabled/configured.

The wireless device may determine a timing of a CSI report, associated with the CSI report configuration, based on subcarrier spacing (SCS) of a first candidate PUCCH cell in the plurality of candidate PUCCH cells. The wireless device may transmit the CSI report in the determined timing. In an example embodiment, the first cell may be the cell via which the CSI report is scheduled to be transmitted.

The wireless device may determine a timing of a CSI report, associated with the CSI report configuration, based on subcarrier spacing (SCS) of a first candidate PUCCH cell in the plurality of candidate PUCCH cells. The wireless device may transmit the CSI report in the determined timing. In an example embodiment, the first cell may be a primary cell (PCell) or a primary secondary cell (PSCell, e.g., in case the CSI report is transmitted via a cell in the secondary cell group (SCG)), or a PUCCH SCell (e.g., in case the CSI report is transmitted via a cell in a secondary PUCCH group) or a reference cell.

In an example embodiment as shown in FIG. 34, a wireless device may receive one or more messages (e.g., one or more RRC messages) comprising configuration parameters. The configuration parameters may comprise first CSI report configuration parameters of a first CSI report configuration for transmission of CSI report via a first cell (e.g., via PUCCH resources of a first cell). The first CSI report configuration parameters may comprise a report config type parameter indicating that CSI reports, associated with the first CSI report configuration, are SP CSI reports or indicating that CSI report, associated with the first CSI report configuration, are periodic CSI reports. The configuration parameters may comprise at least one configuration parameter of a time/timing pattern indicating timings that each candidate PUCCH cell, in a plurality of candidate PUCCH cells, is a PUCCH cell. Each candidate PUCCH cell, in the plurality of candidate PUCCH cells, may be configured with PUCCH resources and the wireless device may determine, based on the time/timing pattern indicated by the at least one configuration parameter, timings that each candidate PUCCH cell, in the plurality of candidate PUCCH cells, is a PUCCH cell. The reception of the at least one configuration parameter may indicate that PUCCH cell/carrier switching is configured for the wireless device. The at least one configuration parameter may indicate that PUCCH cell/carrier switching between the plurality of candidate PUCCH cells is enabled/configured. The plurality of candidate PUCCH cells may comprise a first cell and a second cell.

The wireless device may determine a first scheduled timing of a first CSI report. The wireless device may determine (e.g., based on the time/timing pattern indicated by the at least one configuration parameter) that the first scheduled timing of the first CSI report is while the first cell is the PUCCH cell. The wireless device may perform CSI channel measurements based on the first CSI report configuration while the second cell is a PUCCH cell. The wireless device may perform CSI channel measurements based on the first CSI report configuration regardless of whether the first cell is the PUCCH cell or the second cell is the PUCCH cell when performing the CSI channel measurements.

In an example embodiment, a wireless device may receive first channel state information (CSI) report configuration parameters, second CSI report configuration parameters and at least one third configuration parameter of a timing pattern. The first CSI report configuration parameters may be of a

first CSI report configuration for transmission of CSI reports via a first cell (e.g., via physical uplink control channel (PUCCH) resources of the first cell). The second CSI report configuration parameters may be of one or more second CSI report configurations for transmission of CSI reports via a second cell (e.g., via PUCCH resources of a second cell). The at least one third configuration parameter may be of a timing pattern indicating first timings/durations that the first cell is a PUCCH cell and second timings/durations that the second cell is the PUCCH cell. The at least one third configuration parameter may indicate that PUCCH cell/carrier switching between a plurality of cells, comprising the first cell and the second cell, is enabled/configured. The first CSI report configuration parameters may comprise a first configuration parameter with a first value. Based on receiving the at least one third configuration parameter, the one or more second CSI report configurations may comprise at least one second CSI report configuration comprising a second configuration parameter, corresponding to the first configuration parameter, with the first value. Based on the PUCCH cell/carrier switching between the first cell and the second cell being enabled/configured, the one or more second CSI report configurations may comprise at least one second CSI report configuration comprising a second configuration parameter, corresponding to the first configuration parameter, with the first value.

In an example, the at least one third configuration parameter may indicate that PUCCH carrier switching between the first cell and the second cell is enabled/configured.

In an example, the wireless device may receive first CSI measurement configuration parameters, wherein the first CSI report configuration parameters may be included in the first received CSI measurement configuration parameters. The wireless device may receive second CSI measurement configuration parameters, wherein the second CSI report configuration parameters may be included in the second received CSI measurement configuration parameters. In an example, the wireless device may receive first configuration parameters of the first cell, wherein the first CSI measurement configuration parameters may be included in the first configuration parameters of the first cell. The wireless device may receive second configuration parameters of the second cell, wherein the second CSI measurement configuration parameters may be included in the second configuration parameters of the second cell.

In an example, the first configuration parameter may indicate an identifier of a first serving cell configured with CSI resources/reference signals for channel measurement. The second configuration parameter indicates the identifier of the first serving cell configured with CSI resources/reference signals for channel measurement. In an example, the first configuration parameter may be a first carrier information element (IE) indicating a serving cell index of the first serving cell configured with CSI resources/reference signals for channel measurement. The second configuration parameter may be a second carrier IE indicating the serving cell index of the first serving cell configured with CSI resources/reference signals for channel measurement. In an example, the first CSI report configuration parameters may further comprise a third parameter indicating an identifier of a first CSI resource configuration of the CSI resources/reference signals. Based on receiving the at least one third configuration parameter, the one or more second CSI report configurations may comprise at least one second CSI report configuration comprising a fourth parameter indicating the identifier of the first CSI resource configuration of the CSI resources/reference signals. Based on the PUCCH carrier

switching between the first cell and the second cell being enabled/configured, the one or more second CSI report configurations may comprise at least one second CSI report configuration comprising a fourth parameter indicating the identifier of the first CSI resource configuration of the CSI resources/reference signals.

In an example, the first CSI report configuration parameters comprise a first report config type parameter indicating that the CSI reports are periodic CSI reports. The second CSI report configuration parameters may comprise a second report config type parameter indicating that the CSI reports are periodic CSI reports.

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports are semi-persistent (SP) CSI reports for transmission via PUCCH. The second CSI report configuration parameters comprise a second report config type parameter indicating that the CSI reports are SP CSI reports for transmission via PUCCH. In an example, the wireless device may receive a first activation command indicating activation of the first CSI report configuration. The wireless device may receive one or more second activation commands indicating activation of the one or more second CSI report configurations. In an example, the first activation command may be a first SP CSI reporting on PUCCH Activation/Deactivation MAC CE. The one or more second activation commands may be one or more second SP CSI reporting on PUCCH Activation/Deactivation MAC CEs.

In an example, transmitting the one or more CSI reports may be via PUCCH.

In an example, the first cell is a primary cell (PCell) or a primary secondary cell (PSCell) or a PUCCH secondary cell.

In an example, the first cell may be a secondary cell.

In an example, the second cell is a primary cell (PCell) or a primary secondary cell (PSCell) or a PUCCH secondary cell.

In an example, the second cell may be a secondary cell.

In an example embodiment, a wireless device may receive first channel state information (CSI) report configuration parameters, second CSI report configuration parameters and at least one third configuration parameter of a timing pattern. The first CSI report configuration parameters may be of a first CSI report configuration for transmission of CSI reports via a first cell (e.g., via physical uplink control channel (PUCCH) resources of the first cell). The second CSI report configuration parameters may be of one or more second CSI report configurations for transmission of CSI reports via a second cell (e.g., via PUCCH resources of a second cell). The at least one third configuration parameter may be of a timing pattern indicating first timings/durations that the first cell is a PUCCH cell and second timings/durations that the second cell is the PUCCH cell. The at least one third configuration parameter may indicate that PUCCH cell/carrier switching between a plurality of cells, comprising the first cell and the second cell, is enabled/configured. The first CSI report configuration parameters may comprise at least one first parameter indicating an identifier of a first serving cell configured with CSI resources/reference signals for channel measurement. Based on receiving the at least one third configuration parameter, the one or more second CSI report configurations may comprise at least one second CSI report configuration comprising at least one second parameter indicating the identifier of the first serving cell configured with the CSI resources/reference signals for channel measurement. Based on the PUCCH cell/carrier switching

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between the first cell and the second cell being enabled/configured, the one or more second CSI report configurations may comprise at least one second CSI report configuration comprising at least one second parameter indicating the identifier of the first serving cell configured with the CSI resources/reference signals for channel measurement. The wireless device may transmit one or more CSI reports based on the first CSI report configuration parameters and the second CSI report configuration parameters.

In an example, the at least one third configuration parameter may indicate that PUCCH carrier switching between the first cell and the second cell is enabled/configured.

In an example, the wireless device may receive first CSI measurement configuration parameters, wherein the first CSI report configuration parameters may be included in the first received CSI measurement configuration parameters. The wireless device may receive second CSI measurement configuration parameters, wherein the second CSI report configuration parameters may be included in the second received CSI measurement configuration parameters. In an example, the wireless device may receive first configuration parameters of the first cell, wherein the first CSI measurement configuration parameters may be included in the first configuration parameters of the first cell. The wireless device may receive second configuration parameters of the second cell, wherein the second CSI measurement configuration parameters may be included in the second configuration parameters of the second cell.

In an example, the first CSI report configuration parameters further comprise at least one third parameter indicating an identifier of a first CSI resource configuration of the CSI resources/reference signals. Based on receiving the at least one third configuration parameter, the one or more second CSI report configurations comprise at least one second CSI report configuration comprising at least one fourth parameter indicating the identifier of the first CSI resource configuration of the CSI resources/reference signals. Based on the PUCCH carrier switching between the first cell and the second cell being enabled/configured, the one or more second CSI report configurations may comprise at least one second CSI report configuration comprising at least one fourth parameter indicating the identifier of the first CSI resource configuration of the CSI resources/reference signals.

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports are periodic CSI reports. The second CSI report configuration parameters may comprise a second report config type parameter indicating that the CSI reports are periodic CSI reports.

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports are semi-persistent (SP) CSI reports for transmission via PUCCH. The second CSI report configuration parameters may comprise a second report config type parameter indicating that the CSI reports are SP CSI reports for transmission via PUCCH. In an example, the wireless device may receive a first activation command indicating activation of the first CSI report configuration. The wireless device may receive one or more second activation commands indicating activation of the one or more second CSI report configurations. In an example, the first activation command may be a first SP CSI reporting on PUCCH Activation/Deactivation MAC CE. The one or more second activation commands may be one or more second SP CSI reporting on PUCCH Activation/Deactivation MAC CEs.

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In an example, transmitting the one or more CSI reports may be via PUCCH.

In an example, the first cell may be a primary cell (PCell) or a primary secondary cell (PSCell) or a PUCCH secondary cell.

In an example, the first cell may be a secondary cell.

In an example, the second cell may be a primary cell (PCell) or a primary secondary cell (PSCell) or a PUCCH secondary cell.

In an example, the second cell may be a secondary cell.

In an example, the at least one first parameter may comprise a first carrier IE indicating a serving cell index of the first serving cell configured with CSI resources/reference signals for channel measurement.

In an example, the at least one second parameter may comprise a second carrier IE indicating a serving cell index of the first serving cell configured with CSI resources/reference signals for channel measurement.

In an example embodiment, a wireless device may receive first channel state information (CSI) report configuration parameters and at least one configuration parameter of a timing pattern. The first CSI report configuration parameters may be of a first CSI report configuration for transmission of CSI reports via a first cell (e.g., via PUCCH resources of a first cell). The timing pattern may indicate a first timing/time duration that the first cell is a physical uplink control channel (PUCCH) cell and a second timing/time duration that a second cell is the PUCCH cell. The wireless device may transmit a first CSI report, via the first cell, in a first scheduled timing of the first CSI report based on/in response to the first scheduled timing overlapping with the first timing/time duration. The wireless device may transmit a second CSI report, via the second cell, in a second scheduled timing of the second CSI report based on/in response to the second scheduled timing overlapping with the second timing/time duration.

In an example, the first CSI report configuration parameters may comprise a first reportSlotConfig parameter indicating a first periodicity and slot offset wherein the first scheduled timing of the first CSI report may be based on the first periodicity and slot offset. In an example, the first scheduled timing may further be based on a subcarrier spacing (SCS) configuration of an active uplink bandwidth part (e.g., an active uplink bandwidth part of the first cell) configured with PUCCH resources. In an example, the wireless device may determine the first scheduled timing based on the first periodicity and slot offset and/or the SCS configuration. In an example, the second scheduled timing of the second CSI report may be based on the first periodicity and slot offset indicated by the first reportSlotConfig parameter. In an example, the second scheduled timing may further be based on a subcarrier spacing (SCS) configuration of an active uplink bandwidth part (e.g., an active uplink bandwidth part of the first cell or an active bandwidth part of the second cell) configured with PUCCH resources. In an example, the wireless device may determine the second scheduled timing based on the first periodicity and slot offset and/or the SCS configuration.

In an example, the first scheduled timing may be in one or more first symbols of a first slot of the first cell (e.g., an active uplink bandwidth part of the first cell). The second scheduled timing may be in one or more second symbols of a second slot (e.g., an active uplink bandwidth part of e.g., the second cell).

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports are periodic CSI reports.

In an example, the first CSI report configuration parameters comprise a first report config type parameter indicating that the CSI reports are semi-persistent (SP) CSI reports. The wireless device may receive an activation command (e.g., SP CSI reporting on PUCCH Activation/Deactivation MAC) indicating activation of the CSI report configuration.

In an example, the transmitting the first CSI report may be based on/via first PUCCH resources of the first cell. The transmitting the second CSI report may be based on/via second PUCCH resources of the second cell.

In an example, the transmitting the first CSI report via the first cell may be based on the first CSI report configuration parameters.

In an example, the wireless device may receive second CSI report configuration parameters of a second CSI report configuration for transmission of CSI reports via the second cell (e.g., via PUCCH resources of the second cell). The transmitting the second CSI report via the second cell (e.g., in the second scheduled timing) may be based on the second CSI report configuration parameters. In an example, the second CSI report configuration parameters may comprise a second reportSlotConfig parameter indicating a second periodicity and slot offset. The second scheduled timing of the second CSI report may be based on the second periodicity and slot offset. In an example, the second scheduled timing may further be based on a subcarrier spacing (SCS) configuration of an active uplink bandwidth part (e.g., an active uplink bandwidth part of the second cell) configured with PUCCH resources. In an example, the wireless device may determine the second scheduled timing based on the second periodicity and slot offset and/or the SCS configuration. In an example, the transmitting the second CSI report via the second cell (e.g., in the second scheduled timing) may be based on one or more configuration parameter of the second CSI report configuration parameters having same values as corresponding configuration parameter of the first CSI report configuration parameters. In an example, the one or more configuration parameters may comprise a carrier information element (IE) indicating a serving cell index of a first serving cell comprising CSI resources/reference signals for channel measurement. In an example, the one or more configuration parameters may comprise an identifier a CSI resource configuration of the CSI resources/reference signals.

In an example embodiment, a wireless device may receive first channel state information (CSI) report configuration parameters and at least one configuration parameter of a timing pattern. The first CSI report configuration parameters may be of first CSI report configuration for transmission of CSI reports via a first cell (e.g., via PUCCH resources of a first cell). The timing pattern may indicate a first timing/time duration that the first cell is a physical uplink control channel (PUCCH) cell and a second timing/time duration that a second cell is the PUCCH cell. The wireless device may transmit a first CSI report, via the first cell, in a first scheduled timing of the first CSI report based on/in response to the first scheduled timing overlapping with the first timing/time duration. The wireless device may drop a second CSI report based on/in response to a second scheduled timing of the second CSI report overlapping with the second timing/time duration.

In an example, the first CSI report configuration parameters may comprise a first reportSlotConfig parameter indicating a first periodicity and slot offset wherein the first scheduled timing of the first CSI report may be based on the first periodicity and slot offset. In an example, the first scheduled timing may further be based on a subcarrier

spacing (SCS) configuration of an active uplink bandwidth part (e.g., an active uplink bandwidth part of the first cell) configured with PUCCH resources. In an example, the wireless device may determine the first scheduled timing based on the first periodicity and slot offset and/or the SCS configuration. In an example, the second scheduled timing of the second CSI report may be based on the first periodicity and slot offset indicated by the first reportSlotConfig parameter. In an example, the second scheduled timing may further be based on a subcarrier spacing (SCS) configuration of an active uplink bandwidth part (e.g., an active uplink bandwidth part of the first cell or an active bandwidth part of the second cell) configured with PUCCH resources. In an example, the wireless device may determine the second scheduled timing based on the first periodicity and slot offset and/or the SCS configuration.

In an example, the first scheduled timing may be in one or more first symbols of a first slot of the first cell (e.g., an active uplink bandwidth part of the first cell). The second scheduled timing may be in one or more second symbols of a second slot of the second cell (e.g., an active uplink bandwidth part of the second cell).

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports are periodic CSI reports.

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports are semi-persistent (SP) CSI reports. In an example, the wireless device may receive an activation command (e.g., SP CSI reporting on PUCCH Activation/Deactivation MAC CE) indicating activation of the CSI report configuration.

In an example, transmitting the first CSI report may be based on/via PUCCH resources of the first cell.

In an example, the transmitting the first CSI report via the first cell may be based on the first CSI report configuration parameters.

In an example, the wireless device may receive second CSI report configuration parameters of a second CSI report configuration for transmission of CSI reports via the second cell (e.g., via PUCCH resources of the second cell). The dropping the second CSI report may be based on one or more configuration parameter of the second CSI report configuration parameters having different values from corresponding configuration parameter of the first CSI report configuration parameters. In an example, the one or more configuration parameters may comprise a carrier information element (IE) indicating a serving cell index of a first serving cell comprising CSI resources/reference signals for channel measurement. In an example, the one or more configuration parameters may comprise an identifier a CSI resource configuration of the CSI resources/reference signals.

In an example embodiment, the wireless device may receive first channel state information (CSI) report configuration parameters and at least one configuration parameter of a timing pattern. The first CSI report configuration parameters may be of a first CSI report configuration for transmission of CSI reports via a first cell (e.g., via PUCCH resources of a first cell). The timing pattern may indicate a first timing/time duration that the first cell is a physical uplink control channel (PUCCH) cell and a second timing/time duration that a second cell is the PUCCH cell. The wireless device may transmit a first CSI report, via the first cell, in a first scheduled timing of the first CSI report based on/in response to the first scheduled timing overlapping with

the first timing/time duration. The wireless device may suspend CSI reporting during the second timing/time duration.

In an example, the first CSI report configuration parameters may comprise a first reportSlotConfig parameter indicating a first periodicity and slot offset wherein the first scheduled timing of the first CSI report may be based on the first periodicity and slot offset. In an example, the first scheduled timing may further be based on a subcarrier spacing (SCS) configuration of an active uplink bandwidth part (e.g., an active uplink bandwidth part of the first cell) configured with PUCCH resources. In an example, the wireless device may determine the first scheduled timing based on the first periodicity and slot offset and/or the SCS configuration.

In an example, the first CSI report configuration parameters may comprise a first report config type parameter indicating that the CSI reports are periodic CSI reports.

In an example, the first CSI report configuration parameters comprise a first report config type parameter indicating that the CSI reports are semi-persistent (SP) CSI reports. In an example, the wireless device may receive an activation command (e.g., SP CSI reporting on PUCCH Activation/Deactivation MAC CE) indicating activation of the CSI report configuration.

In an example, transmitting the first CSI report may be based on/via PUCCH resources of the first cell.

In an example, the transmitting the first CSI report via the first cell may be based on the first CSI report configuration parameters.

In an example, the wireless device may receive second CSI report configuration parameters of a second CSI report configuration for transmission of CSI reports via the second cell (e.g., via PUCCH resources of the second cell). The suspending the CSI reporting may be based on one or more configuration parameter of the second CSI report configuration parameters having different values from corresponding configuration parameter of the first CSI report configuration parameters. In an example, the one or more configuration parameters may comprise a carrier information element (IE) indicating a serving cell index of a first serving cell comprising CSI resources/reference signals for channel measurement. In an example, the one or more configuration parameters comprise an identifier a CSI resource configuration of the CSI resources/reference signals.

In an example embodiment, a wireless device may receive channel state information (CSI) report configuration parameters of a CSI report configuration and at least one configuration parameter of a timing pattern. The CSI report configuration parameters may be for transmission of semi-persistent (SP) CSI reports. The timing pattern may indicate timings that each of a plurality of cells is a physical uplink control channel (PUCCH) cell. The wireless device may receive, via a downlink shared channel, a transport block comprising an activation command indicating activation of the CSI report configuration. The wireless device may determine a timing of a first slot based on a subcarrier spacing configuration for a PUCCH of a cell, in the plurality of cells, that is the PUCCH cell at the time that the transport block is received. Transmission of the SP CSI reports may be after the first slot.

In an example, the CSI configuration parameters may comprise a report config type parameter indicating that the CSI report configuration is for transmission of SP CSI reports.

In an example, the activation command may be based on a SP CSI reporting on PUCCH Activation/Deactivation MAC CE.

In an example, the wireless device may transmit the SP CSI reports after the first slot.

In an example, transmitting a SP CSI report, in the SP CSI reports, may be based PUCCH resources of a cell in the plurality of cells.

In an example, the CSI report configuration parameters may comprise a reportSlotConfig parameter indicating a periodicity and slot offset. The timings of the SP CSI reports may be (e.g., may be determined) based on the periodicity and slot offset.

In an example embodiment, a wireless device may receive channel state information (CSI) report configuration parameters of a CSI report configuration and at least one configuration parameter of a timing pattern. The CSI report configuration parameters may be for transmission of semi-persistent (SP) CSI reports. The timing pattern may indicate timings that each of a plurality of cells is a physical uplink control channel (PUCCH) cell. The wireless device may receive, via a downlink shared channel, a transport block comprising an activation command indicating activation of the CSI report configuration. The wireless device may determine a timing of a first slot based on a subcarrier spacing configuration for the PUCCH of a cell, in the plurality of cells, that is the PUCCH cell at the time that an acknowledgement associated with the transport block is transmitted. Transmission of the SP CSI reports may be after the first slot.

In an example, the CSI configuration parameters may comprise a report config type parameter indicating that the CSI report configuration is for transmission of SP CSI reports.

In an example, the activation command may be based on a SP CSI reporting on PUCCH Activation/Deactivation MAC CE.

In an example, the wireless device may transmit the SP CSI reports after the first slot.

In an example, transmitting a SP CSI report, in the SP CSI reports, may be based PUCCH resources of a cell in the plurality of cells.

In an example, the CSI report configuration parameters may comprise a reportSlotConfig parameter indicating a periodicity and slot offset. The timings of the SP CSI reports may be (e.g., may be determined) based on the periodicity and slot offset.

In an example embodiment, a wireless device may receive channel state information (CSI) report configuration parameters of a CSI report configuration and at least one configuration parameter of a timing pattern. The CSI report configuration parameters may be for transmission of semi-persistent (SP) CSI reports. The timing pattern may indicate timings that each of a plurality of cells is a physical uplink control channel (PUCCH) cell. The wireless device may receive, via a downlink shared channel, a transport block comprising an activation command indicating activation of the CSI report configuration. The wireless device may determine a timing of a first slot based on a subcarrier spacing configuration for the PUCCH of a primary cell (PCell)/primary secondary cell (PSCell)/PUCCH SCell/a reference cell. Transmission of the SP CSI reports may be after the first slot.

In an example, the CSI configuration parameters may comprise a report config type parameter indicating that the CSI report configuration is for transmission of SP CSI reports.

In an example, the activation command may be based on a SP CSI reporting on PUCCH Activation/Deactivation MAC CE.

In an example, the wireless device may transmit the SP CSI reports after the first slot.

In an example, transmitting a SP CSI report, in the SP CSI reports, may be based PUCCH resources of a cell in the plurality of cells.

In an example, the CSI report configuration parameters may comprise a reportSlotConfig parameter indicating a periodicity and slot offset. The timings of the SP CSI reports may be (e.g., may be determined) based on the periodicity and slot offset.

In an example embodiment, a wireless device may receive channel state information (CSI) report configuration parameters and at least one configuration parameter of a timing pattern. The CSI report configuration parameters may be of a CSI report configuration for transmission of CSI reports. The timing pattern may indicate timings that each of a plurality of cells is a physical uplink control channel (PUCCH) cell. The wireless device may determine a timing of a CSI reports based on a subcarrier spacing configuration of a first cell in the plurality of cells. The wireless device may transmit the CSI report in the determined timing.

In an example, the first cell may be the cell via which the CSI report is scheduled to be transmitted.

In an example, the first cell is a primary cell (PCell) or a primary secondary cell (PSCell) or a PUCCH SCell or a reference cell.

In an example embodiment, a wireless device may receive first channel state information (CSI) report configuration parameters and at least one configuration parameter of a timing pattern. The first CSI report configuration parameters may be of a first CSI report configurations for transmission of CSI reports via a first cell (e.g., via PUCCH resources of a first cell). The timing pattern may indicate timings that each of a plurality of cells, comprising the first cell and a second cell, is a physical uplink control channel (PUCCH) cell. Based on a first scheduled timing of a first CSI report being while the first cell is the PUCCH cell, the wireless device may perform channel measurements based on the first CSI report configuration regardless of whether the first cell is the PUCCH cell or the second cell is the PUCCH cell when performing the channel measurements. The wireless device may transmit the first CSI report.

In an example embodiment, a wireless device may receive first channel state information (CSI) report configuration parameters and at least one configuration parameter of a timing pattern. The first CSI report configuration parameters may be of a first CSI report configurations for transmission of CSI reports via a first cell (e.g., via PUCCH resources of a first cell). The timing pattern may indicate timings that each of a plurality of cells, comprising the first cell and a second cell, is a physical uplink control channel (PUCCH) cell. A first scheduled timing of a first CSI report may be while the first cell is the PUCCH cell. The wireless device may perform channel measurements based on the first CSI report configuration while the second cell is a PUCCH cell. The wireless device may transmit the first CSI report.

In an example embodiment as shown in FIG. 36, a wireless device may receive one or more messages (e.g., one or more RRC messages) comprising configuration parameters. The wireless device may be configured with a plurality of cells comprising a first cell and a second cell. The configuration parameters may comprise first configuration parameters of the first cell and the second cell. In an example, the plurality of cells may be provided by one base

station (e.g., in case of a carrier aggregation scenario). In an example, the plurality of cells may be provided by a plurality of base stations (e.g., a first base station and a second base station in case of dual connectivity).

The plurality of cells configured for the wireless device may be grouped into one or more PUCCH groups. The first cell and the second cell may be in the same PUCCH group, e.g., in a primary PUCCH group or in a secondary PUCCH group. In case of the primary PUCCH group, the first cell may be a primary cell and the second cell may be a PUCCH switching secondary cell that is configured on top of the primary cell for PUCCH transmission in the primary PUCCH group (e.g., for transmission of UCI associated with the cells within the primary PUCCH group). In case of the secondary PUCCH group, the first cell may be a PUCCH secondary cell (PUCCH SCell) and the second cell may be a PUCCH switching secondary cell that is configured on top of the PUCCH SCell for PUCCH transmission in the secondary PUCCH group (e.g., for transmission of UCI associated with the cells within the secondary PUCCH group). In an example, the configuration parameters may comprise a parameter indicating that the second cell is a PUCCH switching secondary cell, e.g., an alternative PUCCH cell for PUCCH cell switching in the primary PUCCH group (e.g., in case the first cell and the second cell are in the primary PUCCH group) or the secondary PUCCH group (e.g., in case the first cell and the second cell are in a secondary PUCCH group). For example, the first cell and the second cell may be in a primary PUCCH group, the first cell may be the primary cell and the second cell may be a PUCCH switching secondary cell and configured on top of primary cell in the primary PUCCH group for PUCCH transmission. For example, the first cell and the second cell may be in a secondary PUCCH group, the first cell may be a PUCCH SCell and the second cell may be a PUCCH switching secondary cell and may be configured on top of the PUCCH SCell in the secondary PUCCH group for PUCCH transmission.

The configuration parameters may further comprise CSI report configuration parameters for CSI reporting via the first cell (e.g., via PUCCH resources of the first cell). For example, an IE CSI-ReportConfig may be used to configure CSI reporting (e.g., periodic or semi-persistent CSI reporting) on PUCCH of the first cell. In case of semi-persistent CSI reporting, the wireless device may further receive a MAC CE indicating activation of CSI reporting via PUCCH of the first cell and the wireless device may transmit CSI reports semi-persistently in response to receiving the MAC CE and based on a configured periodicity. In case of periodic CSI reporting, the wireless device may transmit periodic CSI reports based on a configured periodicity. The configuration parameters may comprise a second configuration parameter that indicates a cell switching pattern, e.g., a time domain pattern indicating first timings that the first cell is an applicable PUCCH cell for PUCCH transmission and second timings that the second cell is the applicable PUCCH cell for PUCCH transmission. The wireless device may determine the first timings and the second timings based on the second configuration parameter. The wireless device may determine the first timings and the second timings based on a reference numerology (e.g., a reference subcarrier spacing or a reference slot duration), e.g., a reference numerology associated with a reference cell. For example, the reference cell may be the first cell (e.g., the primary cell in case the first cell and the second cell are in the primary PUCCH group or the PUCCH SCell in case the first cell and the second cell are in the secondary PUCCH group).

The wireless device may determine a first report timing of a first scheduled CSI report via the first cell and a second report timing of a second scheduled CSI report via the first cell. The first report timing and the second report timing may be and/or the wireless device may determine the first report timing and the second report timing based on the CSI report configuration parameters. For example, the CSI report configuration parameters may comprise a parameter indicating a periodicity and slot offset. The first report timing and the second report timing may be and/or the wireless device may determine the first report timing and the second report timing based on the periodicity and slot offset. The wireless device may determine the first report timing and the second report timing based on a numerology (e.g., subcarrier spacing or slot duration) associated with an active BWP (e.g., active uplink BWP) of the first cell. The first report timing may be within the first timings (e.g., within the timings that first cell is the applicable PUCCH cell for PUCCH transmission) and the second report timing may be within the second timings (e.g., within the timings that the second cell is the applicable cell for PUCCH transmission). The wireless device may transmit the first scheduled CSI report via the first cell. The wireless device may transmit the first scheduled CSI report based on the first report timing being within the first timing that the first cell is the applicable cell for PUCCH transmission. The wireless device may drop (e.g., may not transmit/may cancel transmission of) the second scheduled CSI report. The wireless device may drop the second scheduled CSI report based on the second report timing being within the second timings that the second cell is the applicable PUCCH cell for PUCCH transmission and the first cell is not the applicable PUCCH cell for PUCCH transmission.

In accordance with various exemplary embodiments in the present disclosure, a device (e.g., a wireless device, a base station and/or alike) may include one or more processors and may include memory that may store instructions. The instructions, when executed by the one or more processors, cause the device to perform actions as illustrated in the accompanying drawings and described in the specification. The order of events or actions, as shown in a flow chart of this disclosure, may occur and/or may be performed in any logically coherent order. In some examples, at least two of the events or actions shown may occur or may be performed at least in part simultaneously and/or in parallel. In some examples, one or more additional events or actions may occur or may be performed prior to, after, or in between the events or actions shown in the flow charts of the present disclosure.

FIG. 37 shows an example flow diagram in accordance with several of various embodiments of the present disclosure. At 3710, a wireless device may receive: first configuration parameters of a first cell and a second cell; channel state information (CSI) report configuration parameters for CSI reports via the first cell; and a second configuration parameter indicating first timings that the first cell is an applicable physical uplink control channel (PUCCH) cell for PUCCH transmission and second timings that the second cell is the applicable PUCCH cell for PUCCH transmission. At 3720, the wireless device may determine that: a first report timing of a first scheduled CSI report via the first cell is within the first timings; and a second report timing of a second scheduled CSI report via the first cell is within the second timings. At 3730, the wireless device may transmit the first scheduled CSI report and may drop the second scheduled CSI report.

In an example embodiment, the CSI report configuration parameters, received at 3710, may be for periodic CSI reports or semi-persistent CSI reports. In an example embodiment, the wireless device may receive a medium access control (MAC) control element (CE) indicating activation of a semi-persistent CSI configuration. In an example embodiment, at least one of the first CSI report and the second CSI report may be based on the semi-persistent CSI configuration.

In an example embodiment, the CSI report configuration parameters, received at 3710, may be for transmission of CSI reports via a PUCCH of the first cell.

In an example embodiment, the second configuration parameter, received at 3710, may indicate a cell switching pattern for PUCCH transmission. In an example embodiment, the cell switching pattern may be a time-domain pattern.

In an example embodiment, the first report timing and the second report timing may be based on the CSI report configuration parameters. In an example embodiment, the CSI report configuration parameters comprise a parameter indicating a periodicity and slot offset. The first report timing and the second report timing may be based on the periodicity and slot offset.

In an example embodiment, the first report timing and the second report timing may be based on a subcarrier spacing of an active bandwidth part of the first cell.

In an example embodiment, the first timings and the second timings may be based on a reference subcarrier spacing. In an example embodiment, the reference subcarrier spacing may be a first subcarrier spacing associated with the first cell.

In an example embodiment, the first cell and the second cell may be in the same PUCCH group. In an example embodiment, the PUCCH group may be a primary PUCCH group or a secondary PUCCH group.

FIG. 38 shows an example flow diagram in accordance with several of various embodiments of the present disclosure. At 3810, a wireless device may determine: first timings that a first cell is an applicable cell for physical uplink control channel (PUCCH) transmission; and second timings that a second cell is the applicable cell for PUCCH transmission. At 3820, the wireless device may transmit a first channel state information (CSI) report in a first report timing via the first cell based on the first report timing being within the first timings. At 3830, the wireless device may drop a second scheduled CSI report, scheduled for transmission in a second report timing via the first cell, based on the second report timing being within the second timings.

In an example embodiment, the wireless device may receive a configuration parameter, wherein the determining may be based on the configuration parameter. In an example embodiment, the configuration parameter may indicate a cell switching pattern for PUCCH transmission.

In an example embodiment, the wireless device may receive CSI report configuration parameters for CSI reports via the first cell.

Various exemplary embodiments of the disclosed technology are presented as example implementations and/or practices of the disclosed technology. The exemplary embodiments disclosed herein are not intended to limit the scope. Persons of ordinary skill in the art will appreciate that various changes can be made to the disclosed embodiments without departure from the scope. After studying the exemplary embodiments of the disclosed technology, alternative aspects, features and/or embodiments will become apparent to one of ordinary skill in the art. Without departing from the

scope, various elements or features from the exemplary embodiments may be combined to create additional embodiments. The exemplary embodiments are described with reference to the drawings. The figures and the flowcharts that demonstrate the benefits and/or functions of various aspects of the disclosed technology are presented for illustration purposes only. The disclosed technology can be flexibly configured and/or reconfigured such that one or more elements of the disclosed embodiments may be employed in alternative ways. For example, an element may be optionally used in some embodiments or the order of actions listed in a flowchart may be changed without departure from the scope.

An example embodiment of the disclosed technology may be configured to be performed when deemed necessary, for example, based on one or more conditions in a wireless device, a base station, a radio and/or core network configuration, a combination thereof and/or alike. For example, an example embodiment may be performed when the one or more conditions are met. Example one or more conditions may be one or more configurations of the wireless device and/or base station, traffic load and/or type, service type, battery power, a combination of thereof and/or alike. In some scenarios and based on the one or more conditions, one or more features of an example embodiment may be implemented selectively.

In this disclosure, the articles “a” and “an” used before a group of one or more words are to be understood as “at least one” or “one or more” of what the group of the one or more words indicate. The use of the term “may” before a phrase is to be understood as indicating that the phrase is an example of one of a plurality of useful alternatives that may be employed in an embodiment in this disclosure.

In this disclosure, an element may be described using the terms “comprises”, “includes” or “consists of” in combination with a list of one or more components. Using the terms “comprises” or “includes” indicates that the one or more components are not an exhaustive list for the description of the element and do not exclude components other than the one or more components. Using the term “consists of” indicates that the one or more components is a complete list for description of the element. In this disclosure, the term “based on” is intended to mean “based at least in part on”. The term “based on” is not intended to mean “based only on”. In this disclosure, the term “and/or” used in a list of elements indicates any possible combination of the listed elements. For example, “X, Y, and/or Z” indicates X; Y; Z; X and Y; X and Z; Y and Z; or X, Y, and Z.

Some elements in this disclosure may be described by using the term “may” in combination with a plurality of features. For brevity and ease of description, this disclosure may not include all possible permutations of the plurality of features. By using the term “may” in combination with the plurality of features, it is to be understood that all permutations of the plurality of features are being disclosed. For example, by using the term “may” for description of an element with four possible features, the element is being described for all fifteen permutations of the four possible features. The fifteen permutations include one permutation with all four possible features, four permutations with any three features of the four possible features, six permutations with any two features of the four possible features and four permutations with any one feature of the four possible features.

Although mathematically a set may be an empty set, the term set used in this disclosure is a nonempty set. Set B is a subset of set A if every element of set B is in set A.

Although mathematically a set has an empty subset, a subset of a set is to be interpreted as a non-empty subset in this disclosure. For example, for set $A = \{\text{subcarrier1}, \text{subcarrier2}\}$, the subsets are $\{\text{subcarrier1}\}$, $\{\text{subcarrier2}\}$ and $\{\text{subcarrier1}, \text{subcarrier2}\}$.

In this disclosure, the phrase “based on” may be used equally with “based at least on” and what follows “based on” or “based at least on” indicates an example of one of plurality of useful alternatives that may be used in an embodiment in this disclosure. The phrase “in response to” may be used equally with “in response at least to” and what follows “in response to” or “in response at least to” indicates an example of one of plurality of useful alternatives that may be used in an embodiment in this disclosure. The phrase “depending on” may be used equally with “depending at least on” and what follows “depending on” or “depending at least on” indicates an example of one of plurality of useful alternatives that may be used in an embodiment in this disclosure. The phrases “employing” and “using” and “employing at least” and “using at least” may be used equally in this in this disclosure and what follows “employing” or “using” or “employing at least” or “using at least” indicates an example of one of plurality of useful alternatives that may be used in an embodiment in this disclosure.

The example embodiments disclosed in this disclosure may be implemented using a modular architecture comprising a plurality of modules. A module may be defined in terms of one or more functions and may be connected to one or more other elements and/or modules. A module may be implemented in hardware, software, firmware, one or more biological elements (e.g., an organic computing device and/or a neurocomputer) and/or a combination thereof and/or alike. Example implementations of a module may be as software code configured to be executed by hardware and/or a modeling and simulation program that may be coupled with hardware. In an example, a module may be implemented using general-purpose or special-purpose processors, digital signal processors (DSPs), microprocessors, microcontrollers, application-specific integrated circuits (ASICs), programmable logic devices (PLDs) and/or alike. The hardware may be programmed using machine language, assembly language, high-level language (e.g., Python, FORTRAN, C, C++ or the like) and/or alike. In an example, the function of a module may be achieved by using a combination of the mentioned implementation methods.

What is claimed is:

1. A method comprising:

transmitting, by a base station:

first configuration parameters of a physical uplink control channel (PUCCH) group comprising a first cell and a second cell;

channel state information (CSI) report configuration parameters for CSI reports via the first cell; and

a second configuration parameter of a time domain pattern indicating first timings that the first cell is the PUCCH cell for PUCCH transmission in the PUCCH group and second timings that the second cell is the PUCCH cell for PUCCH transmission in the PUCCH group;

receiving a first CSI report, scheduled for transmission via the first cell in a first report timing, based on the first report timing being within the first timings;

wherein

a second CSI report, scheduled for transmission via the first cell in a second report timing is dropped, based on the second report timing being within the second timings.

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2. The method of claim 1, wherein the CSI report configuration parameters are for periodic CSI reports or semi-persistent CSI reports.

3. The method of claim 1, wherein the CSI report configuration parameters are for transmission of CSI reports via a PUCCH of the first cell.

4. The method of claim 1, wherein the time domain pattern indicates a cell switching pattern for PUCCH transmission.

5. The method of claim 1, wherein the first report timing and the second report timing are based on the CSI report configuration parameters.

6. The method of claim 1, wherein the first report timing and the second report timing are based on a subcarrier spacing of an active bandwidth part of the first cell.

7. The method of claim 1, wherein the first timings and the second timings are based on a reference subcarrier spacing.

8. The method of claim 7, wherein the reference subcarrier spacing is a first subcarrier spacing associated with the first cell.

9. The method of claim 1, wherein the PUCCH group is a primary PUCCH group.

10. The method of claim 1, wherein the PUCCH group is a secondary PUCCH group.

11. A base station comprising:

one or more processors; and memory storing instructions that, when executed by the one or more processors, cause the base station to: transmit: first configuration parameters of a physical uplink control channel (PUCCH) group comprising a first cell and a second cell; channel state information (CSI) report configuration parameters for CSI reports via the first cell; and a second configuration parameter of a time domain pattern indicating first timings that the first cell is the PUCCH cell for PUCCH transmission in the PUCCH group and

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second timings that the second cell is the PUCCH cell for PUCCH transmission in the PUCCH group; receive a first CSI report, scheduled for transmission via the first cell in a first report timing, based on the first report timing being within the first timings;

wherein

a second CSI report, scheduled for transmission via the first cell in a second report timing is dropped, based on the second report timing being within the second timings.

12. The base station of claim 11, wherein the CSI report configuration parameters are for periodic CSI reports or semi-persistent CSI reports.

13. The base station of claim 11, wherein the CSI report configuration parameters are for transmission of CSI reports via a PUCCH of the first cell.

14. The base station of claim 11, wherein the time domain pattern indicates a cell switching pattern for PUCCH transmission.

15. The base station of claim 11, wherein the first report timing and the second report timing are based on the CSI report configuration parameters.

16. The base station of claim 11, wherein the first report timing and the second report timing are based on a subcarrier spacing of an active bandwidth part of the first cell.

17. The base station of claim 11, wherein the first timings and the second timings are based on a reference subcarrier spacing.

18. The base station of claim 17, wherein the reference subcarrier spacing is a first subcarrier spacing associated with the first cell.

19. The base station of claim 11, wherein the PUCCH group is a primary PUCCH group.

20. The base station of claim 11, wherein the PUCCH group is a secondary PUCCH group.

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